

ABSTRACT

Title of proposal: Inferential Robustness for
Defeasible and Pragmatic NLP Systems

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The construction and evaluation of the first generation of NLP systems have mainly focused on the criterion of accuracy, where leaderboards rank models or systems based on their proportion of correct predictions (Wang et al., 2019; Hendrycks et al., 2021). While work remains to further push the accuracy of these systems, the saturation of these benchmarks suggests that the next generation of these systems must satisfy more profound and complex user expectations (Kahr et al., 2025). Among these expectations is the ability to draw *inferences* consistently and appropriately across the diverse ways users may express themselves: through paraphrases, evolving or complex contexts, or through poorly-posed user inputs, such as those with incorrect assumptions or ambiguity. Accurately and consistently drawing inferences lies at the heart of many tasks in language processing.

This thesis defines and develops tools to study the *inferential robustness* of NLP systems, or the ability to draw inferences consistently across varied, complex, or even faulty input. Systems must appropriately adapt to contextual changes and reason about flawed or ambiguous user input to be effective in diverse contexts. We study inferential robustness in service of building systems that are both capable of *defeasible* reasoning—updating beliefs

as new evidence emerges—as well as *pragmatic* reasoning—reasoning about user input in context. Each work examines progressively more complex and realistic scenarios in which systems must remain robust.

We begin by presenting a method to quantify the sensitivity of natural language reasoning models to different phrasings of input. We define the metric of **paraphrastic consistency**, or the likelihood of a model’s prediction remaining consistent under different phrasings, and find that LLMs still have room to improve their paraphrastic consistency.

Then, we explore the **context sensitivity** of LLMs tasked with solving reasoning problems. We examine whether they genuinely rely on context when drawing inferences or whether they simply exploit spurious correlations in the input. We also examine their *logical consistency* by decomposing complex reasoning problems into sub-problems and measuring whether models’ predictions on individual components align with their overall judgments. Then, we introduce the metric of *inferential consistency* in defeasible reasoning settings which measures whether models make consistently correct or incorrect predictions about the same underlying fact across different contexts.

We then move to analyzing *naturalistic* but faulty user input, where models must draw inferences about the **information need of users**. First, we look at cases where users embed false beliefs in consumer health questions. Here, models must extract and assess the factuality of those beliefs in order to mitigate potential harm. Then, we shift to more generally defining and detecting **poorly-posed information-seeking questions**, where humans and models must leverage context to draw inferences about the intended interpretation of the question.

Overall, our studies show that NLP systems still exhibit various issues with inferential robustness on a variety of different tasks and inputs. We present three proposed directions to bridge these gaps. First, we propose a “*defeasible investigator*”, or a module designed to explore the underlying factors that mediate defeasible inferences, allowing us to not only better evaluate whether systems have a *holistic* understanding of situations when drawing inferences, but also construct more balanced training data that covers the full space of situational variables. Next, we propose the construction of a more diverse, contextual multi-step

defeasible reasoning dataset, δ -NLI++ designed to track the evaluation of inferences across *sequences* of information in three settings: (1) news article revisions, (2) incremental trivia (quiz bowl) questions, and (3) moral reasoning discussions. This dataset will allow us to more rigorously evaluate whether LLMs can appropriately update their beliefs with accumulating, and potentially conflicting, evidence. Finally, we propose studying the answering *strategies* that humans leverage to respond to poorly-posed questions, and comparing these strategies to those that machines use when attempting such questions, teasing apart the context-sensitivity of such strategies across different question answering communities.

Together, this thesis lays the foundation for building and evaluating the next generation of NLP systems that can meet the evolving expectations users in complex and diverse contexts.

Inferential Robustness for Defeasible and Pragmatic NLP Systems

by

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Chapter 1: Introduction

When we interact with a natural language processing (NLP) system such as a search engine, a chatbot, or a reasoning system, we typically have several expectations that, if satisfied, build trust (Jacovi et al., 2021). First and foremost, we expect the system to be *accurate*: correctly answer questions, retrieve relevant documents, or accurately reason about situations. The construction and evaluation of the first generation of NLP systems have been primarily focused on this criterion of accuracy, where leaderboards rank models or systems based on their proportion of correct predictions (Wang et al., 2019; Hendrycks et al., 2021). While work remains to increase the accuracy of these systems, the saturation of benchmarks suggests that the next generation of these systems must satisfy more profound and complex user expectations (Kahr et al., 2025).

For example, users may expect that systems make *consistent* predictions in semantically equivalent contexts, such as when querying language models for factual knowledge (Elazar et al., 2021a). Or, they may expect the system to exhibit *logical* consistency, such as symmetric consistency (e.g. if the system outputs “*Paris is the capital of France*”, then when asked about France’s capital, it should output Paris (Asai and Hajishirzi, 2020)). When using models for summarization or document-based question answering, they may hope that the system is *faithful*, or factually consistent with its input (Kryscinski et al., 2020). Additionally, since models are often considered “experts” in information tasks (Li et al., 2023), users might extend to them the exact expectations that they have of *human* experts, such as the ability to detect problematic assumptions, probe information gaps in queries, and ask relevant follow-up questions.

We can think of each of these properties as a **contract**. Following Hawley (2014)’s “commitment account” of trust, as well as Tallant (2017)’s terminology of “contractual trust”, we can consider trust in AI as contractual (Jacovi et al., 2021): the *trustor* (user) expects the AI system to maintain specific functionality or behavior, and these commitments constitute **contracts**. Contractual trust then amounts to trusting that these specific commitments or contracts will hold. Each chapter in this thesis examines progressively more complex and realistic scenarios in which NLP systems must uphold a particular contract (labeled as **Contract** in each chapter opening).

When a model upholds more “rudimentary” contracts such as obtaining a high accuracy score on a leading benchmark, it exhibits *trustworthiness* (Wright, 2010) **with respect to that contract** (Jacovi et al., 2021), and users begin to build trust with the system (Kahr et al., 2025), using it for increasingly complex tasks or in diverse situations. The usage of NLP systems in more complex scenarios, such as crisis communication (Hong et al., 2025)

or medical decision making (He et al., 2022b), gives rise to more demanding expectations. For example, we may expect the system to gracefully handle ambiguity (Min et al., 2020), asking follow-up (Meng et al., 2023) or clarification questions to disambiguate possible interpretations of the question (Lee et al., 2023). As we deploy these systems in contexts such as medical decision making, mental health support, or legal analysis, we implicitly expect them to exhibit the kind of nuanced, contextually aware, and robust reasoning that defines human intelligence. Importantly, we hope that the system recognizes its own limitations, *abstaining* from answering a user query when it does not have sufficient evidence to provide an accurate answer (Brill et al., 2002) or when the query is unanswerable (Asai and Choi, 2021). We expect that these systems are reasoning about language use *in context* (*pragmatic reasoning*): when a pregnant patient asks whether they *can* eat something, they are typically asking whether it is *safe* to do so. We expect systems to infer our informational needs from our input, sometimes despite its vagueness or ambiguity. Moreover, we expect that these systems are sensitive to context, and able to reason under uncertainty. For example, upon learning that the pregnant patient has a specific medical condition or set of circumstances, the system should be able to *update* its recommendations appropriately, weighing new evidence against prior inferences or assumptions in a principled way (called *defeasible reasoning* (Reiter, 1980)). As systems acquire a wider user base, we may even expect the system to draw upon personal preferences (Hwang et al., 2023) or past behavior (Ageev et al., 2013) to provide maximally relevant responses.

Central to upholding many of these contracts is the ability of NLP systems to draw **inferences**, or derive information from a set of premises to arrive at some conclusion (Swoyer, 2003). Accurately and consistently drawing inferences lies at the heart of many tasks in language processing, including question answering, recognizing textual entailment, text simplification, or summarization. These inferences may span many different types, from simple lexical inferences such as word sense disambiguation (Ide and Véronis, 1998) to semantic or pragmatic inferences such as entailment (Giampiccolo et al., 2007) or implicature (Grice, 1975).

Generalized language models deployed in complex scenarios must not only *adeptly* draw these inferences, but they must do so robustly. Robustness typically refers to the ability of a system to maintain its performance when faced with adversarial perturbations or out-of-distribution inputs (Hendrycks and Dietterich, 2019; Jia and Liang, 2017). For inference tasks, robustness means that systems must adapt their inferential behavior to the input they receive.

We formalize this notion as ***inferential robustness***: the ability of systems to appropriately modulate their likelihood of drawing an inference when faced with varied, complex, or even faulty input. For example, if a user asks “*Is it safe to eat sushi while pregnant?*” versus “*Can I have sushi during pregnancy?*”, a robust system should draw the same inference about food safety. **This thesis defines particular facets of inferential robustness in natural language inference and question answering, proposing metrics and methodologies to measure the extent to which LLMs uphold these capabilities in order to be effective in diverse contexts. In other words, systems that exhibit**

inferential robustness must uphold a particular set of *contracts* with users, a subset of which we define and measure in this thesis.

Of course, when we build an NLP system, it is impossible to know the whole gamut of forms user input could take. As researchers, we inevitably make assumptions about how these inputs are structured or phrased, and build benchmarks around these assumptions. These assumptions become embedded into the way we train systems, starting from data collection and annotation guidelines all the way to evaluation metrics, where the prevailing paradigm involves ranking a list of systems based on their accuracy on a set of curated benchmarks built by researchers. As researchers, we often assume that these evaluations are sufficiently exhaustive or rigorous to claim that the systems that perform well are trustworthy. However, in practice, it is often the case that users experience unexpected, undesirable, or unstable behavior from the system (Shen et al., 2023) depending on their input, violating their expectations of these systems. Not only do these experiences make it difficult for users to build trust in these systems, but they also cast doubt on the benchmarks that researchers build, since LLMs may perform well on controlled evaluations but fail to meet user expectations in real-world scenarios.

The disconnect between in-lab success and practical failure suggests that our current evaluation paradigms may be insufficient for capturing the kind of inferential robustness that users implicitly expect. This thesis proposal explores inferential robustness in two foundational NLP tasks, natural language inference (Chapters 3 and 4) and information-seeking question answering (Chapters 5 and 6). **We set our sights on building systems for question answering and commonsense reasoning that are both defeasible and pragmatic.** Importantly, we arm ourselves with insights from linguistic theory to explore the constructs (*What is the process of asking a question?*) and phenomena (*How are inferences overturned in defeasible reasoning problems?*) underlying these tasks.

Contracts in this thesis are introduced as desired behaviors (e.g. *maintaining* the likelihood of drawing an inference when the input is paraphrased) with a unifying theme of inferential robustness. The work we describe in each chapter uses a mix of theory and computational tools to define, understand, and measure the ability of models to uphold the contract. **Chapter 3** begins by exploring the sensitivity of defeasible and abductive reasoning models to paraphrasing as a way to better characterize their knowledge, contextualize their performance on benchmarks, and evaluate room for improvement on the basis of consistency. We introduce a metric of paraphrastic consistency, or the likelihood of a model’s prediction remaining consistent under different phrasings, and an accompanying dataset, ParaNLU, in order to understand the types of surface-form changes that models are sensitive to. Unlike other paraphrase-based studies, we admit sentences that are not semantically equivalent but that are functionally equivalent under the context of the reasoning problem (e.g. “*The man got a discount*” may be functionally equivalent to “*The man saved 10% with a coupon*”, even though they are not semantically equivalent). However, this study is primarily limited to laboratory-based syntactic or commonsense-based semantic perturbations, a first stop on the road to evaluating model robustness during inference.

We then turn to another core component of inferential robustness: **deep context un-**

derstanding. **Chapter 4** begins by exploring the sensitivity of NLI models to changes in context when there are spurious “giveaways”, or *annotation artifacts*. Using a dataset of manual perturbations, we show that while models *could* ignore context and rely solely on spurious correlations in inference, they actually do condition on contextual information.

Chapter 4 also further explores model understanding of situations evoked by context by decomposing complex sentences into atomic propositions to examine how models reason about granular sub-inferences as a part of their overall reasoning process in textual entailment. We introduce the concept of logical consistency, measuring whether a model’s predictions on component sub-problems align with the logical rules that should govern the overall inference. We then decompose hypothesis sentences in defeasible reasoning problems to better understand the intricacies of defeasible inference, introducing a method rooted in the linguistic theory of questions under discussion to identify the exact pieces of knowledge evaluated in defeasible inference examples. We introduce the metric of inferential consistency, designed to measure the probability that a model makes consistently correct or incorrect predictions about the same fact across different contexts.

Stepping out of the lab and into naturalistic user data, **Chapter 5** turns to domain-specific question answering. We analyze user questions with false assumptions and implications asked to a maternal-infant health question answering system, a setting in which a robust language model must proactively infer the user’s dangerous beliefs from the faulty input question and intervene in order to mitigate harm. To do so, the system must reason about the question in the context in which it was asked. We use theories of presupposition and implicature as a framework to answer questions about what linguistic form false assumptions are likely to take in user questions, as well as how strongly askers are committed to those assumptions that are linguistically embedded in their questions. We show that human experts naturally address false assumptions when answering questions, and that this behavior can be induced in retrieval-augmented generation systems.

Chapter 6 finally explores inferential robustness in the most unconstrained setting thus far: open-domain information seeking question answering. Here, we explore user questions that are poorly-posed, requiring an answerer to infer the user’s underlying information need to best answer their question. We leverage theories from question semantics to study how humans and models infer an interpretation from the user’s original expression that may exhibit a variety of problematic properties such as ambiguity or vagueness, and propose a metric of poorly-posedness inspired by [Ciardelli \(2018\)](#)’s account of question semantics. We show that when dealing with faulty questions, models struggle to correctly infer a user’s information, and when they do converge on an interpretation, it often differs from that selected by humans.

Our studies thus far suggest that we are still far from satisfying many of the expectations that users may have when interacting with NLP systems, but have helped us better understand where exactly models continue to struggle with robust defeasible and pragmatic inference. We propose three directions towards better understanding and reasoning with diverse human input. First, we propose developing a defeasible investigator, or a system designed to determine the factors that may influence a particular inference (factors influ-

encing whether or not a glass breaks may include the *height* from which it was dropped or the *material* it was dropped on), encouraging models to build more holistic understanding of situations (Chapter 4). Then, we build on our findings from Chapter 4 to propose the construction of a new defeasible inference benchmark from three data sources: developing news stories, incremental question answering, and Reddit threads describing social situations. Lastly, as a followup to our studies in Chapters 5 and 6, we propose the construction of a QA system designed to answer poorly-posed questions informed by analysis of the kinds of strategies and objectives that humans naturally use to respond to such questions.

1.1 Roadmap

Each chapter introduces one or more completed works, each designed to define and understand a particular contract. We propose three new works across Chapters 4 and 6 designed to further address these contracts.

Ch.	Task	Title (Inferential Robustness...)	Contract
3	NLI	...to Paraphrastic Variations	Paraphrastic Consistency : Systems should maintain the likelihood of drawing an inference given semantically equivalent phrasings p_i and p_j .
4	NLI	...Through Deep Context Understanding	Context Sensitivity : When the context in a reasoning problem changes in a way that affects the validity of an inference, systems should appropriately modulate their likelihood of drawing that inference. Logical Consistency : When a reasoning problem is decomposed into smaller sub-problems, systems should make predictions on each component that align with their overall judgment in a logically coherent way. Inferential Consistency : When reasoning about the same underlying fact or inference across different contexts, systems should maintain consistent predictions, either by correctly drawing the inferences in both contexts or failing in both, as opposed to succeeding in one and failing in another.
5	QA	...with Presuppositionally Problematic Questions	Making Critical Inferences : When user input contains false assumptions, systems should detect and address these problematic beliefs in order to mitigate potential harm, rather than ignoring it or simply answering the surface-level question.
6	QA	...to Poorly-Posed Information-Seeking Questions	Understanding Information Needs : When faced with poorly-posed questions that exhibit ambiguity, vagueness, or under-specification, systems should infer the user’s underlying information need rather than attempting to answer only the surface-level question.

Chapter 2: Background

The NLP community has transitioned away from “deeper” abstract semantic representations (e.g., FrameNet (Baker et al., 1998)) towards “shallower” representations (e.g., Universal Dependencies (Nivre et al., 2016)) which retain attributes of their original surface form. The culmination of this trend is to use natural language as a semantic representation itself to interact and evaluate language models. As a meaning representation, natural language can be ambiguous. To parse, understand, and communicate in natural language, humans often draw *inferences*; that is, we very frequently derive or conclude some information from a set of premises or from situational context. Inference underlies much of the interactions between humans and language models, as well as many of the tasks and applications researchers consider in traditional natural language processing, including machine translation (Vickrey et al., 2005), information extraction, and question answering (Girju, 2003).

This chapter reviews some of the prerequisites for understanding and positioning the work discussed in this thesis proposal. We discuss relevant types of inferences or reasoning (e.g. monotonic reasoning, non-monotonic reasoning, pragmatic reasoning), while weaving in computational approaches or resources for the evaluation of language models on these reasoning types. Many NLP tasks such as question answering or natural language inference employ a *range* of the few reasoning types we discuss here.

2.1 Textual Entailment

Many fundamental tasks in natural language processing involve explicitly or implicitly drawing inferences, including *semantic entailment*, which we discuss here. The inferential abilities of systems are often implicitly and unevenly evaluated during task-specific evaluation (e.g. question answering or summarization), precluding researchers from uniformly benchmarking a core set of linguistic inference capabilities of models. Here, we discuss an attempt to isolate this ability in the form of the task of textual entailment, as much of the work in this thesis proposal measures the robustness of models when performing natural language inference.

Consider the following utterance:

- (1) *Ann waltzed with Bob.*

If someone uttered this statement to us, way may conclude many things, including:

- (2) Ann danced.
- (3) Bob danced.
- (4) Ann and Bob danced together.

Entailment. We intuitively know that sentence 2–4 *must* hold true in any scenario in which (1) holds true. This example demonstrates *semantic entailment* (Heim and Kratzer, 1998), a fundamental concept in linguistics and logical reasoning which relates one or more *premise* statements to a series of *conclusion* statements that are the logical consequences of the premise(s). More precisely, S *entails* S' (denoted $S \models S'$) if and only if every possible world in which S is true is also a world in which S' is true: a speaker committing to the truth of S is then committing to the truth of S' . Entailment is monotonic, meaning that the addition of information in the premise does not change an entailment relationship from entailment to neutral or contradiction.

Logical Contradiction. *Contradiction* is the opposite of linguistic entailment: committing to the truth of statement S suffices for committing to the truth of $\neg S'$. In other words, there are no possible worlds in which the statement S and the statement S' are both conceivably true, and it is illogical or incoherent to commit to the truth of both S and S' . For example, if we commit to the truth of (1), the statement *Ann waltzed with Bob*, we cannot commit to the truth of the statement *Ann sat with Bob*, since one cannot waltz and sit at the same time.

Non-Entailment. Non-entailment is a concept weaker than contradiction: there may be possible worlds in which both statement S and statement S' hold true, but there may be at least one possible world where statement S holds, but S' does not. For example, consider S as statement (4), *Ann and Bob danced together*, and S' as statement (1), *Ann waltzed with Bob*. While it is possible that in worlds where Ann danced with Bob, the two danced a waltz, it is also conceivable that the two danced bachata or tango.

Recognizing Textual Entailment (RTE) Many fundamental tasks in natural language processing involve explicitly or implicitly drawing some of the linguistic inferences discussed above. For example, producing a grounded summary from a document D involves producing statements that are entailed from D and that do not result in any contradictions. Or, answering a multiple choice reading comprehension question may involve eliminating answer choices that are direct contradictions of the text snippet. In order to deconfound the ability of language models to perform such linguistic inference from their abilities to perform a particular task or application such as summarization, Dagan et al. (2005) introduce the task of **Recognizing Textual Entailment (RTE)** (Giampiccolo et al., 2007). The task of RTE involves an *applied* notion of linguistic semantic entailment: a text T (*premise*) entails a *hypothesis* H if, “typically, a human reading T would infer that H is most likely true.”

Text: Time Warner is the world’s largest media and Internet company.

Hypothesis: Time Warner is the world’s largest company.

Textual Entailment Relation: False

Text: Google files for its long awaited IPO.

Hypothesis: Google goes public.

Textual Entailment Relation: True

Early approaches to the task of recognizing textual entailment involved measuring the lexical similarity between the hypothesis and the original text (Jijkoun et al., 2005) or matching the predicate-argument structure of the text and the hypothesis (Romano et al., 2006). MacCartney and Manning (2007) introduced *natural logic* for the task of textual entailment, a logical inference system designed to operate over natural language. Their system, *NatLog*, incrementally transformed premises into hypotheses by way of *atomic edits* over spans of word tokens in both pieces of texts. They train a model for atomic entailment classification and then predict an entailment relation for each atomic edit, reporting results on RTE, as well as on the FRACAS test suite, a collection of natural language inference problem carefully designed to test the semantic inference capabilities of systems.

Understanding and Characterizing Textual Entailment. While the task of textual entailment as defined above is indeed reminiscent of (in spirit, perhaps equivalent to) logical entailment or deductive consequence, these concepts are insufficient to adequately capture the scope of RTE. Consider the example below:

Text: There are people who speak Portuguese in Goa.

Hypothesis: There are people who speak Portuguese in India.

Textual Entailment Relation: True

Using logical reasoning tools alone does not allow us to easily conclude that the hypothesis is logically entailed from the text, since the inference relies on a piece of background knowledge: that Goa is in India. Korman et al. (2018) argue that making sense of the textual entailment task by way of concepts in logic (material implication, strict implication, etc.) is not viable, as there is a mismatch in many of the definitional aspects of these concepts and the phenomenon present in the RTE data. They argue that *inferential* approaches to analyzing and defining the task of textual entailment are much closer to capturing the scope of the task. For example, *pragmatic* norms (Grice, 1975) may influence the nature of the inferences we draw when determining textual entailment labels. We cover some of these types of inferences in Section 2.3. In addition, an inference-based approach allows us to admit cases of textual entailment that rely on common knowledge, including commonsense reasoning, geographic, or lexical knowledge. This is evident in the second example in the RTE section above where annotators labeled the hypothesis as entailed from the text, demonstrating that entailment systems must have knowledge of the fact that “IPO” is synonymous with “going public.”

In some cases, it is possible that textual entailment can fail to be strictly monotonic:

Text: The spaghetti at Corner Bistro in New Jersey was incredible. If you enjoy eating cardboard drenched in ketchup.

Hypothesis: The spaghetti at Corner Bistro in New Jersey is good.

Textual Entailment Relation: False

Using logical reasoning tools, we may be inclined to conclude that the hypothesis does indeed entail from the text, since if something is incredible, it is inherently good. However, a human would be justifiable in inferring that the last sentence in the text is clearly sarcasm and may label the hypothesis as not textually entailed. After a series of arguments, [Korman et al. \(2018\)](#) propose redefining textual entailment as: T textually entails H if and only if typically, a human reading T would be justified in inferring the proposition expressed by H from the proposition expressed by T .

This modified definition embodies one of the main concerns of this thesis: computational systems must draw nuanced inferences that are governed by the norms and realities of human communication, including those that require pragmatic reasoning and non-monotonic reasoning. This is vital to building systems that are not only adept at the (arguably, synthetic) task of textual entailment, but that are generally able to respond in human-like ways across a variety of applications.

Natural Language Inference (NLI). [Bowman et al. \(2015\)](#) modernize the task of RTE, arguing for its importance as a tool for general purpose evaluation of semantic representations. The formulation of textual entailment as presented in [Dagan et al. \(2005\)](#) and in the FraCas test suite ([Cooper et al., 1996](#)) involved a two-way classification of test items as *entailment* or *not entailment*. [Bowman et al. \(2015\)](#) rebrand textual entailment as the task of *natural language inference*, or the task of inferring the relationship between a *premise* sentence and a *hypothesis*. Namely, the task involves determining whether the premise P entails, contradicts or is neutral in relation to a **hypothesis** H ([Bowman et al., 2015](#)). Their definition and task framing implicitly (and in some cases, explicitly) draw on many of the points argued by [Korman et al. \(2018\)](#), farming out the process of determining that a hypothesis is entailed from a premise as a high likelihood of a human drawing the inference given the premise.

As such, they presented SNLI, the first large-scale human-written corpus of sentence pairs, designed to reflect naturalistic language usage. Since the inferential process may depend on the annotator, SNLI crowdsources multiple human judgments of entailment for each example, and adopts the majority label as gold:

Premise: A smiling costumed woman is holding an umbrella.

Hypothesis: A happy woman in a fairy costume holds an umbrella.

Label: Neutral

Crowdsourced Judgments: N N E C N

Premise: A soccer game with multiple males playing.

Hypothesis: Some men are playing a sport.

Label: Entailment

Crowdsourced Judgments: E E E E E

Premise: A man inspects the uniform of a figure in some East Asian country.

Hypothesis: The man is sleeping.

Label: Contradiction

Crowdsourced Judgments: C C C C C

The task of NLI is a useful tool for a wide variety of applications and analyses in other areas of NLP including claim and fact checking (Scirè et al., 2024), the verification of answers in QA (Chen et al., 2021) as well as for contract review (Koreeda and Manning, 2021) or false presupposition detection (Kim et al., 2021). However, it has been shown that annotators sometimes struggle to agree on entailment labels (Pavlick and Kwiatkowski, 2019; Jiang and Marneffe, 2022), highlighting some of the complexities or challenges of the task (Nie et al., 2020b). Some have proposed shifting to a numerical values to capture the plausibility (Chen et al., 2020) of entailments in SNLI.

Williams et al. (2018a) present an improved corpus of NLI examples, MULTI-NLI, spanning ten different genre of both spoken and written English. Several others have proposed other resources and datasets for NLI, such as adversarial NLI (Nie et al., 2020a) or cross-lingual NLI (Conneau et al., 2018). The work we discuss in Chapter 4 is largely concerned with the robustness of inferences that models must draw in the task of natural language inference.

Artifacts and Partial-Input Baselines in NLI. Gururangan et al. (2018) and Poliak et al. (2018) observe that the crowdsourcing protocols adopted by Bowman et al. (2015) and others lead to the creation of data with annotation artifacts that enable partial-input baselines (e.g., hypothesis-only baselines) to perform well above a majority-class baseline. For SNLI, Poliak et al. (2018) report that an InferSent (Conneau et al., 2017) hypothesis-only baseline surpasses a majority baseline by 35 points. A similar effect is observed by Rudinger et al. (2020a) in the δ -NLI data, with an update-only RoBERTa (Liu et al., 2019) model achieving 15 points above a majority baseline. However, low accuracy of partial-input baselines does not necessarily entail that the dataset is free of artifacts (Feng et al., 2019).

Others have shown the brittleness of systems trained on these datasets (Naik et al., 2018; Glockner et al., 2018a), attributing such behavior to the construction of sentence pairs with large word overlap in the premise and hypothesis or strong negation words. In Chapters 3 and 4, we study how *sensitive* language models are to context when drawing inferences, evaluating their ability to overcome such artifacts in their input.

2.2 Beyond Textual Entailment: Non-Monotonic Inference

Defeasible Inference. This thesis proposal argues that accurate and effective NLP systems must be able to modulate their likelihood of drawing inferences in the on varied or

complex input in service of making sure they can perform more complex forms of reasoning. Semantic entailment, discussed in the previous section, is monotonic, meaning that the addition of information in the premise does not change an entailment relationship from entailment to neutral or contradiction. Whenever a hypothesis is a logical consequence of a premise, the hypothesis is a consequence of the superset of the premise. Consider the example below:

Premise: A dog is chasing a tennis ball in the park.

Hypothesis: An animal is in the park.

Label: Entailment

Here, we see that the hypothesis is entailed from the premise due to hyponymy (dog $\models$ animal). This inference would still hold if we considered a superset of information from this premise, e.g. “A dog is chasing a tennis ball in the park on a rainy day”: the inference is not weakened or defeated (canceled or overturned) by the addition of more information in the premise.

On the other hand, *defeasible reasoning* is a mode of reasoning in which inferences or conclusions may be altered or withdrawn in light of new evidence (Reiter, 1980). For example, given the context “A group of people perform on a stage”, a natural conclusion “A group performs for an audience” may be weakened upon learning that the group is at a rehearsal.

Humans engage in defeasible reasoning all the time: we may start with *plausible* assumptions (*normally*, statement S holds), or *default* assumptions, but upon learning more information or details, this assumption or inference may need revisiting. For example, if we see the lights out in a shared lab workspace on a weekend, we may infer or assume that no one is present. However, we may be forced to revisit that assumption if we encounter a wet umbrella outside the entrance of the lab. Defeasible reasoning often entails entertaining multiple pieces of information that sometimes conflict with each other.

Many kinds of inferences we draw are defeasible, including *pragmatic* inferences such as implicature (Grice, 1975) or *abductive* inferences (Peirce, 1934). Implicatures, or unstated inferences drawn in light of conversational context (“I ate some of the cake” may invite the inference “I did not eat all of the cake”) may be *cancelled* with the addition of more information (e.g. “In fact, it was so good that I ate all of it!”).

Defeasible Reasoning in AI. General-purpose intelligent systems must be capable of a wide variety of reasoning modes, such as probabilistic reasoning or inductive reasoning. In order for AI systems to be deployed in diverse or high-risk contexts such as healthcare or the legal domain, they must also be capable of defeasible reasoning. For example, when interacting with patients in clinical settings, language models providing medical recommendations must revisit earlier guidance upon learning more about relevant parts of the patient’s medical history. Or, when providing legal advice, a system must revisit earlier issued guidance upon learning that the user resides in a different country.

Reiter (1980) introduce *default logic*, a logic framework designed to accommodate commonly occurring default assumptions. First order logic (FOL) is monotonic, and his logic supports nonmonotonic reasoning by way of defaults, to represent propositions like *most* birds fly. Others have proposed slightly different approaches to formalizing and framing defeasible reasoning in classical AI including the definition of *warranted* propositions to represent “ideal” reasoners (Pollock, 1987) or the extension of the logic of *belief* (Levesque, 1990).

To support the study and development of defeasible reasoning in language models, Rudinger et al. (2020b) introduce the task of defeasible NLI. Traditionally, NLI involves determining whether a **premise** P entails, contradicts or is neutral in relation to a **hypothesis** H (Giampiccolo et al., 2007). Defeasible NLI studies how introducing *new* information about the situation by way of a third *update* sentence, can affect the inference both in terms of magnitude and direction. Specifically, when the premise P and hypothesis H of an NLI pair are *neutral* in relation to one another, defeasible NLI studies whether a third **update** (U) sentence strengthens or weakens H . They employ a working definition of this task as:

Given premise P , a hypothesis H is defeasible if there exists an update U (consistent with P) such that a human would find H less likely to be true after learning U . Specifically, an update U is called a **weakener** if, given a premise P and hypothesis H , a human would most likely find H less likely to be true after learning U ; if they would find H more likely to be true, then we call U a **strengtheners**.

For example, consider this example taken from the SNLI split of δ -NLI:

Premise:	In front of a decorated Christmas tree, four adults and two children sit or stand amongst gifts, and one of the adults gives a child a drink of water.		
Hypothesis:	One of the adults gives a child a drink of water from a sippy cup.		
Update 1:	The child is in a stroller.	Update 1 Type:	Strengtheners
Update 2:	The child spills the water everywhere.	Update 2 Type:	Weakener

In essence, defeasible NLI studies how introducing *new* information about the situation by way of the update sentence, can affect the original entailment inference both in terms of magnitude and direction. The δ -NLI dataset has been used to develop systems that ask clarification questions in moral and social situations (Pyatkin et al., 2023), improve the defeasible reasoning abilities of large language models (Madaan et al., 2021), as well measure the ability of models to provide rationales that explain model predictions (Brahman et al., 2021).

Others have investigated the role of nonmonotonic reasoning in characterizing language model knowledge of generics (Allaway et al., 2023; Allaway and McKeown, 2025; Leiding et al., 2024) or of causal reasoning (Cui et al., 2024).

This thesis proposal studies the robustness of models tasked with drawing defeasible inferences, including pragmatic and abductive inferences.

Abductive Inference. Abductive inferences, or inference to the most plausible explanation (Peirce, 1974), are also defeasible and may be defeated in light of additional information. Such inferences are hypotheses that can best fit one or more incomplete observations. For example, if we wake up one morning and see that the lawn outside our apartment is wet, we may infer that it rained last night. However, upon seeing a bulletin in the lobby that the maintenance team installed new sprinklers yesterday, we are forced to cancel that inference. Abductive inference requires systems to reason about potential *implicit* contexts beyond the provided observations. Hobbs et al. (1993) posit that abductive inference is at the core of how readers interpret text, introducing a framework to describe the interpretation of a sentence implemented in a discourse processing system. Bhagavatula et al. (2019) introduce abductive NLI, an abductive reasoning task formulated binary classification. α -NLI examples consist of two observations O_1 and O_2 (where O_2 occurs some point in time after O_1), h^+ , a plausible hypothesis, and h^- , an implausible hypothesis. Given both observations, the task is to determine which hypothesis is more plausible.

Many researchers have investigated the performance of transformer-based architectures on this task (Kadiķis et al., 2022), including developing systems designed to generate text explaining given observations in natural language (Zandie et al., 2023) or designing learning methods to adapt language models to reason about the plausibility of explanations (Zhao et al., 2023). Others have constructed automated reasoning systems designed to use abductive reasoning to generate natural language proofs (Sprague et al., 2022).

We study the *paraphrastic* robustness of large language models when drawing abductive inferences in Chapter 3.

2.3 Pragmatic Inference

Pragmatics is largely concerned with understanding the “effects of context” (Korta and Perry, 2024) in language. Let us assume your friend hands you a cup of tea and says to you: “*The tea is warm.*” Upon hearing this, you may infer that the tea is not boiling, considering that your friend would have used a stronger term like “hot” or “boiling” if that were the case. Such inferences are called *scalar implicatures* (Hirschberg, 1985), and typically emerge when listeners consider alternative expressions the speaker might have chosen, but deliberately avoided. Here, we cover a few types of pragmatic phenomena relevant to the works in this thesis proposal: *presupposition*, *implicature*, and *information need*. We also briefly discuss computational resources and approaches to analyzing these phenomena.

Presupposition. Presuppositions are implicit assumptions in utterances taken for granted by discourse participants (Beaver, 1997). The question “*What vitamins should I stop taking after becoming pregnant?*” presupposes “*I was taking vitamins before becoming pregnant.*” Presuppositions can often be detected solely by the presence of a lexical or syntactic trigger (Levinson et al., 1983). In the example above, the word *stop* presupposes that an activity was already in motion.

Domain or world knowledge is often needed to capture presuppositions in real-world data

that are not apparent from lexical or syntactic cues (Abusch, 2002). For example, the question “*Are multiple ultrasounds dangerous for my baby?*” does not directly result in non-trivial trigger-based presuppositions. However, the asker of the question does presuppose that the effects of an ultrasound are *additive* and hence asks about whether that additive effective is *dangerous*. Presuppositions *project* out of negations (Heim, 1983): both the utterance “*She finished graduate school*” and its negation “*She didn’t finish graduate school*” trigger the presupposition “*She attended graduate school*”. While it is largely accepted that presuppositions can be conventional properties of particular lexical items (e.g. *stop* or *both*), it has been argued that at least *some* presuppositions can be derived conversationally (Simons, 2013), since certain presuppositions exhibit *contextual defeasibility*. Listeners express different levels of confidence when drawing presuppositions based on different triggers (e.g. *only* as in “These muffins *only* have blueberries in them” resulted in a lower median projectivity rating than *know* as in the sentence “Billy *knows* that Martha has a new BMW”). Some triggers are *hard* triggers (e.g. *both*), requiring the presupposition to hold in order for the sentence to be considered well-formed, while other triggers are considered *soft*, and can be canceled (De Marneffe et al., 2019).

Understanding and analyzing language model behavior with presupposition has recently received sustained focus in the NLP community. Jeretic et al. (2020) find that language models trained on NLI corpora recognize presuppositions only arising from certain triggers such as *only* or possessives. Echoing this finding, Parrish et al. (2021b) find that while LLMs are able to correctly draw inferences in simple cases, they struggle to recognize more complex presuppositions in situations where human judgments also reveal trigger and context complexity. Others release resources and analysis designed to support the training and evaluation of presupposition recognition by language models (De Marneffe et al., 2019; Ross and Pavlick, 2019; Jiang and de Marneffe, 2019; Sravanthi et al., 2024; Azin et al., 2025; Atwell et al., 2025). Questions often carry presuppositions, and many works analyze cases of *false* presuppositions. Kim et al. (2021) present the first study of presuppositions in Google search queries using the Natural Questions (NQ) dataset (Kwiatkowski et al., 2019a) that are unanswerable due to false presuppositions. Other work has looked at Google queries with *questionable* assumptions (Kim et al., 2022) and false presuppositions in open-domain Reddit questions (Yu et al., 2022) as well as in other application contexts such as medical QA (Kaur et al., 2024).

Implicature. Implicature is a type of pragmatic inference that is *suggested* by an utterance as opposed to part of its literal meaning (Grice, 1975). Consider the question “*Do most babies fit in newborn clothes?*” While the speaker understands that newborn clothes fit *some* babies, their question *implies* that not all babies fit in newborn clothes.

Some implicatures are related to lexical items or syntactic structure of utterances. For example, the statement “*These vitamins are in gumdrop form, but are healthy*” implies that gumdrops are usually *not* healthy. Others are a function of a speaker’s intent, beliefs, and other contextual elements (Zheng et al., 2021). While they are a part of the content of an utterance, these implicatures are not *at-issue* (e.g. the main point under discussion (Potts,

2004; Koev, 2018)) and are not encoded by the linguistic properties of a sentence (Allott, 2018). Consider the question “*Should I be worried if my child still can’t read by age seven?*”. Reasoning about asker belief, we may conclude that they are implying that most children can read by that age, and that their child’s development may be delayed.

From a linguistic perspective, presupposition and implicature represent different levels of an asker’s commitment to the propositional content of the inference (Peters, 2016). Presuppositions are already a part of an asker’s world model. In contrast, implicatures are *likely* beliefs that may be negated in an asker’s subsequent utterances. Consider the question “*Is it normal for my baby to move more than usual when closer to due date?*” with both the presupposition “*There are factors that contribute to changes in fetal movement as the due date approaches*” and the implicature “*It may not be necessary to be concerned if there is a significant increase in fetal movement close to the due date.*” While both are false, the presupposition is stronger, and is clearly in need of addressing in a potential answer. As illustrated, these distinct phenomena must be dealt with differently when answering a question.

Computational studies of implicature have mainly focused scalar implicature (Schuster et al., 2020; Zheng et al., 2021; Kabbara and Cheung, 2022; Jeretic et al., 2020). It has been shown that LLMs such as ChatGPT (OpenAI, 2024) and Cohere’s Command (Cohere et al., 2025) struggle with drawing inferences involving implicatures (Ruis et al., 2023; Nizamani et al., 2024).

Common Ground. Pragmatic reasoning often involves reasoning about the mental state of discourse participants. For example, conversational participants often contribute and rely on *common ground*, or the body of statements treated as mutual knowledge by all conversational participants (Stalnaker, 2002). Presuppositions are vital to the maintenance of common ground (Stalnaker, 1975): speakers may add content to the common ground by way of presupposition (e.g. “*I need to pick my kids up from school*” presupposes that I have children). By presupposing something, a speaker is signaling that they believe it is already a part of the common ground. If the presupposition remains unchallenged, it confirms that all participants share that common knowledge. Understanding what is and is not part of the common ground from speaker utterances is vital when drawing pragmatic inferences. Chapters 5 and 6 discuss work where inferring what is and what is not a part of the common ground from utterances can improve LLM responses to questions.

2.4 Modeling Primer

Here, we quickly cover some of the details of the natural language processing models that serve as the objects of study in this thesis proposal. Importantly, models are designed to create distributed *representations* of text that can be leveraged downstream.

Transformers. All of the language models we study in this thesis proposal are developed using the *transformer* architecture (Vaswani et al., 2017). Unlike predecessors such as the

LSTM (Hochreiter and Schmidhuber, 1997), the transformer uses a *self-attention mechanism*, designed to compute attention weights **between all pairs of tokens in an input sequence**. This allows the model to capture longer-range dependencies across a wider context in its distributed representation of the input.

For a given input sequence, the model projects each token into query (Q), key (K), and value (V) vector representations, and computes attention as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2.1)$$

where d_k is the dimension of key vectors. Transformers utilize **multi-head attention**, which applies the attention mechanism in parallel:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (2.2)$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$ and W^O is a learned output projection matrix. Each Transformer layer consists of multi-head attention followed by a position-wise feed-forward network, with residual connections and layer normalization applied around each sub-layer. Attention is computed over the whole input.

Encoder-Only Models. Encoder-only models, such as BERT (Devlin et al., 2019a) and RoBERTa (Liu et al., 2019) produce contextual representations of input by using only the encoder component of the Transformer module. Unlike autoregressive decoder models, encoders can use *bidirectional* attention, allowing each token to attend to both preceding and following tokens in the sequence. These models are typically pretrained using masked language modeling objectives, where random tokens are masked and the model learns to predict them from surrounding context.

Decoder-Only Models. Decoder-only models use only the *decoder* module of the Transformer architecture to generate text autoregressively. These models employ unidirectional attention: each token can only attend to the previous tokens in the sequence, preventing the leakage of information from future token positions. Decoder models are pretrained using a language modeling objective, learning to predict the next token given all previous tokens.

Retrieval-Augmented Generation Language models that rely solely on parametric knowledge for text generation have been shown to *hallucinate*, or produce factually inaccurate content (Li et al., 2024). In addition to parametric knowledge, it has been shown that *retrieving* information prior to generation can help mitigate hallucinations (Lewis et al., 2020). This is called retrieval-augmented generation (RAG). RAG models use an input sequence x to retrieve text documents z and subsequently use z as additional context when generating the target sequence y . These models involve two components: a *retriever*, such as Dense Passage Retriever (DPR) (Karpukhin et al., 2020b) or Contriever (Tizacard et al.,

2022) and a *generator*, which can be any decoder or encoder-decoder model. The retriever and generator are trained jointly. Chapter 5 uses a RAG architecture to produce higher quality text for questions asked in health context.

In-Context Learning and Prompting. Supervised finetuning was the primary way to adapt large-scale pretrained language models in the early ages of large language models. Language models such as BERT (Devlin et al., 2019b) were trained further with adaptive or task-specific finetuning (Howard and Ruder, 2018). However, it has been shown that modern LLMs such as GPT-3 (Brown et al., 2020) are able to adapt and perform tasks through *in-context learning*, where the model can learn to perform a new task based solely on instructions and examples provided in an input *prompt*. This learning process is performed without any parameter updates to the underlying model. By scaling to billions of parameters, models have been shown to perform tasks such as arithmetic reasoning, translation (Zhang et al., 2023), and question answering from a few demonstrations, and sometimes, from instructions of tasks alone (Kojima et al., 2022). Researchers have developed methods to carefully craft input prompts (Schulhoff et al., 2024), a process instrumental to effectively interacting with LLMs. For example, frameworks like *dspy* (Khatab et al., 2024) remove some of the randomness around constructing a working prompt through manual trial and error, instead introducing a programming model to learn prompts that optimize defined metrics.

Chapter 3: Inferential Robustness to Paraphrastic Variations¹

We begin our study of inferential robustness by defining and studying a basic contract that NLP systems must uphold: paraphrastic robustness. Users may naturally vary the phrasing they use when asking NLP systems to answer a question or reason about their input, and when systems are presented with *semantically* equivalent contexts, they must draw similar inferences. Thus, inferential robustness in this scenario amounts to **maintaining** the same likelihood of drawing the particular inference the reasoning problem requires *in the face of different phrasings*. This chapter presents a method to quantify the sensitivity of natural language reasoning models to different phrasings of input.

Contract: Paraphrastic Consistency. NLP systems should maintain the likelihood of drawing a particular inference in a reasoning problem given semantically equivalent phrasings p_i and p_j .

3.1 Introduction

As discussed in Chapter 2, natural language as a meaning representation is highly ambiguous (Schubert, 2015). While versatile and compact, it leaves open the possibility that systems are not robust to *different ways* of expressing the same meaning in natural language.

Benchmark evaluation datasets such as SNLI (Chapter 2) consist of a collection of reasoning problems designed to probe particular aspects of commonsense knowledge, with each example represented by a *singular* linguistic expression. When a system gets a particular example correct, it is only evidence that it was able to correctly reason *for the particular phrasing used in the example*, allowing for the possibility of systems that can correctly solve one form of a reasoning problem, but not others. Conversely, if a model gets a question wrong, how can we tell if the error was due to a failure in language understanding or a failure in reasoning?

Consider the defeasible reasoning example in Figure 3.1. A language model finetuned on the δ -NLI dataset (Rudinger et al., 2020b) may correctly predict that the original update sentence *strengthens* a human’s belief in the hypothesis sentence. However, different linguistic expressions of that same update sentence may yield high variance in a model’s predictions. If

¹Neha Srikanth, Marine Carpuat, and Rachel Rudinger. 2024a. [How often are errors in natural language reasoning due to paraphrastic variability?](#) *Transactions of the Association for Computational Linguistics*, 12:1143–1162

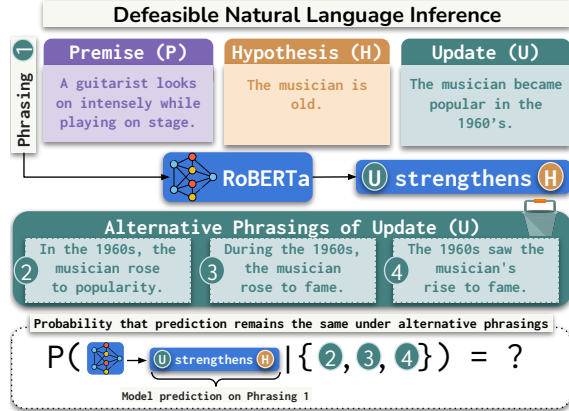


Figure 3.1: δ -NLI instance with a set of paraphrased update sentences. We study **paraphrastic consistency**, or the probability that a model’s prediction for two phrasings of the same problem match.

models stay *consistent* in the face of paraphrastic variability, we may conclude that correctly reasoning about one expression is indicative of an understanding of that *reasoning problem*, a desirable property of teaching machines to reason entirely in natural language.

We explore the sensitivity of natural language reasoning models to paraphrasing as a way to better characterize their knowledge and reasoning ability, contextualize their performance on evaluation sets, and evaluate room for improvement on the basis of consistency. Under the assumption that world and linguistic knowledge are separable, our study attempts to disentangle the two by generating examples that hold the required world knowledge of a reasoning problem constant while modifying its surface form, a problem formulation with connections to causality (Stolfo et al., 2023) and counterfactual invariance (Veitch et al., 2021; Kaushik et al., 2020).

To study this, we build on top of two NLU datasets—Abductive NLI (Bhagavatula et al., 2019) and Defeasible NLI (Rudinger et al., 2020b) by collecting paraphrases of reasoning problems using *label-preserving* paraphrasing, a functional change of traditional paraphrasing that preserves the semantics of the *core reasoning problem*. Our dataset, PARANLU, contains 7,782 human-elicited and 7,295 model-elicited paraphrases across 1000 reasoning problems spanning both datasets. We select diverse examples to paraphrase ranging in difficulty (Sakaguchi et al., 2021) and model confidence. Our dataset is *entirely manually validated*, ensuring semantic equivalence while maximizing paraphrase diversity.

We measure **paraphrastic consistency** (P_C), or the likelihood of model’s prediction remaining consistent under different phrasings to understand the types of surface-form changes that models are sensitive to. We study the relationship between consistency and various data conditions and modeling paradigms, exploring factors such as data source, example difficulty, model complexity, and training dynamics. *For their given accuracy level*, we find that models still have room to improve on paraphrastic consistency. Since no model demonstrates high accuracy *and* high paraphrastic consistency, we conclude that attempts to measure their

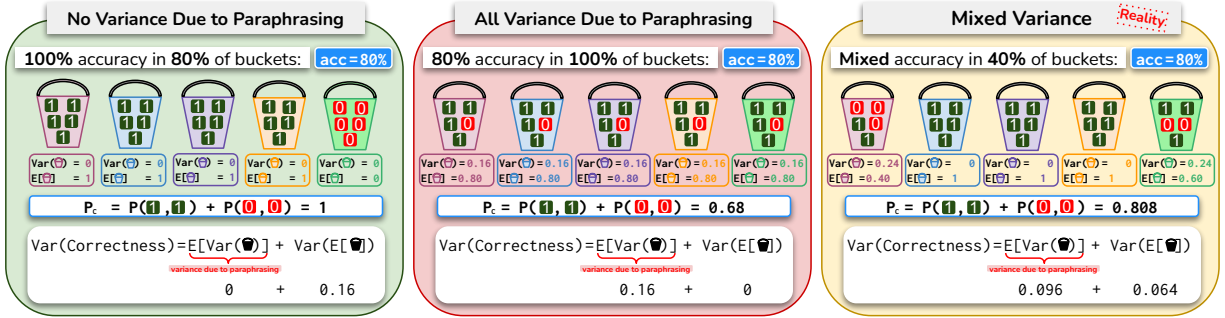


Figure 3.2: Three scenarios, all with equivalent overall accuracy of 80%, illustrating different distributions of variance in model predictions leading to different P_C values. Buckets represent underlying commonsense reasoning problems. Numbers within buckets represent model correctness on 5 paraphrases. Most models achieve a mix of accuracy within and across buckets of paraphrased examples.

reasoning abilities will be confounded by inconsistencies in their *linguistic* abilities.

3.2 Paraphrastic Consistency

In principle, natural language reasoning tasks like abductive NLI and defeasible NLI require the ability to linguistically *decode* the meaning of the text that expresses an underlying problem, as well as the knowledge and reasoning capabilities to *solve* the underlying problem.

By analogy, consider evaluating a child’s understanding of the concept of addition. Instead of simply presenting them with a mathematical expression (say, $7 + 7$), we write a *word problem* that can be answered by (1) understanding the situation in natural language, (2) recognizing that the answer corresponds to the mathematical reasoning problem $7 + 7$, and finally, (3) solving $7 + 7$. If the child answers incorrectly, we must figure out whether they did not understand the goal of the word problem or were not able to perform the arithmetic in order to evaluate their mathematical reasoning ability.

For models tasked with natural language reasoning problems, teasing apart these two failure modes (namely, deficiencies in language understanding versus deficiencies in knowledge or reasoning) requires more than reporting test set *accuracy*. The design of natural language reasoning test sets does not facilitate this type of analysis: if a test set contains 100 different natural language reasoning problems, and a model correctly answers 80% of them, which failure mode should we attribute the 20% of errors to?

For practitioners, it is useful to characterize performance by measuring *paraphrastic consistency* alongside accuracy: how likely is it that a model’s prediction for a natural language reasoning problem will remain the same given a *different phrasing* of the problem? We collect a dataset that changes the *language* of NLI examples while maintaining the *underlying logic* of the problem to tease the two apart. For a test example, x , we collect a set of paraphrases $\{x'_1, x'_2, \dots\}$, which we call a *bucket* (🗑️). After collecting many such buckets, we can directly

estimate the probability that a model’s prediction for any two paraphrases belonging to the same bucket will be the same.

3.2.1 Measuring Paraphrastic Consistency

When authoring a test example for a natural language reasoning task, a crowdworker has many linguistic choices for expressing the underlying reasoning problem. If the purpose of the resulting test set is to evaluate a model’s ability to perform the underlying reasoning task, then ideally the crowdworker’s choice of phrasing would have no effect on the model’s performance. In practice, however, it is known that language models exhibit some degree of sensitivity to paraphrastic variation (Verma et al., 2023; Jiang et al., 2021). To quantify this effect, we pose the following counterfactual question: **If a given test question had been written differently, what is the probability that a model would still receive the same credit for its prediction on the paraphrased question?**

Quantitatively, we introduce a metric of *paraphrastic consistency*, P_C , defined as the probability that a model’s predictions for two paraphrases of the same problem, x'_i and x'_j , are either both correct or both incorrect (provided the ground truth labels of x'_i and x'_j match). To formalize P_C , we define the following terms:

- $R_{\blacksquare}$: a discrete random binary variable indicating whether a model’s prediction of a paraphrased reasoning problem, x' , is correctly predicted (1) or incorrectly predicted (0).
- $\theta_{\blacksquare}$: $\mathbb{E}[R_{\blacksquare}]$, or the average correctness (i.e., accuracy) of the paraphrased problems ($x'_1, x'_2, \dots$) in a particular bucket.
- A : Overall accuracy of model M over a set of natural language reasoning problems. This is equivalent to $\mathbb{E}[\theta_{\blacksquare}]$ across all buckets.

For a binary classification task, where $y \in 0, 1$ denotes the gold label, we then define P_C as:

$$P_C = \underbrace{P(M(x_i) = y, M(x_j) = y)}_{\text{prob. of both predictions correct}} + \underbrace{P(M(x_i) \neq y, M(x_j) \neq y)}_{\text{prob. of both predictions incorrect}} \quad (3.1)$$

P_C can be estimated from the accuracies of paraphrase buckets, $\theta_{\blacksquare}$, as follows:

$$P_C = \mathbb{E}[\theta_{\blacksquare}^2] + \mathbb{E}[(1 - \theta_{\blacksquare})^2] \quad (3.2)$$

Informally, the *paraphrastic consistency* of a natural language reasoning model is its ability (or lack thereof) to make the same predictions on paraphrased inputs that, in principle, should yield the same answer. We note that the metric P_C *cannot be computed from a standard test set containing only one phrasing per test example*; to estimate P_C we collect paraphrases of test examples, as described in §3.4.

Figure 3.2 illustrates the computation of P_C for different patterns of model behavior. If $P_C = 1$, the model is either entirely correct or incorrect on all paraphrases of the same problem, as in the left-most panel of Figure 3.2 where each bucket contains only 1’s or only 0’s. In this case, no errors can be attributed to paraphrastic variance. The minimum P_C occurs when every paraphrase bucket has the same accuracy, as shown in the middle panel; in this case *all* errors are likely due to paraphrastic variability. In practice, P_C lies between these two extremes and *some*, but not all, errors are due to paraphrasing, as in the right-most panel. Modern NLP evaluation sets usually consist of a collection of independent reasoning problems each represented by singular natural language expression, and practitioners often make claims about the reasoning capabilities or world knowledge given a model’s accuracy on such evaluation sets. However, as depicted in Figure 3.2, accuracy presents an incomplete picture of performance: in all three scenarios, the overall accuracy remain 80%, but only the first scenario, in which the model makes equivalent predictions given many alternate phrasings of a reasoning problem, results in a high P_C .

P_C in Practice. P_C can be interpreted as the probability that given two phrasings of the same reasoning problem, the model’s two predictions are either both correct or both incorrect. This summary metric allows us to capture the *reliability* of a model’s prediction: how confident can we be that a model’s prediction would remain correct (or incorrect) if it had been phrased differently? While P_C is lower-bounded by a function of accuracy, we design it as a metric *complementary* to accuracy in order to better characterize model performance, diagnose modeling errors, and benchmark the linguistic reasoning capabilities. For example, when two models achieve similar accuracies, computing their respective paraphrastic consistencies may help as an additional point of comparison. Just as desired model accuracy may be dependent on the application in which it is deployed, different settings may mandate varying appropriate P_C values and practitioners must decide on tolerable thresholds based on their use case.

3.2.2 Proportion of Variance Attributable to Paraphrasing

Decomposing the *total variance in correctness* across all paraphrased examples in all buckets gives us a clearer picture of the two failure modes described in the opening of §3.2. Using the law of total variance (Weiss et al., 2006), we decompose the total variance of the correctness across all *paraphrased* examples (the variance of R for all paraphrased problems in all buckets) into two terms: the average variance of correctness within a bucket, $\mathbb{E}[\text{Var}(R_{\bullet})]$, and the variance in mean correctness (accuracy) across buckets, $\text{Var}(\mathbb{E}[(R_{\bullet})])$, or simply, $\text{Var}(\theta_{\bullet})$.

$$\text{Var}(R) = \underbrace{\mathbb{E}[\text{Var}(R_{\bullet})]}_{\text{variance from paraphrasing}} + \text{Var}(\theta_{\bullet}) \quad (3.3)$$

This breakdown allows us to better identify the *source* of the variance in correctness. The first term, commonly known as *unexplained variance*, measures variance attributable to paraphrasing (henceforth denoted as **vap**) within a bucket. The second, commonly known

as *explained variance*, represents variance *across buckets* due to inherent differences in latent characteristics (e.g. difficulty) of different problems. If paraphrasing has no effect, the variance of correctness in each bucket ($\text{Var}(R_{\blacksquare})$) is zero and consequently, the **vap** should also be zero. Most evaluation paradigms cannot directly measure **vap**, since during data collection, multiple surface forms of the same reasoning problem are not collected. We replicate the conditions under which original examples were produced to simulate different linguistic expressions annotators *could* have chosen.

Mathematical manipulation yields:

$$P_C = 1 - \underbrace{P(M(x_i) = y, M(x_j) \neq y)}_{\text{prediction flips to incorrect}} - \underbrace{P(M(x_i) \neq y, M(x_j) = y)}_{\text{prediction flips to correct}} \quad (3.4)$$

$$P_C = 1 - \underbrace{\mathbb{E}[\theta_{\blacksquare} \cdot (1 - \theta_{\blacksquare})]}_{\text{prediction flips to incorrect}} - \underbrace{\mathbb{E}[(1 - \theta_{\blacksquare}) \cdot \theta_{\blacksquare}]}_{\text{prediction flips to correct}} \quad (3.5)$$

$$P_C = 1 - 2 \cdot \mathbb{E}[\theta_{\blacksquare} \cdot (1 - \theta_{\blacksquare})] = 1 - 2 \cdot \underbrace{\mathbb{E}[\text{Var}(R_{\blacksquare})]}_{\text{vap}} \quad (3.6)$$

Although the primary metric we report throughout the paper is P_C , Equation 3.6 highlights the direct negative relationship between P_C and variance attributable to paraphrasing (**vap**).

Related to **vap** is the *proportion* of total variance attributable to paraphrasing (**pvap**), which is simply **vap** divided by the total variance:

$$\text{PVAP} = \text{VAP} / \text{Var}(R) \quad (3.7)$$

Model Confidence and P_C . We choose to characterize paraphrastic consistency via variance in *correctness* instead of variance in *model confidence*. Confidence represents a model’s *overall* estimate in the correct answer, and thus conflates confidence in linguistic understanding and problem solving. A low confidence in the *correct label* may indicate that the model understood what the problem was asking, but was unable to reach the answer (or vice versa). Models trained to optimize for *accuracy* are not calibrated to explicitly encode confidence in linguistic decoding or problem-solving ability.

3.3 Reasoning Tasks

We study paraphrastic consistency across two commonsense reasoning tasks: defeasible reasoning (§3.3.1) and abductive reasoning (§3.3.2).

3.3.1 Defeasible Reasoning

Defeasible reasoning is a mode of reasoning in which inferences or conclusions may be altered or withdrawn in light of new evidence (Reiter, 1980). For example, given the context

Context (<i>C</i>)	Original Target (<i>T</i>)	Admissible Paraphrase	Inadmissible Paraphrase
O1: My cell phone lanyard broke on Wednesday O2: My wife was able to repair the lanyard so I did not have to buy one.	h^+: It looked like it was ripped or torn. h^-: It didn't look like it was ripped or torn.	h^+: It seemed to be split and frayed. h^-: It did not appear to be damaged.	h^+: The screen had wear and tear. h^-: It looked like it was in good shape, but apparently it wasn't.
O1: Mike was graduating high school later in the day. O2: When he got back home, Mike regretted skipping graduation.	h^+: Mike's parents said they couldn't go, but they did. h^-: Mike's parents didn't show up.	h^+: Despite Mike's parents' claims that they couldn't, they went. h^-: Mike's parents were absent from the graduation ceremony.	h^+: His parents mentioned to him they couldn't go. h^-: Nobody showed up to see Mike's parents.
P: PersonX stops eating fast food. H: PersonX is seen as better.	U: PersonX's high school friends think this is strange.	U: This was considered weird by PersonX's friends from high school.	U: PersonX's alumni from high school find this to be peculiar.
P: Two men walking down the sidewalk carrying skateboards. H: Two people are walking to the skate park.	U: Both men are adjusting their wheels as they walk.	U: As they stride along, both men are changing the settings on their wheels.	U: While walking the men re-wax their boards.
H: It's okay to cancel an job interview.	U: You are too lazy to get out of bed.	U: You lack the motivation to leave the comfort of your bed.	U: You're feeling sick and can't make yourself rise from bed.

Table 3.1: Examples of paraphrased abductive NLI (rows 1 and 2) and defeasible NLI (rows 3 – 5) problems in PARANLU. We show paraphrases written by crowdworkers that are admissible under our definition of label-preserving paraphrasing, and rejected paraphrases that did not meet the label-preserving criteria.

“A group of people perform on a stage”, the natural conclusion “A group performs for an audience” may be weakened upon learning that the group is at a rehearsal.

To study defeasible reasoning in language models, [Rudinger et al. \(2020b\)](#) introduce the task of defeasible natural language inference (NLI). Traditionally, NLI involves determining whether a **premise** *P* entails, contradicts or is neutral in relation to a **hypothesis** *H* ([Giampiccolo et al., 2007](#)). When the premise *P* and hypothesis *H* are neutral in relation to one another, defeasible NLI studies whether a third **update** (*U*) sentence strengthens or weakens *H*. Namely, a human may determine *H* more likely to be true when *U* is a strengthener, and less likely to be true when *U* is a weakener.

Premise: A woman in shorts throwing a bowling ball down a bowling alley.
Hypothesis: A woman is getting a strike!
Update: The bowling ball falls in the gutter first. **Update Type:** Weakener

[Rudinger et al. \(2020b\)](#) also introduce δ -NLI, a dataset that extends three existing natural language datasets: SNLI ([Bowman et al., 2015](#)), SOCIAL-CHEM-101 ([Forbes et al., 2020a](#)), and ATOMIC ([Sap et al., 2019](#)). For each premise-hypothesis pair (or just hypothesis, in the case of δ -SOCIAL), crowdworkers write multiple strengthening and weakening updates, ensuring a balance. Defeasible NLI is a binary classification task that involves predicting whether the update sentence is a strengthener or a weakener (e.g. the original update in Row 5 of Table 3.1 is a weakener).

Adopting the terminology from [Srikanth and Rudinger \(2022\)](#), we distinguish between *context* parts of examples, and *target* parts of examples, with our study of consistency concerning *target* portions of examples. For δ -NLI, we consider the premise P and hypothesis H as *context* sentences, and the update U sentence as the target (Table 3.1).

3.3.2 Abductive Reasoning

Abduction is inference to the most likely explanation ([Peirce, 1974](#)). Such inferences are hypotheses that can best fit one or more incomplete observations.

Observation 1:	George decided to buy a TV.
Observation 2:	It turned out just as he'd hoped.
Hypothesis -:	The TV had a cracked screen.
Hypothesis +:	Upon taking it home and unpacking it, he placed it where he wanted it.

Given the example above, any human would most likely infer the second hypothesis over the first to explain the two observations. [Bhagavatula et al. \(2019\)](#) introduce abductive NLI, an abductive reasoning task formulated binary classification. α -NLI examples consist of two observations O_1 and O_2 (where O_2 occurs some point in time after O_1), h^+ , a plausible hypothesis, and h^- , an implausible hypothesis. Given both observations, the task is to determine which hypothesis is more plausible. We treat O_1 and O_2 as context (C) and h^+ and h^- as the target (T) portion of the example.

3.4 Constructing PARANLU

We study paraphrastic consistency by paraphrasing the target portions (T) of examples from α -NLI and all three data sources in δ -NLI (δ -SNLI, δ -ATOMIC, and δ -SOCIAL). For an original example x , we collect a *bucket* of paraphrased examples x_i such that the context portions C and gold label l remain identical, while the target portions T are rewritten as *label-preserving paraphrases*. This allows us to modify the surface form of the example while retaining the underlying commonsense reasoning problem.

Label-Preserving Paraphrases. We construct quasi-paraphrases ([Bhagat and Hovy, 2013](#)) of *target* sentences in reasoning problems by loosening the requirement of semantic equivalence. Given a natural language reasoning task L , a textual instance x of task L with label $\ell_L(x)$, we say that x' is a *label-preserving* paraphrase of x if:

1. $\ell_L(x) = \ell_L(x')$
2. x' does not contradict any context (C) in x .
3. x' remains consistent with the situation evoked by x .

Label-preserving paraphrases are *functionally* equivalent: target sentences (hypotheses in α -NLI and updates in δ -NLI) may introduce small bits of information as long as the same scenario is plausibly described. Consider the following δ -NLI example: **P:** *A man stands*

	# original	# paraphrases	mean # paraphrases/ex
α -NLI	250	2098	8.4 ± 1.2
δ -SNLI	250	1980	7.9 ± 1.4
δ -ATOMIC	250	1869	7.5 ± 1.6
δ -SOCIAL	250	1835	7.3 ± 1.8

Table 3.2: We sample 250 problems from α -NLI and δ -NLI datasets. Total number and mean number of paraphrases depended on validation.

in front of a cashier and kiosk at a grocery store **H:** *He is smiling* **U:** *“The man got a discount.”* While the sentence *“The man saved 10% with a coupon”* is not semantically equivalent to the strengthening update sentence, it is a valid, label-preserving paraphrase since it retains the logic of the problem and the label. Table 3.1 shows examples of admissible and inadmissible paraphrases under our definition. Label-preserving paraphrases represent alternative, but equivalent expressions annotators *could* have chosen when writing the original problem that employ similar world knowledge. This is different from *label-altering* edits proposed by Gardner et al. (2020), where minimal human edits that shift the target label were used to create examples for measuring linguistic robustness.

We describe our example selection process for annotation (§3.4.1) and crowdsourcing protocol for collecting paraphrases and summary statistics of our dataset (§3.4.2).

3.4.1 Original Example Selection

We adopt a stratified sampling strategy to obtain diverse examples for annotation that vary in difficulty. To obtain such examples, we leverage AFLITE (Sakaguchi et al., 2021), an adversarial filtering (Zellers et al., 2018) algorithm designed to partition datasets based on difficulty using pre-computed dense embeddings of examples fed into an ensemble of n logistic regression classifiers. At each iteration of AFLITE, members of the ensemble are trained on random partitions of m examples and evaluated on the remaining validation examples. Each validation example is assigned a score, computed by the proportion of correct predictions. The top- k examples with scores above a threshold τ are subsequently added to the easy partition of the dataset and filtered out, and the process repeats until less than k instances are removed in a particular iteration. Including both types of examples, easy and hard, ensures that PARANLU can support analysis that investigates whether models are inconsistent only on certain *classes* of examples (e.g. those filtered out due to lexical artifacts).

Defeasible NLI. We use the train splits of δ -SNLI, δ -ATOMIC, and δ -SOCIAL to source examples for annotation, since they are large enough to meaningfully partition using AFLITE. We partition each train set into 3 sections: (1) examples to finetune the ROBERTA-base models used to embed examples (ROBERTA_{embed}), (2) training examples for models used

for consistency analysis (ROBERTA_{analysis}), and (3) a pool from which examples are sampled for annotation. We partition at the premise-hypothesis (P - H) level to avoid leakage, since multiple examples may contain the same P - H pair, but different updates (U).

For each dataset, we pre-compute example embeddings with the ROBERTA_{embed} model, and run AFLITE with the $n = 64$ linear classifiers, $\tau = 0.75$, $k = 500$, and $m = 5000$.² Examples in the annotation pool with $\tau \leq 0.75$ are added to the *easy* subset, and the those with $\tau > 0.75$ are labeled as *difficult*. We then finetune a separate ROBERTA-large model on ROBERTA_{analysis} and use it to obtain predictions on examples in the annotation pool. Based on the ROBERTA_{analysis} model confidence in the *gold label*, we sample 125 examples from the easy subset, as determined by AFLITE, in a round-robin fashion for each decile between 0 and 1. We repeat this to collect 125 examples from the difficult subset.

Abductive NLI. Since the publicly-released α -NLI dataset only contains examples that survived adversarial filtering (“difficult” examples), we reached out to the authors to obtain *easy* examples that were filtered out. We source our original examples from the test split of α -NLI. We train a ROBERTA-large model on examples from the train split of α -NLI and follow the same stratified sampling protocol on 125 examples from each of the easy and difficult subsets, according to the confidence of the ROBERTA_{analysis} in the correct label.

3.4.2 Paraphrased Example Collection

We obtain 250 examples per dataset (α -NLI, δ -SNLI, δ -ATOMIC, δ -SOCIAL), resulting in 1000 examples for which we collect paraphrases.

Crowdsourcing. We use Amazon Mechanical Turk to collect paraphrases of the target portions of each example. Workers are shown context sentences and must write a paraphrase of the target sentence(s) according to the definition of label-preserving paraphrasing presented earlier.

We abstract the underlying reasoning task away, presenting α -NLI examples as short stories that require paraphrasing middle sentences, and δ -NLI examples as scenarios with weakening or strengthening evidence. In order to encourage diversity of paraphrases, we display the Jaccard similarity between tokens in the original sentence and the paraphrase as workers typed. Figure A.12 shows our annotation interface and instructions for collecting paraphrases of α -NLI and δ -NLI problems respectively.

Workers provide 3 paraphrases of both plausible and implausible hypotheses for α -NLI examples and 3 paraphrases of updates for δ -NLI examples. In the case of α -NLI, we randomly pair together plausible and implausible hypotheses written by the same worker to construct paraphrased examples. Each example was annotated by 3 workers. See Appendix A.17 for more details.

²We keep all hyperparameters unchanged from Sakaguchi et al. (2021) with the exception of m which we reduce from 10000 to 5000 to account for the smaller training set partitions, as in Herlihy and Rudinger (2021a).

Example	O1: George decided to buy a TV.
	h^+ : He took it home, unpacked it, and placed it where he wanted it.
	O2: Things had turned out just as he'd hoped.
Accepted	✓ George unboxed the TV and placed it on his mantel-piece.
	✓ He removed the plastic and positioned the TV where he had planned to put it.
	✓ George brought the new TV home and mounted it on the wall.
Rejected	✗ George came home and drank a cup of coffee.
	✗ He returned the TV.

Table 3.3: An α -NLI example and paraphrases that were accepted and rejected from crowdworkers.

Paraphrase Example Validation. Ensuring semantic equivalence between paraphrased examples and original reasoning problems is essential to our study. Inadvertent removal of the crux of the reasoning problem while paraphrasing may result in invalid examples. Table 3.3 includes an α -NLI example with paraphrases that were both accepted and rejected from crowdworkers based on our definition of label-preserving paraphrasing. The first and third *accepted* paraphrases both introduce new pieces of information (“mantelpiece”, “mounted it on the wall”) but do not violate the situation evoked by the original problem. In contrast, the first *rejected* paraphrase is incompatible with the situation and the second rejected paraphrase does not retain the plausibility of the hypothesis.

We opt to have an author validate all paraphrases, each within the context of the problem.³ A second author and two external annotators annotated a sample of 100 paraphrases, again labeling each as either valid or invalid. We obtain a Fleiss’s Kappa (Fleiss, 1971) value of $\kappa = 0.81$ between all validators—the two authors and two external validators. This measures agreement *on the criterion of label-preservation*, reflecting whether a paraphrase written by a crowdworker was of high enough quality to admit into our dataset, as opposed to a measurement of agreement on the correct label given paraphrased examples.

Dataset Overview. Our resulting dataset, PARANLU, contains 1000 examples uniformly split across α -NLI, δ -SNLI, δ -ATOMIC, and δ -SOCIAL. Table 3.2 shows the total number of post-validation paraphrased examples per data split, and the statistics of sizes of buckets.

	α -NLI						δ -SNLI						δ -ATOMIC						δ -SOCIAL					
	A_O	A_T	$A_{\blacksquare}$	P_C	$\tilde{A}_{\blacksquare}$	$\tilde{P}_C$	A_O	A_T	$A_{\blacksquare}$	P_C	$\tilde{A}_{\blacksquare}$	$\tilde{P}_C$	A_O	A_T	$A_{\blacksquare}$	P_C	$\tilde{A}_{\blacksquare}$	$\tilde{P}_C$	A_O	A_T	$A_{\blacksquare}$	P_C	$\tilde{A}_{\blacksquare}$	$\tilde{P}_C$
Lexical (BoW)	44.8	52.4	44.2	100	52.4	100	58.0	55.7	53.7	82.2	52.6	80.9	49.2	53.6	51.4	76.5	53.2	76.5	57.6	61.8	51.4	78.2	54.9	78.5
BiLSTM	53.5	51.6	54.2	100	51.6	100	62.0	68.0	57.6	73.2	60.4	74.2	52.8	67.4	54.2	73.1	61.1	74.8	60.4	72.0	52.4	71.7	59.7	72.8
RoBERTa	53.6	83.5	56.4	69.8	81.5	86.3	51.2	86.7	53.8	74.8	84.6	90.1	53.6	82.6	54.8	76.2	77.9	87.1	51.6	90.9	56.9	74.3	87.8	91.9
DeBERTa-V3	85.6	90.6	73.5	78.4	77.3	79.7	76.8	91.2	70.4	82.8	80.5	84.1	70.4	88.1	66.8	82.7	81.8	87.3	78.4	94.1	71.9	82.2	78.6	83.7
Unified RoBERTa	—	—	—	—	—	—	66.0	85.7	59.9	78.0	81.5	88.6	65.6	84.5	62.8	80.7	78.8	86.8	70.8	90.4	65.7	77.7	83.3	87.7
GPT-3 Curie	46.8	53.5	46.2	80.1	51.4	79.1	52.8	48.9	51.1	89.3	51.3	88	52.0	49.5	52.8	90.2	50.6	89.6	55.6	49.9	57.4	91.3	53.0	90.9

Table 3.4: Consistency across modeling architectures. A_O is accuracy on original examples in PARANLU, A_T is full test set accuracy, $A_{\blacksquare}$ is accuracy on paraphrased examples ($\tilde{A}_{\blacksquare}$ is corrected), and P_C is paraphrastic consistency ($\tilde{P}_C$ is corrected). No model achieves both high $\tilde{A}_{\blacksquare}$ and $\tilde{P}_C$ (columns highlighted in blue). Optimizing $\tilde{A}_{\blacksquare}$ may come at the cost of $\tilde{P}_C$, or vice versa. We highlight in red the set of *Pareto-optimal* points, or those that are not strictly dominated in $\tilde{A}_{\blacksquare}$ or $\tilde{P}_C$ by any other model.

3.5 Consistency on Human Paraphrases

We first examine several different models’ behavior on PARANLU to measure robustness to different linguistic expressions. While language models such as RoBERTa are trained on vast amounts of text that may instill some paraphrastic consistency, especially given label-preserving paraphrases that not be semantically equivalent, other non-pretrained models without access to such knowledge may falter. We characterize the progress of models with respect to P_C to understand whether factors such as training setups (training from scratch, supervised finetuning, prompting) and model complexity (ranging from bag-of-words representations to GPT-3) affect consistency.

3.5.1 Model Variants

We train 5 different types of models per data source. For all models, we use the same set of examples that were used to finetune the RoBERTa_{analysis} models introduced in §3.4.1.

Bag of Words. We train bag-of-words models (BoW) using `fasttext`, an off-the-shelf text classification library, with a maximum of 4-grams (Joulin et al., 2017a) for 5 epochs with the default learning rate of 0.1.

BiLSTM. We train end-to-end BiLSTM models using the architecture from Conneau et al. (2017) and initialize them with GloVe embeddings (Pennington et al., 2014). We use 3 fully connected layers for classification with max pooling. After tuning on the development sets, models are trained for 10 epochs with early stopping and a batch size of 64.

RoBERTa. We use the RoBERTa_{analysis} models in §3.4.1, and add one more setting for defeasible examples in which we finetune a RoBERTa-large model on all combined data

³We explored using NLI models to automatically ensure semantic equivalence, but found it too strict of a formulation to capture the spirit of label-preserving paraphrasing.

α -NLI	Given a story with a beginning and an end , select from two choices the middle sentence that is most plausible and best explains the beginning and end of the story.
δ -NLI	Given a Premise sentence, a Hypothesis sentence is defeasible if there exists an Update sentence (consistent with the Premise) such that a human would find the Hypothesis less likely to be true after learning the Update . An Update is called a weakener (abbreviated W) if given a Premise and Hypothesis , a human would most likely find the Hypothesis less likely to be true after learning the Update ; if they would find the Hypothesis more likely to be true, then the Update is called a strengthener (abbreviated S). Given a Premise , a Hypothesis , and an Update sentence, assign either W or S to the Update sentence.

Table 3.5: Few-shot prompts for α -NLI and δ -NLI tasks. Both prompts were presented to GPT-3 along with 36 in-context examples.

across the 3 data sources in δ -NLI, which we refer to as a unified ROBERTA model. All ROBERTA-large models were finetuned for 2 epochs with a learning rate of 2e-5 and a batch size of 32.

DeBERTa. We finetune DeBERTa-v3-large (He et al., 2022a) for 2 epochs with a learning rate of 5e-6 and a batch size of 16.

GPT-3. Lastly, we experiment with prompting GPT-3 (Brown et al., 2020) using TEXT-CURIE-001 (prompts in Table 3.5). For α -NLI, we randomly sample 36 examples from the training set and include instructions derived from those shown to the crowdworkers that annotated the α -NLI dataset. For δ -NLI, we randomly sample 12 examples per dataset (36 in-context examples total) and include the task definition from Rudinger et al. (2020b) in the prompt. Since we cannot reliably extract a softmax distribution over binary classes for our tasks (GPT-3 is not a classification model), we calculate model confidence in a particular class by extracting log probabilities associated with the tokens for both labels and normalize them.

3.5.2 Results

For all models, we compute P_C (Equation 4.2). In addition, we undo the biasing effects of the stratified sampling according to model confidence (§3.4.1) and report a corrected version of paraphrastic consistency, $\tilde{P}_C$, by weighting the expectations in Equation 4.2 according to the distribution of model confidences in the correct label of the corresponding test set. We compute four accuracy metrics: (1) accuracy on original examples in the PARANLU (A_O), (2) accuracy on the test set of the original dataset (A_T), (3) accuracy on all paraphrases

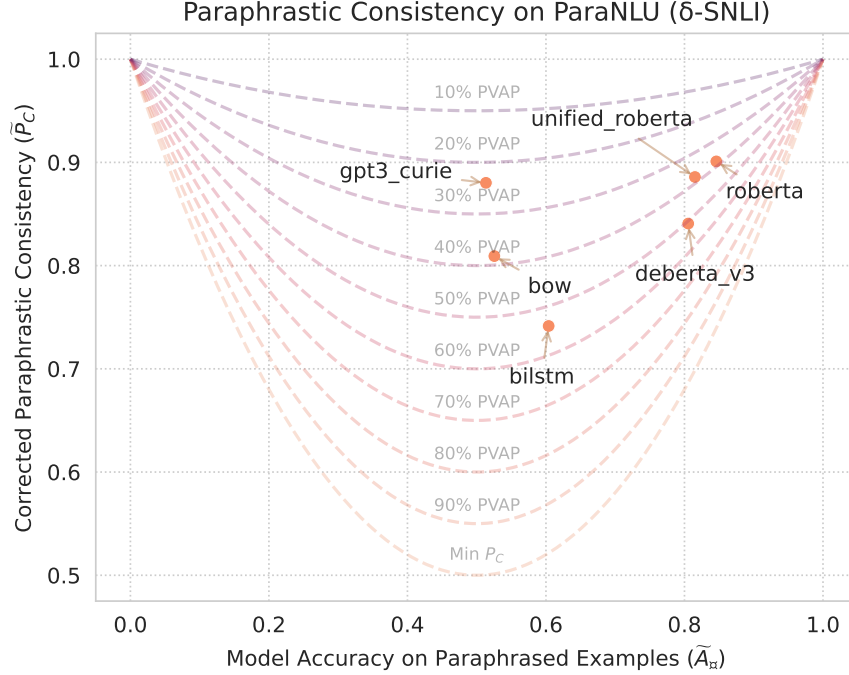


Figure 3.3: Paraphrastic consistency ($\tilde{P}_C$) of different models on δ -SNLI paraphrased examples. All models still have room for improvement in $\tilde{P}_C$ for their accuracy level. Here, we add supporting lines to denote varying levels of the proportion of variance attributable to paraphrasing, or **pvap**.

across all buckets ($A_{\blacksquare}$), and (4) *corrected* accuracy on all paraphrases across all buckets ($\tilde{A}_{\blacksquare}$) which is weighted in the same manner as $\tilde{P}_C$ to undo stratification effects.

Table 3.4 shows these accuracy metrics along with P_C and $\tilde{P}_C$ for all models across all data sources. We highlight $\tilde{A}_{\blacksquare}$ and $\tilde{P}_C$, as they are *complementary* metrics meant to jointly assess model performance. Some models optimize $\tilde{A}_{\blacksquare}$ at the cost of $\tilde{P}_C$, or vice versa. To capture models that balance both, we highlight $(\tilde{A}_{\blacksquare}, \tilde{P}_C)$ points that are *Pareto-optimal* for each dataset. The highest performing model according to $\tilde{A}_{\blacksquare}$, ROBERTA, earns a $\tilde{P}_C$ of around 0.9, indicating room to improve on its paraphrastic consistency **for its accuracy level**. We observe that a GPT-3-CURIE model with minimal prompt engineering (we simply use the definition of defeasible inference directly from Rudinger et al. (2020b)) along with a handful of in-context examples has a $\tilde{P}_C$ value competitive with a ROBERTA model finetuned on thousands of examples. A stronger GPT-3 variant may better perform defeasible reasoning while maintaining a competitive P_C .

Figure 3.3 visualizes the relationship between accuracy and $\tilde{P}_C$ for models on δ -SNLI examples (Figure A.13 shows similar plots for δ -ATOMIC and δ -SOCIAL examples). For each model, we plot $\tilde{A}_{\blacksquare}$ on the x-axis and $\tilde{P}_C$ on the y-axis along with two types of supporting curves for the δ -SNLI split of PARANLU. The curve with the lowest minima (labeled **Min P_C**) indicates the theoretical lower bound for P_C **given a particular accuracy level**: if all the

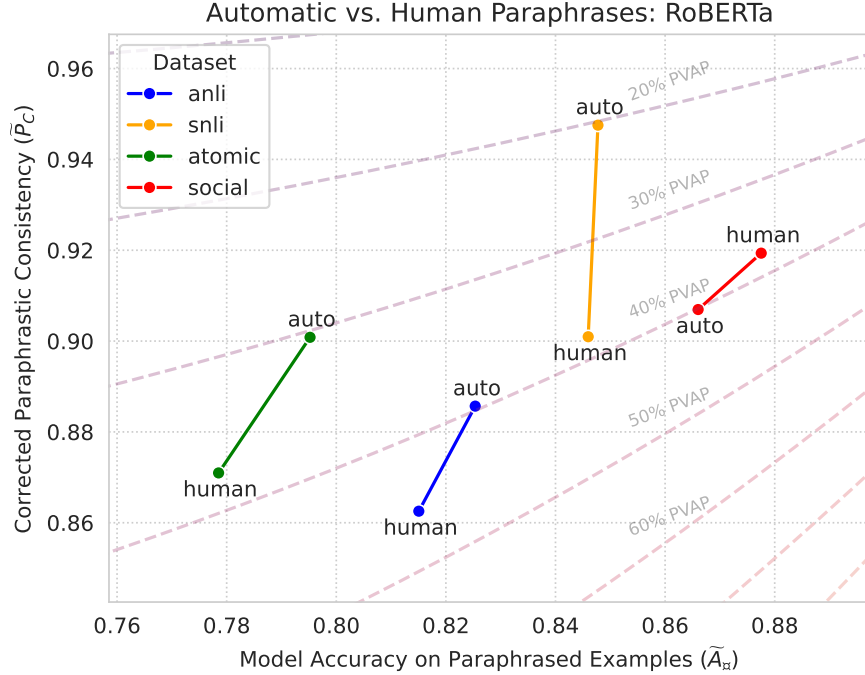


Figure 3.4: ROBERTA-large model on automatic versus human paraphrases. Models are more consistent on automatic than human paraphrases. Dashed lines indicate varying levels of **pvap**.

variance in model correctness (Equation 3.3) is attributable to the paraphrasing variance term present in Equation 3.6, then the minimum possible value for P_C is $1 - 2 * (Acc * 1 - Acc)$, where $Acc * (1 - Acc)$ is the variance of a Bernoulli random variable with probability Acc . In addition to this theoretical lower bound, we plot curves indicating the proportion of total variance attributed to paraphrasing (labeled %PVAP). As P_C increases, less variance is attributed to variance within buckets due to phrasing. Visualizing this relationship makes it clear that accuracy alone provides an incomplete picture of model performance. A perfect model would reside in the top right of Figure 3.3, not only achieving high accuracy but also high paraphrastic consistency. Models with similar accuracies may have largely different $\tilde{P}_C$ values, indicating to practitioners how sensitive they are to problem phrasing.

We now turn to a series of experiments to better characterize paraphrases in PARANLU as well as contributing factors to model’s P_C value using our best performing model, ROBERTA.

3.6 Paraphrase Source

Human-written paraphrases in PARANLU span all of the transformations delineated by Bhagat and Hovy (2013), sometimes involving more complex reasoning that falls between linguistic and world knowledge. Such paraphrases were elicited by providing humans

the entire example, encouraging them to engage with both the reasoning problem itself and the wide scope of possible meaning-preserving transformations. To understand the utility of label-preserving paraphrases, we compare a ROBERTA model’s behavior on our human-written paraphrases with paraphrases generated automatically, as previous studies have explored (Verma et al., 2023).

Using our ROBERTA models (§3.5.1), we probe the relationship between paraphrastic consistency ($\tilde{P}_C$) and the *source* of paraphrased examples. Are models more robust to the paraphrastic transformations produced by automatic paraphrase generation models, or do they only struggle with the more complex, example-aware transformations made by humans?

Since human paraphrases and model paraphrases are generated by different processes, and are thus drawn from different distributions, they may exhibit different properties. Paraphrase generation models are predisposed to biases arising from n-gram frequency effects.⁴ However, reasoning models should exhibit consistency *regardless* of whether correct answers are phrased as high-probability sentences under a language model.

We use two models (§3.6.2) to automatically paraphrase target sentences in original examples and compare model P_C on automatic and human paraphrases.

3.6.1 Experimental Setup

For each original example in PARANLU, we sample paraphrases of targets from generation models. As with humans, we elicit paraphrases of target sentences: update sentences for δ -NLI and both hypothesis sentences for α -NLI. In contrast to the human elicitation process, however, we do *not* provide any context sentences to generation models. While this limits the scope of possible paraphrases, it allows us to gauge the value of exposure to context during paraphrasing.

We adopt a generate-then-validate scheme and have an author again validate all target paraphrases to ensure that their consistency with our definition of label-preserving paraphrasing. In the case of α -NLI examples, where there are multiple target sentences, we randomly pair valid paraphrases together, resampling where necessary when numbers of valid generated hypotheses are unequal.

3.6.2 Paraphrase Generation Models

Quality-Controlled Paraphrase Generation. We use a QCPG model (Bandel et al., 2022), a controllable paraphrase generation system that conditions on a 3-dimensional vector encoding semantic similarity and lexical and syntactic distances. We pool all paraphrases from a per-example sweep of these hyperparameters.

GPT-3. In addition to a supervised, explicitly controllable paraphrase generation model, we elicit paraphrases from GPT-3 using 10 in-context examples of paraphrases randomly

⁴For example, a generation model is less likely to paraphrase “*The camera zooms out to show the man spraying the car with soap*” to “*The camera zooms out to servicemen sprinkling the automobile with soap*”.

		lex	syn	sem
α -NLI	<i>automatic</i>	25.0	20.3	74.3
	<i>human</i>	35.3	26.8	64.0
δ -SNLI	<i>automatic</i>	24.1	18.5	76.3
	<i>human</i>	34.4	24.6	67.4
δ -ATOMIC	<i>automatic</i>	30.4	19.2	66.7
	<i>human</i>	36.9	22.4	58.7
δ -SOCIAL	<i>automatic</i>	30.8	22.2	70.1
	<i>human</i>	40.0	24.5	60.1

Table 3.6: Lexical (**lex**) and syntactic (**syn**) diversity of human and automatic paraphrases and semantic similarity (**sem**) as compared to original target sentences. Human paraphrases are more diverse than automatic ones: they exhibit higher lexical and syntactic diversity and lower semantic similarity on all datasets (bolded).

sampled from PARANLU. Setting temperature to 0.7, we sample 9 paraphrases from TEXT-DAVINCI-002 per target sentence from original examples and similarly validate them to ensure label preservation.

3.6.3 Results

In total, we generate 7,295 valid paraphrased examples across 1000 examples by pooling together all *valid* examples from both QCPG and GPT-3 and evaluate our ROBERTA models on these examples. Figure 3.4 plots $\tilde{A}_{\blacksquare}$ on the x-axis and $\tilde{P}_C$ on the y-axis for human-generated and automatically-generated paraphrased examples for each dataset. On all datasets with the exception of δ -SOCIAL, we observe that models have a higher $\tilde{P}_C$ value on automatically generated paraphrased examples than on human-elicited paraphrases. We hypothesize that this pattern may not hold for δ -SOCIAL due to the fact that the dataset does not contain premise sentences, and hence has a smaller scope for more complex transformations involving context. This result suggests that reasoning models may be more robust to the simpler, in-distribution types of paraphrase transformations that automatic paraphrase generation models produce than to those written by human annotators, indicating that over-reliance on evaluation using synthetically generated data may be misleading.

Paraphrase Diversity. To dissect this result further, we measure lexical diversity, syntactic diversity, and semantic similarity (Bandel et al., 2022) of target paraphrases and original target sentences. *Lexical distance* is measured by the normalized character-level minimal edit distance between the bag of words (Bandel et al., 2022), and *syntactic distance* is computed as the normalized tree edit distance between the third level constituency parse trees of the original target and the paraphrased target (Iyyer et al., 2018). We measure semantic similarity using BLEURT (Sellam et al., 2020) as in Bandel et al. (2022). Across all four data

	artifacts...	partial-input			full-input			P_C	$\tilde{P}_C$
		A_O	$A_{\blacksquare}$	$\tilde{A}_{\blacksquare}$	A_O	$A_{\blacksquare}$	$\tilde{A}_{\blacksquare}$		
δ -SNLI	<i>likely</i>	100	81.8	54.6	54.3	56.6	85.8	74.7	91.4
	<i>unlikely</i>	0	20.7	6.9	45.5	48.8	79.6	75.1	84.0
δ -ATOMIC	<i>likely</i>	100	77.6	53.9	55.6	58.3	78.0	76.4	87.4
	<i>unlikely</i>	0	21.3	8.3	50.9	49.9	79.2	75.9	86.5
δ -SOCIAL	<i>likely</i>	100	77.2	58.9	52.9	55.5	85.8	73.9	90.9
	<i>unlikely</i>	0	28.8	8.4	50	58.7	90.7	74.7	93.5

Table 3.7: Paraphrastic consistency on examples which are likely and unlikely to contain artifacts, as predicted by a partial-input baseline. Inconsistency on both types of examples indicates P_C is attributable to factors *beyond* artifacts.

splits in PARANLU, human-elicited paraphrases are more lexically and syntactically diverse, as well as less semantically similar to original examples than automatically generated paraphrases (Table 3.6). In addition, we find that automatically generated paraphrases are 3–4% more likely to be bidirectionally entailed than human-written paraphrases, as detected by a ROBERTA-large model finetuned on SNLI (Bowman et al., 2015), MNLI (Williams et al., 2018a), ANLI (Nie et al., 2020a), and FEVER (Thorne et al., 2018).

Taken together, these results underscore the benefit of our human annotation to generate PARANLU — evaluation solely on automatically generated paraphrases, as others have done, is insufficient to fully characterize their robustness.

3.7 Do artifacts explain inconsistency?

Many NLP datasets are constructed by crowdworkers writing most or all parts of a natural language reasoning problem. While efficient and scalable, this paradigm can give rise to annotation artifacts (Gururangan et al., 2018), or statistical biases in parts of examples that correlate with the correct label (McCoy et al., 2019). For example, Gururangan et al. (2018) found that negation words (“no”, “nothing”, etc) in NLI examples are strong indicators of the contradiction label.

One way to detect artifacts in datasets is through a partial-input baseline (Poliak et al., 2018; Feng et al., 2019), a setting in which only the *target* portion of an NLI instance (i.e the hypothesis in a traditional NLI setup) is used to train a model to predict entailment. When partial-input models achieve high accuracy, it is often indicative of annotation artifacts.

Full-input models that are trained on datasets with annotation artifacts may learn to rely on such shallow signals instead of performing true inferential reasoning. This may lead to lower paraphrastic consistency in the face of alternative phrasings of examples, since they may no longer contain the annotation artifact the model leveraged. Can we attribute a model’s issues with paraphrastic consistency on PARANLU entirely to the presence of

annotation artifacts? That is, are models only inconsistent on examples with artifacts, since they are relying on spurious correlations? Or, are they inconsistent even when no artifacts are present?

3.7.1 Experimental Setup

Rudinger et al. (2020b) find that partial-input baselines trained on δ -NLI perform at least 10% better than random chance, indicating the presence of annotation artifacts in their dataset. As such, we focus on δ -NLI for our experiments.⁵ For each split of δ -NLI, we train a ROBERTA-large model using only the update sentence as input, keeping the training hyperparameters identical to the full-input ROBERTA models from §3.5.1. Then, we use these partial-input models to partition buckets in PARANLU into two subsets: those on which the partial-input model correctly predicted the label from the update sentence of the *original example*, indicating that an artifact is likely present in the original example, and those incorrectly predicted. Using the full-input ROBERTA models from §3.5.1, we then compute $\tilde{P}_C$ on both example subsets, and the accuracy of partial-input and full-input models on paraphrased and original examples.

3.7.2 Results

Table 3.7 shows the accuracy metrics for both partial-input and full-input models on original and paraphrased examples in PARANLU, as well as paraphrastic consistency metrics on both examples that are likely (+) and unlikely (−) to contain artifacts. While not all original examples that a partial-input model predicts correctly *necessarily* have artifacts, we expect that (1) examples with particularly strong artifacts are grouped in the *likely* (+) category, and (2) the *unlikely* (−) category contains a significantly smaller number of examples with strong artifacts. We observe a dramatic drop in the accuracy of a partial-input baseline on original examples (A_O) and paraphrased examples ($\tilde{A}_{\blacksquare}$), indicating that most artifacts detectable with a partial-input model *do not* project through our label-preserving paraphrase process.

Even on examples *unlikely* (−) to contain artifacts, where a full-input model cannot rely on shallow signals, models do still have room to improve their paraphrastic consistency. These results indicate that issues with paraphrastic consistency are attributable to factors beyond the presence of artifacts in examples.

3.8 Training Dynamics and Paraphrastic Consistency

Lastly, we explore the relationship between different parts of the model training pipeline (e.g. pretraining and finetuning) and paraphrastic consistency. How does consistency *change* as these training processes progress, and does it change in a similar manner as accuracy?

⁵We choose δ -NLI instead of α -NLI for this experiment, since the authors of α -NLI released their dataset *after* running adversarial filtering.

Is it the case that simply increasing the volume of pretraining or finetuning data linearly impacts paraphrastic consistency? We train a series of ROBERTA models and adjust the number of pretraining tokens (§3.8.1) and finetuning examples (§3.8.2) to explore how they impact a model’s consistency.

3.8.1 Pretraining and P_C

Experimental Setup. Using the MINIBERTs (Warstadt et al., 2020), a series of ROBERTA models pretrained from scratch on varying numbers of tokens, we compare models trained on 1M, 10M, 100M, and 1B tokens along with ROBERTA-base, which is pretrained on approximately 30B tokens. All models have the same number of parameters as ROBERTA-base (125M), with the exception of the model trained on 1M pretraining tokens which was scaled down in accordance with the smaller volume of pretraining data, and has 45M parameters. We *finetune* all models on the same data (§3.4.1) and keep all hyperparameters constant (batch size of 64, 2 finetuning epochs, and a learning rate of 5e-6), ensuring direct measurement of the impact of pretraining words on paraphrastic consistency without other confounds.

Results. Figure 3.5 plots model accuracy on paraphrases ($\tilde{A}_{\blacksquare}$) against paraphrastic consistency ($\tilde{P}_C$), along with the same supporting curves as in Figure 3.3 corresponding to decreasing proportions of variance attributable to paraphrasing (PVAP). As expected, pretraining on increasing amounts of data yields both monotonically increasing accuracy and paraphrastic consistency. However, paraphrastic consistency grows more rapidly in the beginning (1M - 100M tokens) as the plots climb steeply between supporting curves and eventually hugs a single PVAP curve past 100M tokens, indicating a slower payoff of more pretraining tokens.

3.8.2 Finetuning and P_C

After pretraining, models are endowed with the ability to represent natural language inputs but do not know how to perform a particular reasoning task. As such, we expect monotonically increasing accuracy as the model is shown a larger volume of finetuning examples. However, it is unclear how *paraphrastic consistency* changes as the model is exposed to more task-specific examples.

Experimental Setup. For each dataset, we finetune a series of fully-pretrained ROBERTA-large models on 1%, 5%, 10%, 50%, and 100% of examples from the training split, sampled at random. δ -ATOMIC has 28.3K training examples, δ -SNLI has 75.2K training examples and δ -SOCIAL has 65.3K examples. We sample at the premise-hypothesis level and include all examples that share the same premise and hypothesis to prevent data leakage during evaluation. We hold all training hyperparameters constant, keeping the same configuration as the finetuning in §3.8.1.

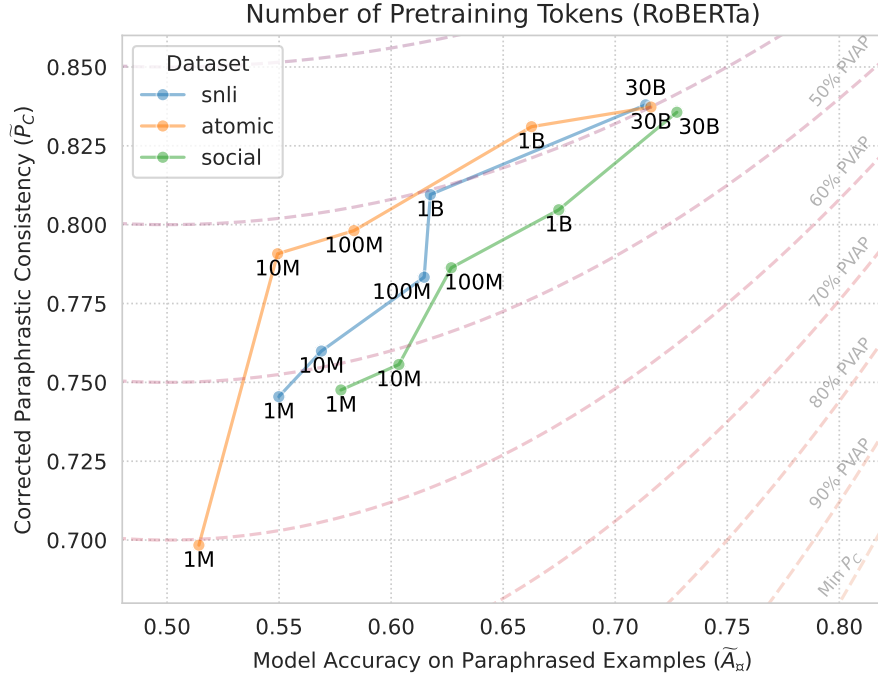


Figure 3.5: Paraphrastic consistency monotonically increases as a model sees more pretraining tokens, but grows rapidly during early pretraining.

Results. Figure 3.6 plots corrected paraphrase accuracy ($\tilde{A}_\pi$) against corrected paraphrastic consistency ($\tilde{P}_C$) for models trained on increasing numbers of finetuning examples across all three datasets in δ -NLI. We observe that as the model starts to learn the task at hand and draw increasingly complex decision boundaries, it is more likely to be inconsistent. Models trained on $\leq 5\%$ of the available training examples are highly consistent since they make the same prediction for all examples (thus earning an accuracy of around 50%). As the model is shown more examples, it makes finer-grained distinctions between examples, in turn impacting its paraphrastic consistency. Though our results show this decrease, it is possible that with even more finetuning data, a model’s paraphrastic consistency will start to increase again. This relationship may also be altered if, during finetuning, models are shown increasing amounts of automatically-generated paraphrased examples in order to learn both the task *and* paraphrastic consistency.

3.9 Related Work

Natural language understanding models may produce different predictions in the face of varying expressions of reasoning problems. A wide range of data generation frameworks have been proposed to study these behaviors in NLP systems. Iyyer et al. (2018) automatically generate paraphrases with specified syntactic templates and measure accuracy on these adversarial examples. Verma et al. (2023) introduce a test set of paraphrases generated with

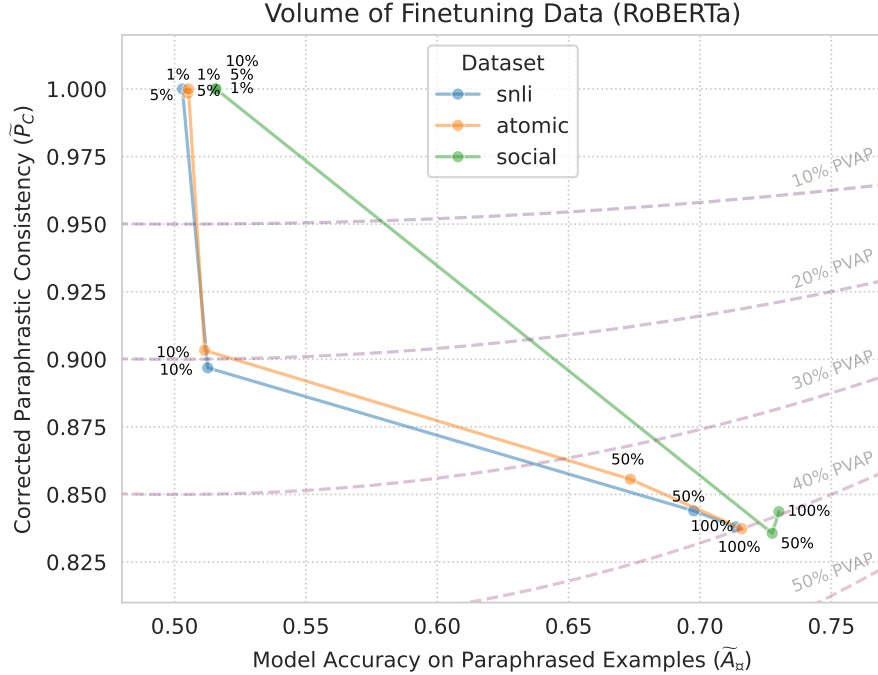


Figure 3.6: Paraphrastic consistency decreases as a model learns increasingly complex decisions boundaries by seeing more finetuning examples.

a finetuned T5 model (Raffel et al., 2020) and measure the accuracy of several models. Hu et al. (2019) generate paraphrases of MNLI examples using lexical constraints and evaluate an NLI model on the paraphrased inputs, finding that paraphrasing leads to degraded accuracy. Arakelyan et al. (2024) measure the semantic sensitivity of NLI models by automatically generating examples with FLAN-T5 (Chung et al., 2022) and verifying the generations with bidirectional entailment predicted by pretrained NLI models. While scalable, our findings illustrate that it is insufficient to evaluate models on automatic paraphrases alone, as human-written paraphrases introduce more semantic and pragmatic diversity (Section 3.6). Moreover, we show that bidirectional entailment as a verification method for generated paraphrases is extremely stringent, precluding us from testing consistency in the face of more challenging label-preserving transformations.

Another body of research studies the creation of adversarial examples to improve model robustness. Nie et al. (2020a) construct an NLI benchmark, Adversarial NLI, by developing a model-in-the-loop framework to iteratively produce examples that models cannot correctly solve. Naik et al. (2018) develop a suite of adversarial examples to “stress test” common failure modes of NLI models, such as phenomena word overlap or negation. In contrast with these studies, our goal is not to generate a test suite of difficult examples that “break” models (Glockner et al., 2018a), but rather to carefully measure the role of *paraphrastic* variability in model performance.

Other approaches to measuring robustness also include counterfactual example generation (Srikanth and Rudinger, 2022; Kaushik et al., 2020). Kaushik et al. (2020) recruit

humans to create counterfactual examples by minimally editing example text in order to flip the gold label and show that models trained on the original datasets perform poorly on counterfactually-manipulated data. Similarly, [Gardner et al. \(2020\)](#) argue for the creation of evaluative contrast sets, or manual minimal perturbations of dataset examples that change the gold label, in order to probe the decision boundary of models. Our work has a related, but distinct, counterfactual flavor: if an original annotator had chosen to phrase the question differently *with the same target label*, what is the probability that a model’s prediction would stay consistent? We aim to estimate, in expectation, the reliability of models when they are faced with different phrasings of the same problem.

Most of these studies measure *accuracy* on adversarial examples as the main determination of robustness. [Elazar et al. \(2021b\)](#) instead measure the *consistency* of models with respect to factual knowledge, evaluating whether extracted information from masked language models is invariant to paraphrasing using an agreement-based consistency metric. Our study is similarly concerned with consistency, however we make precise the relationship between accuracy and consistency on natural language reasoning tasks.

It is possible that human-in-the-loop approaches to data collection ([Wallace et al., 2022](#)) that continuously measure paraphrastic consistency on examples could increase the robustness of reasoning models.

3.10 Conclusion

As more studies investigate the capabilities of LLMs, the ability to disentangle the effects of paraphrastic variability from other target attributes will be an important analytical tool.

This work introduces a new methodology and dataset for measuring paraphrastic consistency, or P_C , of models on natural language reasoning tasks. P_C captures the probability that a model will remain consistent in its prediction given different phrasings of the same underlying reasoning problem. We design P_C as a metric complementary to accuracy, and propose practitioners use it alongside accuracy when diagnosing modeling errors, summarizing a model’s performance, or deciding when a model is ready for deployment to users for a particular application.

The beginning of this chapter introduced the contract of *paraphrastic consistency*. Our results confirm that paraphrastic sensitivity is present in all models, indicating that many of the models that users interact with frequently *do not uniformly uphold this contract*. Future work may consider adapting this approach for tasks with more complex or open-ended evaluations. Because P_C only requires model predictions to be labeled as correct or incorrect, our approach can generalize to any task with binary scoring (and where answers must be invariant to paraphrases).

This chapter focused on robustness when paraphrasing the *target* portions of reasoning problems. Next, we study inferential robustness to the *context* portion of examples and introduce three new related contracts for a more holistic picture of inferential robustness in natural language inference.

Chapter 4: Inferential Robustness Through Deep Context Understanding^{1,2}

In the previous chapter, we established that one of the basic contracts that helps establish broad spectrum inferential robustness is the ability of NLP systems to *maintain* the likelihood of computing a particular inference under semantically equivalent inputs. We measured the probability of a model achieving the same correctness on two paraphrases of the same problem. In particular, we focused on perturbing the *target* portions of reasoning problems, or the specific components of reasoning problems that are evaluated *with respect to* the context (e.g. *hypotheses* in traditional NLI or abductive NLI).

We now turn our attention to a set of contracts revolving around model use of *context* in reasoning. Deep context understanding is essential for language models to consistently draw correct inferences, and if systems are indeed *understanding* and paying attention to this context, they should be able to modulate the inferences they draw when perturbing this context *or* when further probed about the situation evoked by this context.

When we provide an NLP system some context from which we would like it to derive an inference, we expect that it is actually *using* this context to draw the inference as opposed to indexing on shallow features of non-context components (e.g. hypotheses in NLI) that may be correlated with the correct answer, often called annotation artifacts (Gururangan et al., 2018). This chapter begins by discussing work (Section 4.1) in which we identify natural language inference problems that contain annotation artifacts (e.g. an example containing negation words may have a label of **contradiction**) and edit the premises of these problems to *flip* the gold label. Systems that exhibit inferential robustness will then appropriately *decrease* their likelihood of drawing the original inference.

Contract: Context Sensitivity. When the context in a reasoning problem changes in a way that affects the validity of an inference, NLP systems should appropriately modulate their likelihood of drawing that inference.

¹Neha Srikanth and Rachel Rudinger. 2022. [Partial-input baselines show that NLI models can ignore context, but they don’t](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4753–4763, Seattle, United States. Association for Computational Linguistics

²Neha Srikanth and Rachel Rudinger. 2025. [NLI under the microscope: What atomic hypothesis decomposition reveals](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2574–2589, Albuquerque, New Mexico. Association for Computational Linguistics

Next, we examine two additional contracts that reasoning models must uphold, both requiring deep contextual understanding: *logical* and *inferential* consistency (Section 4.2). A reasoning model’s ability to consistently draw correct inferences depends heavily on whether it possesses relevant knowledge about the context and can reason effectively with that knowledge. Consider a scenario involving a person sweeping a sidewalk in front of a garbage truck. For the model to infer the person’s likely occupation (e.g., that they work for the city department), it must understand how each piece of contextual information relates to the target inference. When subsequently asked about individual pieces of information in isolation (the person’s action of sweeping, or the fact that they are in front of a garbage truck), the system must respond in a way that remains logically consistent with its original inference based on the *full* context. If the system is able to do so, we consider it *logically consistent*, that is, when we decompose a natural language reasoning problem into subproblems, we can hold it accountable to maintain consistency between its judgments on each sub-problem and its overall judgment in a logical way. This gives us necessary, but not sufficient, evidence to help signal that it has “understood” the situation. We study logical consistency in traditional NLI reasoning problems.

Contract: Logical Consistency. When a reasoning problem is decomposed into smaller sub-problems, systems should make predictions on each component that align with their overall judgment in a logically coherent way.

Then, we shift to focusing on defeasible NLI, where we define the concept of *inferential consistency*, or the probability that a model’s predictions for two examples e_i and e_j that share the same core inference or fact are either both correct or incorrect. This is a key contract that general purpose language models must uphold: users should be able to trust that when the system they are using seems to draw an inference about a fact in one context, it is able to do it in another. For example, a model correctly predicting whether or not people are friends under some contexts but not others may indicate that it has not fully understood the factors that influence the inference.

Contract: Inferential Consistency. When reasoning about the same underlying fact or inference across different contexts, systems should maintain consistent predictions, either by correctly drawing the inferences in both contexts or failing in both, as opposed to succeeding in one and failing in another.

Lastly, we introduce two proposed works to better evaluate the robustness of defeasible inference models: (1) a defeasible *investigator* designed to probe the factors contributing to an inference, and (2) a new, more contextual multi-step defeasible reasoning dataset that sequentially presents information to the system for evaluation against a particular inference.

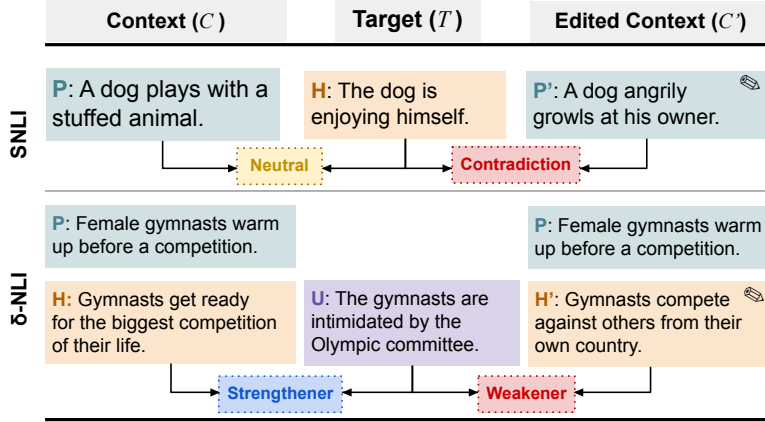


Figure 4.1: Given NLI and δ -NLI instances that partial-input baselines correctly label, we edit their context to induce a different gold label. We use the edited instances to probe a full-input model’s ability to leverage context to deviate from the predicted partial-input label.

4.1 Partial-input baselines show that NLI models can ignore context, but they don’t.

4.1.1 Introduction

Researchers have shown that many NLI datasets contain statistical biases, or “annotation artifacts” (Gururangan et al., 2018; Herlihy and Rudinger, 2021b) that systems leverage to correctly predict entailment. To diagnose such artifacts in datasets and provide a stronger alternative to majority-class baselines, Poliak et al. (2018) introduced partial-input baselines, a setting in which models are provided parts of NLI instances to predict an entailment relation. Poliak et al. (2018), Gururangan et al. (2018), and others posit that datasets containing artifacts may in turn produce models that are incapable of learning to perform true reasoning.

In this work, we re-examine such dismissal of the reasoning capabilities of models trained on datasets containing artifacts. While a competitive partial-input baseline is sufficient to cast doubt on the inferential capabilities of full-input models trained on the same data, it is premature to conclude that full-input models are not capable of such reasoning at all. As a thought experiment, we imagine training a human to learn a task from a set of examples containing artifacts that allow them to cheat. If we force them to learn to perform the task by removing relevant context, they must rely on the artifacts to accurately perform the task. However, provided all the data, they may still learn to perform the task the “right way”.

Through two sets of experiments, we investigate whether NLI models are able to condition on the full input despite learning from artifact-ridden datasets. Section 4.1.3 investigates whether additional context strengthens a full-input model’s confidence in the correct label, despite a partial-input model’s correct prediction. In Section 4.1.4, we introduce an evaluation set to probe whether full-input models are sensitive to changes in context that flip the

gold label in examples containing artifacts. We make our evaluation set and code publicly available.³

Our results indicate that full-input models are still successfully learning to utilize context, overriding strong signal in the partial input. With this finding, we argue that while partial-input baselines are still a useful tool, they do not license the conclusion that models trained on datasets with artifacts do not learn to leverage context from the full input.

4.1.2 Background

This work studies natural language inference and defeasible natural language inference examples, covered in Chapter 2, from examples in SNLI and δ -SNLI (Rudinger et al., 2020b); and annotation artifacts present in these datasets.

As a reminder, defeasible examples have a third *update* sentence that is designed to strengthen or weaken the hypothesis, which already has a neutral relationship to the premise.

Premise:	A man is sitting in a dim restaurant.	Hypothesis:	He is eating food.
Update:	He is browsing a menu.		

In this work, we adopt the terms **context** (C) and **target** (T) for clarity when describing partial-input baselines for the two tasks. In NLI, $C = P$ and $T = H$; in defeasible NLI, $C = (P, H)$, and $T = U$. Thus, for either task, the partial-input baseline we use looks only at T and ignores C .

For the task of *defeasible* NLI, Rudinger et al. (2020a) introduce the δ -NLI dataset, which consists of extensions to three pre-existing English natural language reasoning datasets: SNLI (Bowman et al., 2015), ATOMIC (Sap et al., 2019), and SOCIAL-CHEM-101 (Forbes et al., 2020b). To extend each dataset, instances (e.g., P - H pairs) were presented to crowdworkers who then wrote an update sentence (U) to strengthen or weaken the given hypothesis (as described in Chapter 2). The resulting binary classification task is to predict whether an update is a strengthener or weakener, given a (P, H, U) triple. In this work, we focus our evaluation on δ -SNLI, the SNLI portion of the δ -NLI dataset.

4.1.3 Experiment 1: Context in NLI

An essential component of true reasoning is learning to leverage all parts of an example’s input to make a determination of entailment. Strong partial-input models demonstrate that full-input models do not necessarily need to utilize context to make correct predictions. However, we explore whether they do at all, and how access to such context shifts a full-input model’s confidence in the correct label. If, upon supplying C , a model strengthens its confidence in its prediction, we may conclude that it utilizes both C and T during inference as intended.

³<https://github.com/nehasrikn/context-editing>

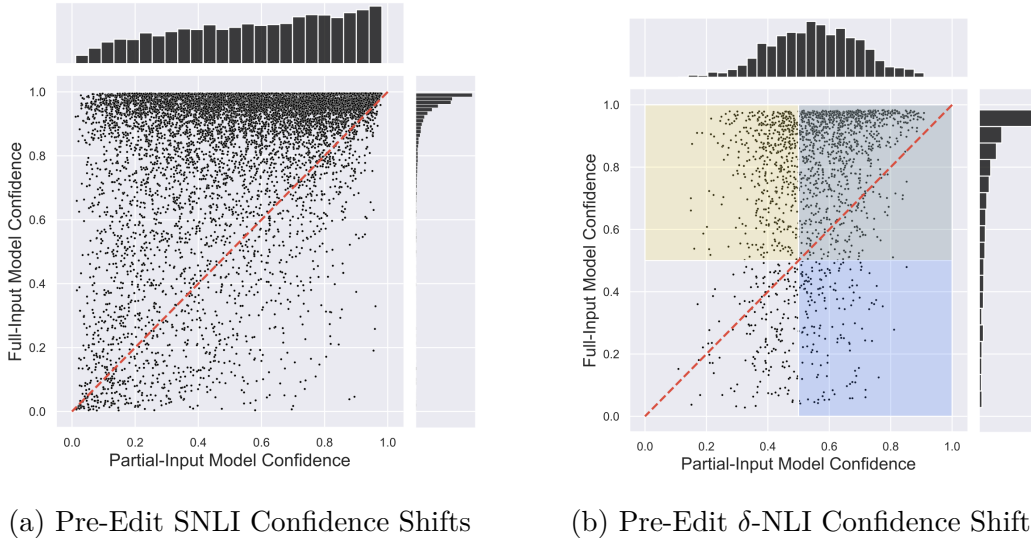


Figure 4.2: Visualizing the role of context as confidence shifts. 4.2a and 4.2b plot ordered pairs of confidence in the correct label l of partial-input vs. full-input RoBERTa models on examples from the SNLI and δ -SNLI test splits. Blue and yellow regions mark correct predictions from the x and y-axis models respectively. Green regions represent correct predictions from both models. Diagonals indicate no shift in confidence.

	SNLI	δ -SNLI
Partial-Input (C)	0.70	0.65
Full-Input (C, T)	0.91	0.82

Table 4.1: Accuracy of both partial-input and full-input models on test splits of SNLI and δ -SNLI.

Experimental Setup. We finetune two sets of RoBERTa (Liu et al., 2019) models (a partial-input and a full-input model) on the train splits of SNLI and δ -NLI (see Appendix A.13 for dataset sizes). We utilize `roberta-base` from the Hugging Face library (Wolf et al., 2020), and finetune each model for two epochs. Appendix A.14 further details our training setup. We then run inference on the test splits from SNLI and δ -SNLI (the SNLI portion of δ -NLI) using each pair of models. Table 4.1 reports accuracy of all models on the corresponding test splits. We calibrate RoBERTa models post-hoc using temperature scaling (Guo et al., 2017) as suggested by Desai and Durrett (2020), and examine confidence shifts in the correct label to understand whether full-input models utilize context at all.⁴

As shown in Fig. 4.2, we plot an ordered pair of each model’s confidence in the *correct label* for examples in the test splits, with the partial-input model’s confidence along the x-axis and the full-input model’s confidence along the y-axis. Density around the diagonal would

⁴We observed minimal differences between pre and post-calibration results.

indicate no change in confidence.

Evidenced by the density *above the diagonal* in Figures 4.2a and 4.2b, full-input models (i.e access to both C and T) are more confident in the correct label than partial-input models.⁵ While this behavior may seem unsurprising, partial-input baselines illustrate that models may show confidence in the correct label without needing to condition on context at all. Our results hint that full-input models may be successfully learning to leverage additional context instead of overgeneralizing on artifacts in the target. To probe this behavior directly, we introduce an evaluation set crafted by editing contexts in the following section.

4.1.4 Experiment 2: Context Editing

We investigate a model’s ability to leverage context despite the presence of artifacts by exploring how sensitive full-input models are to changes in *non-target* components of the input. We present an example modification scheme, illustrated in Figure 4.1, in which we edit context sentences from examples where a model correctly predicts the label l from the target T alone. Namely, while holding T constant, we introduce an edited context sentence C' that induces a different label $l' \neq l$ on the new (C', T) pair. Using this scheme, we construct an evaluation set of 600 examples sourced from SNLI and δ -NLI.

Example Subselection. We select SNLI and δ -SNLI test examples to edit by running the partial-input RoBERTa models from Section 4.1.3 and full-input bag-of-words (BoW) models, implemented via `fasttext` with a maximum of 4-grams (Joulin et al., 2017b). See Appendix A.14 for training details on the bag-of-words models. We select examples to edit for which *either* the partial-input model or the BoW model predicted the correct label. This identifies the subset of examples likeliest to contain artifacts in T , lexical or otherwise.

Editing SNLI Examples. For a given SNLI example (P, H, l) and a new predefined target label l' , we edit P , creating a modified SNLI example (P', H, l') . For each of the six directional pairs of labels (e.g., *entailment* $\rightarrow$ *contradiction*), we randomly sample 50 examples from the subset to edit, resulting in 300 examples evenly distributed across label pairs.

Editing δ -SNLI Examples. Given a δ -SNLI instance (P, H, U, l) , we edit H while holding P and U constant to induce a new label $l' \neq l$, resulting in a modified example (P, H', U, l') . We edit 300 examples total, turning 150 *strengtheners* examples into *weaker* examples, and vice-versa.

Our final evaluation set consists of 600 examples containing edited context-target pairs split evenly across SNLI and δ -SNLI. Figure 4.4 shows examples of editing contexts from both datasets. All examples were manually edited by one author and independently validated by another. During validation, we hide both the l and l' , and ask the annotator to label the text

⁵The distribution difference between SNLI and δ -SNLI may be attributed to the difference in the task difficulty, as well overall lower performance of RoBERTa models on δ -NLI.

		l'		
		entailment	neutral	contradiction
l	entailment	–	0.76	0.76
	neutral	0.42	–	0.78
	contradiction	0.90	0.78	–

Table 4.2: A full-input RoBERTa model’s accuracy on the edited SNLI portion of our evaluation set. l and l' represent the original and target label respectively, before and after editing.

		l'	
		weaker	stronger
l	weaker	–	0.76
	stronger	0.75	–

Table 4.3: A full-input RoBERTa model’s accuracy on the edited δ -SNLI portion of our evaluation set. l and l' again respectively represent the original and induced post-edit label.

pair. Using Cohen’s Kappa (Cohen, 1960), we obtain an agreement measure of $\kappa = 0.78$ and $\kappa = 0.76$ for SNLI and δ -SNLI examples in our test set respectively, indicating substantial agreement (Artstein and Poesio, 2008).

Results. Using the same full-input RoBERTa models trained in Section 4.1.3, we run inference on our edited evaluation set. Tables 4.2 and 4.3 show model performance stratified by original label l and target label l' . Our results show that full-input models are in fact sensitive to context modifications despite the presence of artifacts in T , consistently achieving above 70% accuracy on edited examples. Thus, we conclude that these models are not overgeneralizing on artifacts in the instance, learning to condition on context for prediction.

Analyzing Post-Edit Model Confidence. Similar to the analysis in Section 4.1.3, we inspect shifts in confidence upon editing contexts to shed more light on a full-input model’s utilization of C . For δ -SNLI examples, we plot ordered pairs in Fig. 4.3 of a full-input model’s confidence in the correct label pre-edit, and its confidence in the *same label* post-edit (i.e the confidence in the now-incorrect label). The majority of mass is under the diagonal, indicating that our model is indeed sensitive to changes in context. The green bottom-left quadrant delineates ideal performance (correct before *and* after editing examples). We attribute the small cluster of examples in the blue quadrant (previously highly confident in the correct label and subsequently remained confident, but in the wrong label) to strong, non-lexical artifacts overriding additional signal from the context. Appendix A.15 visualizes shifts in SNLI examples using simplex plots to accommodate the ternary label.

Lexical Model Performance. We do not retrain any of the models on edited examples prior to evaluating with our edited test set, precluding them from picking up on any newly-

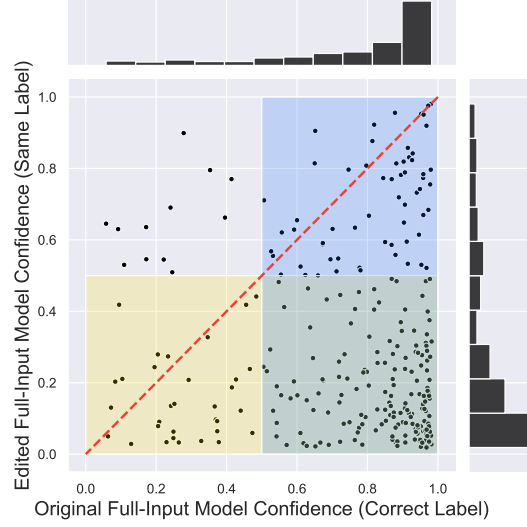


Figure 4.3: Full-input model’s confidence in l before vs. after editing C to induce a different label.

	Context ($C \rightarrow C'$)	Target (T)	Label ($l \rightarrow l'$)
SNLI	Premise: A little girl in a pink hat is in a lush green field walking an ox. Edited Premise 🐘: A little girl in a pink hat sits on an ox carrying her through the middle of the Sahara.	Hypothesis: A little girl is riding her ox in a desert.	Contradiction $\rightarrow$ Entailment
	Premise: A girl in a yellow dress looks at a restaurant building. Edited Premise 🍴: A girl in a yellow dress enjoys some food at her favorite restaurant.	Hypothesis: A girl debates eating at a certain restaurant.	Neutral $\rightarrow$ Contradiction
δ -NLI	Premise: A boy in a red hooded top is smiling whilst looking away from his reflection. Hypothesis: The boy doesn't want to see his reflection. Edited Hypothesis 🐼: The boy wants to see his reflection.	Update: A woman's voice says to him 'Don't be afraid of it, it's harmless'	Strengtheners $\rightarrow$ Weakeners
	Premise: A music concerto is happening on stage, with the lights on spot, while the audience listens in the dark. Hypothesis: The concert is being held at night. Edited Hypothesis 🌞: The concert is being held during the day.	Update: Blackout curtains cover the windows.	Weakeners $\rightarrow$ Strengtheners

Figure 4.4: Editing SNLI and δ -SNLI examples. We edit the premise (SNLI) and the hypothesis (δ -NLI), while holding the target constant. The last column contains the original label l and the new induced label l' .

introduced non-lexical artifacts. However, to validate the absence of trivial lexical features that override the artifacts in T and to ensure sufficient difficulty, we run full-input BoW models on our edited evaluation set. A full-input `fasttext` model, often used for adversarial filtering (Zellers et al., 2018), achieves 16% and 24.3% accuracy on the SNLI and δ -SNLI portions of our evaluation set respectively.

4.1.5 Related Work

In addition to work on partial-input baselines for NLI (Gururangan et al., 2018; Poliak et al., 2018; Tsuchiya, 2018), partial-input models have been studied for story completion (Cai et al., 2017) and reading comprehension (Kaushik and Lipton, 2018). Feng et al. (2019) observe that low-scoring partial-input baselines do not preclude other artifacts and heuristics, while Glockner et al. (2018b) and McCoy et al. (2019) demonstrate lexical and syntactic examples of NLI heuristics. Adversarial editing has been explored for non-NLI tasks as well (Jia and Liang, 2017; Ribeiro et al., 2018). Finally, adversarial filtering has been proposed as a means of removing artifacts from datasets (Zellers et al., 2018; Le Bras et al., 2020).

4.1.6 Discussion and Conclusion

While partial-input models are useful tools for analysis, often leveling fair criticism of datasets, our results show it is hasty to conclude that models trained on such datasets are not capable of reasoning. Even though high-scoring partial-input baselines show that full-input models *could* ignore context, our experiments show that they can leverage this context quite effectively.

We argue that artifacts do not *necessarily* spell disaster for a model’s reasoning capabilities. In particular, our context-editing experiments identify a set of instances that partial-input models fail (by design), but full-input models largely succeed at, displaying the capability of full-input models to leverage context to overcome SNLI and δ -NLI artifacts in many, but not all, cases. Of course, we do not deny that artifacts *can* and *do* lead to models with exploitable heuristics, as demonstrated by Glockner et al. (2018b) and McCoy et al. (2019).

While we do not attempt to define the *sufficient* conditions for a model to perform “true inference,” we demonstrate that these models can and do meet the *necessary* condition of leveraging the full input. Thus, we conclude that partial input baselines should be understood as agnostic warning signs: sufficient to conclude that full-input models *might* not be leveraging critical context, but insufficient to prove that they don’t.

This raises a number of interesting questions for follow-up work. If adversarial filtering (Le Bras et al., 2020) can identify instances containing artifacts, is it beneficial to remove these instances from the training set? Or could they be edited to flip the label and mitigate the spurious correlation? The edits we made in this work were done manually, but another interesting question is whether these edits could be made automatically or semi-automatically. Having a more efficient way of producing these examples would enable both rapid evaluation of models trained on datasets with artifacts (as in this work), as well as expansion of training sets to preemptively mitigate artifacts.

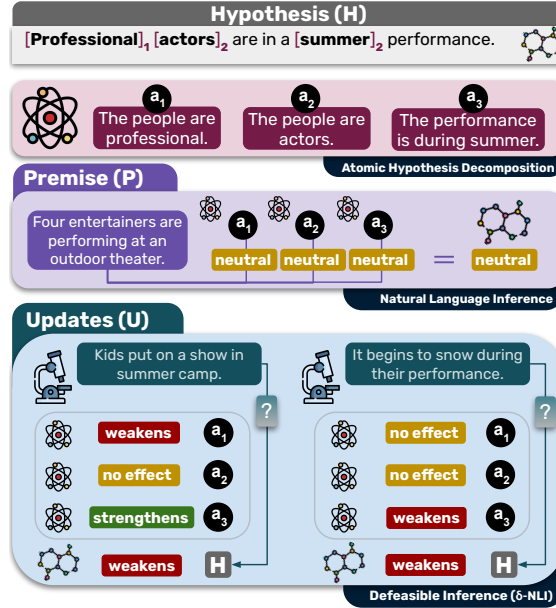


Figure 4.5: Top: Atomic hypothesis decomposition breaks down hypotheses (H) into entailed propositional “atoms” ($a_1 - a_3$). Middle: Pairing the premise (P) with each atom yields a set of NLI sub-problems ($P + a$); the sub-problem labels predict the full NLI problem ($P + H$) label. Bottom: Paired with an update (U), each atom yields a defeasible NLI sub-problem ($P + a + U$); the set of sub-problem labels are predictive of the full problem ($P + H + U$) label, but the non-monotonic relationship is more complex than for traditional NLI.

4.2 What Atomic Hypothesis Decomposition Reveals

4.2.1 Introduction

Atomic decomposition involves breaking sentences down into *atomic propositions*, or granular facts that are explicitly supported by the original text. This style of decomposition has widespread applications, including assessing the factual precision of generated text (Min et al., 2023), claim verification (Chen et al., 2024), and multihop QA (Perez et al., 2020), since it allows for careful, finer-grained inspection of text.

We use atomic decomposition as tool to dive deeper into two types of natural language reasoning: traditional NLI (Giampiccolo et al., 2007) and defeasible inference (Rudinger et al., 2020b), a mode of reasoning where inferences may change in light of new evidence. In both tasks, atomic decomposition of hypotheses into *atoms* breaks complex sentences into granular pieces of information that models must weigh when drawing higher level inferences, producing *atomic sub-problems*. We use these sub-problems not only for better insight into the structure and nuances of NLI and defeasible NLI, but also to assess accompanying benchmarks and to more deeply probe the robustness of models’ situational understanding.⁶

⁶Code and data available at <https://github.com/nehasrikn/nli-atoms>.

Consider the example in Figure 6.1 of a premise and hypothesis with a neutral relationship. When predicting an overall neutral relation, a model must determine the relation between the premise and each of the three atomic propositions that together form the hypothesis — in this case, all of which are neutral. If the model predicts an entailment or a contradiction between the premise and one of the atoms, its understanding of the situation may be called into question.

We first present our two tasks of interest (§4.2.2) and discuss the utility of atomic sub-problems and their construction (§4.2.3). Then, we analyze the behavior of large language models (LLMs) on atomic sub-problems in traditional NLI (§4.2.4) where we evaluate their logical consistency between each original SNLI instance and its corresponding sub-problems. We find that despite high accuracy, LLMs still struggle with logical consistency. Then, we study atomic sub-problems in defeasible NLI (§4.2.5) and propose a framework to pinpoint the inference(s) evaluated in each example by way of the *question under discussion* of examples, a well-studied linguistic phenomenon (Benz and Jasinskaja, 2017; Wu et al., 2023). Finally, we present a method to group defeasible NLI examples based on related atomic sub-problems (§4.2.6) and measure the *inferential consistency* of a model, a metric capturing the likelihood that its prediction for a particular inference will remain **consistently** correct or incorrect under different contexts.

4.2.2 Background

Natural Language Inference

Natural language inference (Giampiccolo et al., 2007; MacCartney, 2009; Bowman et al., 2015) is the task of determining whether a premise P entails, contradicts, or is neutral in relation to a hypothesis, H . For example, the premise “*A little girl in a lush green field*” contradicts the hypothesis “*A little girl rides her ox in a desert*”, as “desert” directly contradicts the “lush green field”.

Snli. The first large-scale NLI dataset, SNLI, uses image captions as premises paired with hypotheses elicited from crowdworkers. Though this benchmark has been well-saturated by LLMs over the past few years, it serves as a useful resource for studies in model robustness (Srikanth and Rudinger, 2022; Kaushik et al., 2019) and annotation artifacts (Gururangan et al., 2018).

Defeasible Inference

Defeasible reasoning is a form of non-monotonic reasoning in which inferences may be altered in light of new evidence (Reiter, 1980). For example, given the premise “*A group of people sitting around a rectangular table*”, the inference “*they have a work meeting*” is weakened upon learning that “*the people are toddlers*.”

δ -NLI. Rudinger et al. (2020b) introduce the task of defeasible natural language inference and an accompanying benchmark, δ -NLI. Given a P and H pair with a *neutral* relation, a third **update** sentence U *strengthens* H if, upon reading U , H is more likely to be true, and *weakens* H if H is less likely to be true upon reading U . Defeasible NLI is then a binary classification task of predicting a strengthener or weakener label for a (P, H, U) set. δ -NLI is built on top of three popular commonsense reasoning datasets: SNLI (Bowman et al., 2015), ATOMIC (Sap et al., 2019), and SOCIAL-CHEM-101 (Forbes et al., 2020b). For a P and H pair (or, in the case of SOCIAL-CHEM-101, just H), crowdworkers write an update sentence for a target label, but are not instructed to target a particular part of H when doing so. The authors ensure that the train, development, and test splits of the data are split at the P - H level to avoid leakage. Crowdworkers may not write updates that directly contradict information in the premise. For simplicity, we focus on the SNLI-derived split of δ -NLI, or δ -SNLI, which selects neutral P - H SNLI pairs.

Related Work

Atomic decomposition has been used in fact checking (Min et al., 2023; Glover et al., 2022; Yuan and Vlachos, 2024), claim verification (Chen et al., 2024), summarization (Nenkova and Passonneau, 2004), and text-to-image generation (Cho et al., 2023) among others. Kamoi et al. (2023) construct a dataset of claims and sub-claims for claim checking where sub-claims, analogous to our atoms, are labeled with respect to evidence. Most relevant to our work, Stacey et al. (2022) train a span-based NLI model to make *span-level* decisions on SNLI and SICK examples that are composed to produce an overall label. In contrast, our work measures the logical consistency of LLMs that may have seen NLI data during pretraining, but that are not explicitly trained to weigh atomic inferences. Their followup (Stacey et al., 2024) *trains* another NLI system using LLM-generated atoms, however their study focuses primarily on *premise* decomposition. To the best of our knowledge, our study is the first to explore atoms in defeasible inference.

4.2.3 Atomic Sub-Problems in NLI

Reasoning about situations often involves weighing multiple pieces of information to draw inferences. Consider the last row in Table 4.4. A human determining that H contradicts P will attribute the contradiction to the fact that P mentions a father and daughter, but H mentions two men. Implicitly, they will have also weighed the fact that H ’s mention of “cutting grass” is entailed by P ’s mention of a lawnmower, and so it does not contribute to the contradiction. We treat these two determinations as distinct *atomic sub-problems*.

Hypotheses in SNLI, and in turn, δ -SNLI, can be complex sentences, and while solving inference problems, models must weigh all pieces of information in both P and H . We expect humans to make inferences about constituent pieces of information in a manner that is consistent with their overall judgment, an equally desirable property in models. Not only does it signal holistic understanding of the situation described in the problem, but it

Premise (P)	Hypothesis (H)	Update (U)	Atoms
A man in a white t-shirt and jeans is holding a mallet and chisel next to his abstract sculpture which stands on several bricks.	A man is trying to finish his sculpture for a church	The man has taken his first strike against the granite. (Weakener)	a_1 : The thing the person is trying to do is finish. (-1) a_2 : The thing the person is trying to finish is a sculpture. (+1) a_3 : The sculpture is for a church. (0)
There is a green trash truck in road with a person sweeping sidewalk.	The garbage man sweeps up where the can spilled.	The person is wearing a city uniform. (Strengtheners)	a_1 : The person is a garbage man. (+2) a_2 : The thing being swept is up. (invalid) a_3 : There is a can. (0) a_4 : The can has spilled. (+1) a_5 : The person is sweeping up a spill. (+2)
Two young men climb a tree overlooking a rural setting, with one of them out far on a limb and clutching a white helmet.	Two brothers are climbing a tree to get down their Frisbee. (N)	N/A	a_1 : There are two people. (E) a_2 : There are two people who are brothers. (N) a_3 : There are people climbing. (E) a_4 : There are people climbing a tree. (E) a_5 : There is a purpose for people climbing a tree. (N) a_6 : The purpose for people climbing a tree is to get something. (N) a_7 : The thing people are trying to get is a Frisbee. (N) a_8 : The thing people are trying to get is down. (invalid)
A father and his daughter are riding a lawn mower down a street while dressed in American colors.	two men cut grass by hand (C)	N/A	a_1 : There are two people. (E) a_2 : There are two people who are men. (C) a_3 : There are people cutting. (E) a_4 : There are people cutting grass. (E) a_5 : There is a method of cutting grass. (E) a_6 : The method of cutting grass is by hand (C)

Table 4.4: δ - δ -SNLI (Rows 1 and 2) and SNLI (Rows 3 and 4) instances, along with atomic decompositions of hypotheses and the label of the sub-problem involving that atom.

can help pinpoint exactly what types of inferences models struggle with. Identifying atomic sub-problems also allows us to understand the granular inferences that are evaluated in benchmark datasets. In turn, this helps to understand the *diversity* in the dataset: despite there being thousands of examples, certain inferences may come up repeatedly.

To identify the constituent sub-problems, we break *hypotheses* in SNLI and δ -SNLI into atomic propositions (Wanner et al., 2024) to use in subsequent analyses. Each atomic decomposition represents a single piece of information. Formally, given an SNLI example with P and H , we generate atomic decompositions of H represented by $a_1 \dots a_n$. Each atomic sub-problem then involves predicting the relation between a (P, a_i) tuple (§4.2.4). Given a δ -NLI example with P , H , and U , atomic sub-problems involve a (P, a_i, U) tuple where the task is to determine whether U strengthens, weakens, or has no effect on a_i (§4.2.5).

Generating Atomic Propositions

To generate atomic propositions, we draw on Neo-Davidsonian event-based semantic representations of sentences (Castañeda, 1967; Parsons, 1990). Sentences can be represented in first-order logical form as conjunctions of predicates representing entities, where actions are explicitly represented with event variables and predicate arguments are mapped to semantic

roles (Dowty, 1991). For example, the sentence “*The juggler performs at a party*” could be represented as:

$$\begin{aligned} \exists x_1 \exists e \text{ (Juggler}(x_1) \wedge \text{Perform}(e) \wedge \text{Agent}(e, x_1) \wedge \\ \exists x_2 \text{ (Party}(x_2) \wedge \text{At}(e, x_2))) \end{aligned}$$

Each conjunct can then be mapped to a natural language expression, called an *atom*. This ensures that both arguments of actions *and* the actions themselves are included as separate atoms.

We draw on this intuition to carefully hand-construct exemplars, a methodology shown to improve the atomicity and groundedness of decompositions (Wanner et al., 2024). We prompt `llama-3-8b-instruct` with these exemplars (Appendix A.1) to generate atoms for each example in the δ -SNLI test set (henceforth, δ -SNLI-TEST), as well as for 1000 randomly sampled examples in the SNLI test set (δ -SNLI-TEST-SAMPLE). See Table A.6 for dataset statistics.

Validating Atomic Decompositions

Valid atomic decompositions of hypotheses must be logically entailed from the hypothesis they were decomposed from. For our experiments on δ -SNLI-TEST-SAMPLE (§4.2.4), we do not validate atom entailment ourselves, letting each model determine whether H entails each a_i itself (§4.2.4), and only measuring consistency on the atomic sub-problems that the model itself admits as “valid”.

However, we *do* validate all generated atoms in δ -SNLI-TEST, since non-monotonic reasoning does not give rise to clear constraints between an original problem and its constituent atomic sub-problems. Our two-step validation process involves pruning decompositions with a strong, finetuned NLI model followed by human validation.

Pruning. For each example in δ -SNLI-TEST, we use a DEBERTA-large model finetuned on popular NLI datasets⁷ and remove all generated atoms that are not entailed by the hypothesis. By design, P - H pairs in δ -SNLI have a neutral relation, and updates strengthen or weaken propositions in the *hypothesis*. Hence, we run a secondary pruning stage to retain only those atoms that are *not* entailed by the premise (see Table A.6). See Appendix A.1 for a discussion of coverage.

Human Validation. An author annotated all atoms that survived pruning as either *invalid* or *valid* (see Table 4.4 for examples of invalid atoms). Valid δ -SNLI atoms had to (1) be grammatical, (2) entail from H , (3) not entail from P . Atoms introducing new information were considered invalid, including those that were *pragmatic* inferences of H (Jeretic et al., 2020; Srikanth et al., 2024b).

⁷MNLI (Williams et al., 2018a), Fever-NLI (Thorne et al., 2018), Adversarial NLI (Nie et al., 2020a), LingNLI (Parrish et al., 2021a), and WANLI (Liu et al., 2022a)

	Full Example Accuracy	Overall Logical Consistency	Consistency on Correct Exs	Consistency on Incorrect Exs	Logical Consistency by Label			Induced Atom Label Accuracy
					entailment	neutral	contradiction	
gpt-4o-mini-2024-07-18	89.8	84.0	86.8	59.8	78.8	90.1	82.6	81.3
gpt-4o-2024-08-06	88.5	87.9	89.6	74.8	88.9	92.7	79.2	81.9
llama-3-70b-it	87.7	84.2	89.5	46.3	82.1	81.3	89.5	84.8
llama-3-8b-it	85.2	81.2	88.2	41.2	76.0	85.0	85.1	82.9
gemma-2-9b-it	84.2	80.9	85.9	54.4	84.0	82.4	73.8	78.9
gpt-35-turbo-0125	82.1	74.4	80.4	46.9	70.4	90.9	64.2	74.9

Table 4.5: Accuracy (col. 1) and logical consistency of LLMs full SNLI examples (col. 2). We also report logical consistency on examples where the full prediction was correct (col. 3) and incorrect (col. 4), as well as stratify by predicted label (cols. 5–7). Finally, we logically compose atomic sub-problem labels to see if this more interpretable method outperforms full example accuracy (col. 8).

95.7% of pruned atoms were determined as valid by the author annotator. An external annotator also annotated a sample of 100 atoms for validity for an agreement of $\kappa = 0.82$ measured by Cohen’s Kappa (Cohen, 1960). The remaining analysis in this work is done on the set of *valid* δ -SNLI atoms.

4.2.4 Atoms in Traditional NLI

While performing strongly on various benchmarks, LLMs still struggle with many types of consistency including paraphrastic consistency (Srikanth et al., 2024a; Verma et al., 2023), hypothetical consistency (Chen et al., 2023), or even preferential consistency (Zhao et al., 2024c). When LLMs make entailment judgments, another desirable property is *logical consistency*. Namely, when an LLM itself deems a set of atoms $a_1 \dots a_n$ entailed by H , we can hold it accountable to maintain consistency between its judgments on each (P, a_i) sub-problem and its overall (P, H) judgment in a logical way. This gives us necessary, but not sufficient, evidence to help signal that it has “understood” the situation.

Atomic and Overall Label Consistency

We construct a set of rules to establish the relationship between atomic sub-problems and overall problem labels.

1. If H is entailed by P : Each valid a_i must be entailed by P .
2. If H contradicts P : At least one valid a_i must contradict P .
3. If H is neutral with respect to P : At least one valid a_i must be neutral with respect to P , all others may be either neutral or entailed.

Experimental Setup

We experiment with six LLMs: gpt-4o (OpenAI, 2024), gpt-4o-mini (OpenAI, 2024), gpt-3.5-turbo-0125 (Ouyang et al., 2022), llama-3-8b-instruct (Dubey et al., 2024), llama-3-70b-instruct (Dubey et al., 2024), and gemma-2-9b-instruct (Team et al., 2024).

First, we benchmark each models’s performance on δ -SNLI-TEST-SAMPLE original examples (Table 4.5) using Prompt A.2.1 adapted from Liu et al. (2023). We use 12 in-context *original* SNLI examples from the dev split evenly distributed over the three NLI labels. Then, for each δ -SNLI-TEST-SAMPLE example, we have each model predict the relation between H and each generated atom a_i from §4.2.3 using the same prompt and exemplar set. For each atom that the LLM predicts as entailed, we have it predict the relation between P and a_i using the same prompt and exemplar set.

We report overall logical consistency as the percent of examples where the full prediction was logically consistent with the predicted labels for sub-problems as dictated by the rules in §4.2.4.

Results. Despite higher *accuracy* numbers, LLMs seem to struggle with *logical consistency* (Table 4.5). Interestingly, a model’s accuracy is not fully indicative of its logical consistency. When models incorrectly predict the full example’s label, they are more prone to logical inconsistencies between atomic sub-problems and the full problem (Table 4.5, Columns 3 and 4). Though not the top performing model, **gpt-4o** outperformed other models on logical consistency **even on examples where its full prediction was incorrect**. Such logical consistency indirectly captures the *reliability* of a model’s full prediction: when two LLMs achieve similar accuracies, logical consistency serves as another point of comparison.

We also stratify our results by the predicted overall label and report logical consistency within each class (Table 4.5, Columns 5—7). All models exhibit consistency gaps *within the 3 labels*, and different models struggle with different example classes.

Lastly, we experiment with *atomic inference* (Stacey et al., 2024), or inducing an overall label via logical rules over the predicted atomic sub-problem labels, to understand whether a setting in which models only provide granular inferences can be more effective. We induce an overall label with similar logical rules to those in §4.2.4: (1) if all a_i are predicted as entailed by P , we predict entailment, (2) if at least one a_i is predicted as contradicting P , we predict a contradiction, (3) otherwise, predict neutral. While this strategy does not yield competitive performance with full example accuracy (Table 4.5, Column 1 versus Column 8), it does offer a more *interpretable* framework for LLMs that otherwise seem to struggle with logical consistency. Atom judgments may more difficult than overall judgments for a variety of reasons (see Appendix A.2.1), and in turn, inducing a label from individual atomic predictions may be less reliable.

4.2.5 Atoms in Defeasible Inference

We now turn to defeasible inference to explore how atomic sub-problems can help us better understand the complexities of the task, model performance, and the knowledge evaluated in the δ -SNLI dataset.

In traditional NLI (§4.2.4), labels of atomic sub-problems function like terms in an equation—the overall relation between P and H can be computed from strict logical rules over relations between P and individual a_i . In contrast, defeasible inference functions akin

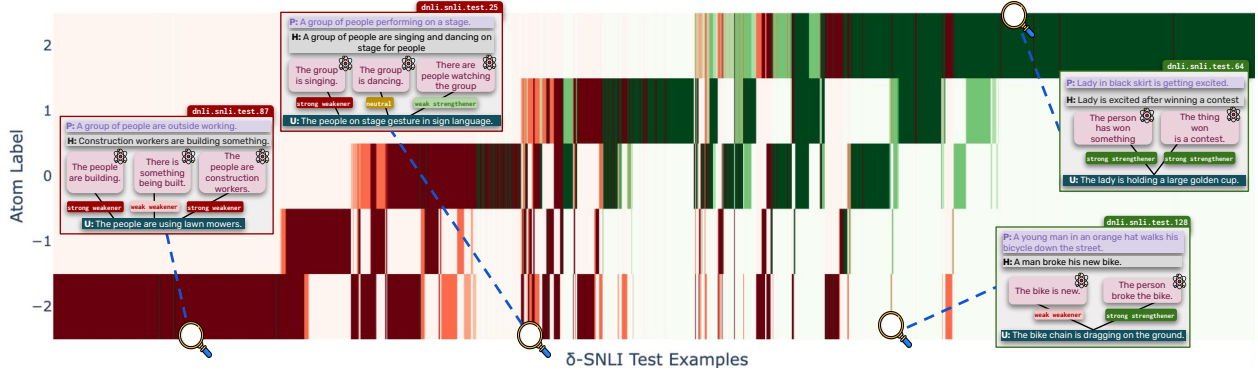


Figure 4.6: A rug plot visualization of 1,761 δ -SNLI instances and their corresponding distribution of atomic sub-problem labels. Each vertical slice represents one full δ -SNLI instance. Slice color (red or green) represents the full instance label (weakeners or strengtheners). For each δ -SNLI problem, we manually label each corresponding atomic sub-problem on a -2 (strongly weakens) to +2 (strongly strengthens) scale. Each vertical slice uses shading (light/dark) to represent the resulting distribution of atomic sub-problem labels (-2 to +2). Slices are ordered left to right by proportion of weakener labels, showing relatively high separation between red and green instances. When atomic sub-problems contain a mix of positive and negative labels, the full problem label may be a strengthener or a weakener, as illustrated by the two center-most exemplars.

to fuzzy logic (Castro et al., 1998). Determining the overall effect of the update U on H involves a softer weighing of the direction and magnitude of its effect on each atom.

For example, consider the second row in Table 4.4. A human reading the update U (“The person is wearing a city uniform”) would conclude that U *strengthens* their belief in H (“The garbage man sweeps up where the can spilled”) as opposed to weakens it. Looking at this problem through the lens of atomic decomposition helps pinpoint why. H consists of five atoms, each representing pieces of evidence *not* present in P ripe for targeting by updates. Two of the five atoms are most strongly supported by U : the person is wearing a city uniform strongly strengthens our belief that they may be a garbage man (a_1) as well as that they may be sweeping up a spill (a_5). However, U has no effect on our belief that there is a can in the scenario (a_3). The co-occurrence of a_1 and a_5 compound the strengthening effect, leading to an overall strengthening effect of U on H .

Breaking down δ -SNLI hypotheses and forming atomic sub-problems in this manner gives us a framework to understand the intricacies of defeasible inference. We begin by benchmarking recent LLMs to understand the state of defeasible inference capabilities of models. Then, we introduce the idea of a *critical atom*, or the primary piece of information an update acts on. Finally, we use critical atoms as a way to better understand and interpret model behavior, as well as argue that critical atoms serve as a useful representation for measuring the type of knowledge evaluated in δ -SNLI.

Understanding Defeasible Inference with Atomic Sub-Problems

Benchmarking LLMs on δ -NLI. We benchmark a suite of recent models on full examples from δ -SNLI-TEST, including encoder models and prompt-based models, open-source and proprietary systems, as well as models of various sizes. We finetune encoder-only models (**roberta-large** and **deberta-v3-large**) on the train set of δ -SNLI for 2 epochs with a learning rate of $2e-5$ and a batch size of 32. For all prompt-based models, we do few-shot evaluation with Prompt A.3.1 and 10 in-context examples evenly split between strengtheners and weakeners.

Many of the models in our suite surpass the human performance benchmarked by [Rudinger et al. \(2020b\)](#), with **gpt-4o** as the top performing model at 92% accuracy (Table 4.6). However, since it remains unclear whether this accuracy is indicative of a holistic understanding of situations in δ -SNLI, we turn to studying performance on the atomic reasoning problems that compose each δ -SNLI example to better contextualize these results.

Annotating Atoms. Each atomic sub-problem in δ -SNLI is (P, a_i, U) tuple capturing the effect of the update on a specific atom a_i . An author annotated all valid (as determined in §4.2.3) atomic sub-problems for each example in δ -SNLI-TEST according to the five-point scale used in [Rudinger et al. \(2020b\)](#) for validation (Table A.6) ranging from *strongly weakens* (-2) to *strongly strengthens* (+2) with a midpoint value of *no effect* (0) for atoms on which U had no effect. The same external annotator from §4.2.3 annotated a random sample of 100 *valid* atomic sub-problems on the same -2 to +2 scale (Appendix A.5), obtaining an agreement of $\tau = 0.79$ with Kendall’s Tau ([Kendall, 1938](#)).

	Full Example Accuracy	Atom Accuracy	Critical Atom Accuracy	P(Full ✓ Critical ✓)	P(Full ✓ Critical ✗)
llama-3-8b-it	80.1	65.3	77.0	90.9	45.9
gpt-3.5-turbo	81.5	66.1	76.4	91.8	49.7
gemma-2-9b-it	82.3	68.5	72.1	91.9	60.5
Human	83.6	—	—	—	—
gpt-4o-mini-2024-07-18	86.9	74.8	79.4	93.1	66.4
roberta-large	87.4	83.4*	87.8*	94.0	44.9
llama-3-70b-it	88.0	73.8	81.9	93.6	64.5
deberta-v3-large	91.1	87.4*	91.5*	95.0	55.5
gpt-4o-08-06	92.6	77.2	83.5	96.5	75.5

Table 4.6: LLM accuracies on full δ -SNLI examples (col. 1), all atomic sub-problems (col. 2), and just critical atom sub-problems (col. 3), using our manual atomic labels. We also report full δ -SNLI example accuracy conditioned on accurate (col. 4) and inaccurate (col. 5) prediction of the corresponding critical atom sub-problem label.

Ground Truth Label Distribution. Figure 4.6 visualizes the label distribution (−2 to +2) of atomic sub-problems of each δ -SNLI-TEST example as a thin vertical strip. Strips are green if the original example is a strengthener and red for weakeners. While some examples

have all atomic labels of the *same* polarity, **a significant chunk of the dataset includes atomic sub-problems with no effect or the *opposite* polarity**. Examples depicted across the spectrum in Figure 4.6 illustrate the non-monotonicity of defeasible inference.

Atomic Sub-Problem Performance. We first measure the performance of models on atomic sub-problems using annotated ground-truth labels. The original δ -SNLI dataset was designed as a binary prediction task. However, as Figure 4.6 depicts, updates may also have no effect on atoms. We adapt Prompt A.3.1 to accommodate this ternary task (Prompt A.3.2) and use atoms in exemplars instead of full hypotheses. Since the train set of δ -NLI only admits binary labels, we reuse the finetuned models (*deberta* and *roberta*), and report atom accuracy on non-neutral atoms (80% of atoms, Figure A.1).

Across the board, models perform worse on atomic sub-problems than on full examples (Table 4.6, Column 2). Since updates often act on multiple parts U (Table 4.4), we hypothesize that this compounding effect may contribute to higher performances on full examples.

We observe that some atomic sub-problems are more critical contributors to the overall effect of U on H . Consider the example in row 2 of Table 4.4. Since U acts most strongly on a_1 and a_5 , **we can assume that they are critical in determining the overall effect of U on H** , and correctly understanding the effect of U on a_3 or a_4 is not essential to the overall problem. We formalize this below.

Critical Atoms as Questions Under Discussion (QUD)

As established, updates vary in which atom they most strongly affect. Consider H and the three updates in Figure 4.7. Each U targets a distinct (or *critical*) atom without having an effect on the others. We formalize this notion by recognizing that hypotheses (or more broadly, sentences in discourse) serve as an answer to a large space of possible questions: all three questions in the right-most column could be answered with H . However, when updates target particular atoms, the *strategy* by which they do so favors a particular question Q , making it more likely that H is the answer to Q as opposed to any other question.

These questions function as *questions under discussion* (QUD), a well-studied linguistic phenomenon (Benz and Jasinskaja, 2017). The update “*The man smiles on his boat*” has no effect on the atom about vest color and hence does not pick out the QUD associated with that atom. We call the atoms above the *critical atom* for each corresponding update, as they uniquely pick out particular QUDs over others. **Critical atoms correspond to the particular inference or piece of knowledge a defeasible NLI example aims to test.** As such, we use them as a framework to measure consistency as well as the diversity of the δ -SNLI dataset. This formulation of QUDs relates to question answering-based semantics (He et al., 2015; Pyatkin et al., 2021; Klein et al., 2022) in which QA pairs capture semantic information, such as semantic roles.

Identifying Critical Atoms of Updates. In order to identify the critical atom for a defeasible NLI example, we identify the subset of its *valid* atoms with the strongest labels

Hypothesis: A man in an orange vest stands on a dock .		
Atom	Example Update	H answers this QUD
a₁ The vest that the person wears is orange .	The vest is made from purple yarn.	What color is the vest that the man is wearing?
a₂ The person stands .	His folded legs rest on the ground.	What is the man doing on the dock?
a₃ The person is on a dock .	The man smiles while on his boat .	Where is the man?

Figure 4.7: Updates (U) may act on the same hypothesis H in different ways by targeting different atoms. Here, each U strongly targets a different atom, while having no effect on the other atoms derived from H (e.g. the U in the first row has no effect on a_3 in the last row). We refer to the atom(s) which an update most strongly affects as the “critical” atom of the (P, H, U) δ -NLI example. Critical atoms help identify the *question under discussion* of the example.

that match its overall polarity. The majority of δ -SNLI-TEST examples have one critical atom (Figure A.2), but it is possible for example to have multiple if the effect is equally strong, as in row 2 of Table 4.4.

Performance on Critical Atomic Sub-Problems versus Full Examples. Across the board, models are stronger on the subset of critical atomic sub-problems than on all atomic problems (Table 4.6). We hypothesize that LLMs may be better at modeling stronger, **direct** inferences such as those in critical sub-problems, but may struggle when the effects are *indirect* or weaker.⁸ Such nuanced distinctions require a robust understanding of the multiple factors that control the underlying inference, a skill that even larger models seem to struggle with.

We also measure the probability that a model correctly predicts the label for the full example *given* that it has correctly (Column 4) and incorrectly (Column 5) solved all critical atomic sub-problems (Table 4.6). Correctly solving all atomic sub-problems is a strong indicator that a model is likely to predict the full problem correctly. However, some models still have as high as a 75% probability of predicting the full answer even having incorrectly predicted critical sub-problems, calling into question the robustness of their reasoning process in the face of such inconsistency.

4.2.6 Measuring Inferential Consistency in δ -NLI

As established, identifying the critical atom(s) of a δ -NLI example allows us to pinpoint the knowledge or fact that the example is designed to evaluate (§4.2.5). The two different examples in Figure 4.8 share the same critical atom, representing two different contexts under which the model must directly evaluate the fact “*The people are friends.*” A model correctly predicting whether or not people are friends under some contexts but not others may

⁸For example, U in Row 2 of Table 4.4 has no effect on a_3 , but that effect is slightly strengthened upon mention of a spill in a_4 , since donning a city uniform is likelier at spill sites.

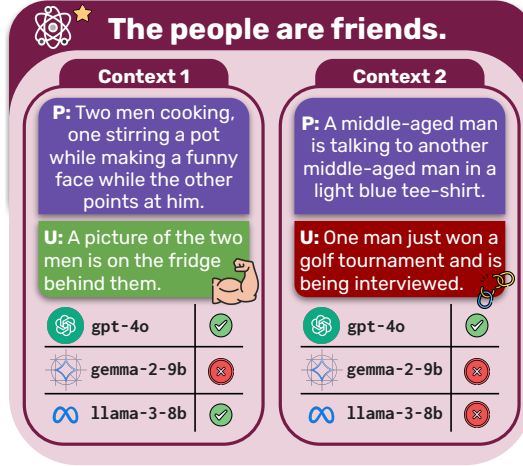


Figure 4.8: Grouping examples by their critical atom(s) allows us to understand under which *contexts* ($P + U$) a model has understood a piece of knowledge. Here, we show two δ -NLI examples that evaluate the same atom (top): one that *strengthens* it (left), and one that *weakens* it (right). A model that truly understands a fact and the factors that influence it (or, conversely does not) should yield consistently correct or incorrect predictions. However, some models have mixed accuracy among examples targeting the same atom, indicating that they only understand the inference under *some* contexts.

indicate that it has not fully understood the factors that influence the inference. Because they contain independent examples, few datasets accommodate measuring a model’s *inferential consistency* (I_C), or the likelihood that its prediction for a particular inference will remain consistently correct or incorrect under different contexts (here, contexts refers to different (P, U) pairs). Correctly drawing an inference under a single context does not guarantee that the model will make a correct prediction for the same inference under a different context.

To quantify this, we group examples in δ -SNLI-TEST by their critical atom and report the inferential consistency of different models.

How many unique critical atoms does δ -Snli-test evaluate? While δ -SNLI contains around 2K test examples and 88K training examples, how many distinct critical atoms underlie those examples? We quantify this by identifying semantically equivalent critical atoms in δ -SNLI-TEST by embedding each atom with NV-Embed-7B (Lee et al., 2024) and then computing pairwise cosine similarity between all critical atoms. We construct a graph G where nodes correspond to critical atoms and edges are drawn between nodes if their similarity is above a threshold ($\theta = 0.75$) and they are bidirectionally entailed as determined by the NLI model in §4.2.3. Finding semantically equivalent groupings then reduces to finding all maximal cliques (Tomita et al., 2006) in G .

The 1,761 δ -SNLI-TEST examples that contain at least one valid atom (from 193 unique

P - H pairs) contain 429 unique atoms.⁹ Following the procedure above yields 349 unique cliques, or **349 unique critical atoms**.¹⁰ We bucket examples by these cliques, resulting in groups of examples that share a critical atom (which we call *critical atom buckets*, or just *buckets*, represented as $\otimes$). We find that certain critical atoms arise frequently: top critical atoms include “The others are friends of the person” (21 examples) or “The person is a man” (20 examples).

Measuring Inferential Consistency (I_C). Srikanth et al. (2024a) introduce *paraphrastic consistency*, a metric capturing the probability that a model’s prediction for two paraphrases of the same NLI problem remain consistent (both incorrect or both correct). We adapt this metric to compute a model’s *inferential consistency*, or the probability that its predictions for two defeasible NLI examples e_i and e_j **that share the same critical atom** are either both correct or incorrect. As in Srikanth et al. (2024a), we define two terms:

- $R_{\otimes}$: a discrete random binary variable representing whether a model’s prediction for a full δ -NLI example is correct (1) or incorrect (0).
- $\theta_{\otimes}$: $\mathbb{E}[R_{\otimes}]$, or the average correctness (i.e., accuracy) of the δ -NLI examples in a particular critical atom bucket.

For a binary classification task ($y \in \{0, 1\}$) and a model M , we define I_C as

$$I_C = \underbrace{P(M(e_i) = y, M(e_j) = y)}_{\text{prob. of both predictions correct}} + \underbrace{P(M(e_i) \neq y, M(e_j) \neq y)}_{\text{prob. of both predictions incorrect}} \quad (4.1)$$

We estimate I_C directly from the accuracies of critical atom buckets as:

$$I_C = \mathbb{E}[\theta_{\otimes}^2] + \mathbb{E}[(1 - \theta_{\otimes})^2] \quad (4.2)$$

Note that δ -NLI examples can have multiple critical atoms (Figure A.2). In these cases, we divide the weight of the example e across all critical example buckets that share e when computing $\theta_{\otimes}$.

Results. All models exhibit room for improvement in inferential consistency (Table 4.7), giving us a sense of how well the LLMs we analyze have internalized the critical atomic facts evaluated in δ -SNLI-TEST. One source of inconsistency we observe arises when certain contexts (premise-update pairs) demand more implicit reasoning or background knowledge. For example, consider two different contexts under which the model must evaluate the same critical atom “The people are tall”:

⁹Note that the number of unique atoms is upper-bounded by the SNLI hypotheses that δ -SNLI builds upon.

¹⁰We also experiment with generating the QUDs (Appendix A.4) for each critical atom and find that δ -SNLI-TEST evaluates 223 unique QUDs, around an 81% coverage of the available atoms.

Model	Inferential Consistency
llama-3-8b-it	74.5
gpt-3.5-turbo	75.8
gemma-2-9b-it	76.5
gpt-4o-mini-2024-07-18	81.7
roberta-large	82.6
llama-3-70b-it	83.5
deberta-v3-large	87.0
gpt-4o-08-06	88.7

Table 4.7: Inferential consistency of models (I_C) on δ -NLI examples. We group examples that share the same critical atom and compute the probability that two examples in the same group were both incorrectly or both correctly predicted by a model.

Context 1: P : Four people standing on a hiking trail in a forest with big tree logs on the ground, U : Their long legs step across several logs at once.

Context 2: P : Two men in orange uniforms stand before a train and do some work, U : They can easily touch the top of the train.

Several models struggle to draw the strengthening inference under the first context, but all models that we analyze successfully draw the strengthening inference under the second.

These analyses help us understand whether models understand pieces of knowledge and the factors that influence them, raising interesting questions about how best to collect updates to increase the coverage of a fact or situation. Defeasible reasoning systems must be able to deftly modulate inferences in a manner that is sensitive to diverse contexts. Future work may leverage the identification of critical atomic sub-problems to nudge annotators toward underrepresented critical atoms or contexts.

4.2.7 Conclusion

Hypothesis decomposition allows us to measure the logical consistency of models as well as better understand the structure of non-monotonic reasoning. We find that labels of atomic sub-problems in defeasible reasoning share a more complex relation to the full problem than in traditional NLI, even within the same label class. We introduce *critical atoms* as the primary fact evaluated by a defeasible NLI example, enabling us to group examples by shared critical atoms and measure the *inferential consistency* of LLMs across different contexts.

4.3 Proposed Work 1: Exploring Influence Factors in Natural Language Inference with a Defeasible Investigator

Here, we describe one proposed work designed to better explore some of the factors that *influence* inferences in the δ -NLI dataset. This work is designed to directly and indirectly target all three contracts presented in this chapter: context sensitivity, logical consistency, and inferential consistency.

4.3.1 Motivation

When drawing an inference in a situation, humans rely on their ability to identify the *factors* that may influence that inference. For example, let’s consider the following scenario:

Premise: A woman drops a glass.

Hypothesis: The glass breaks.

When trying to understand whether or not to believe the hypothesis above, we may identify several underlying *factors* that may influence the outcome: (1) the **height** from which the glass was dropped, (2) the **surface** upon which it fell, (3) the **thickness and durability** of the glass, or (4) whether the glass **landed** on its base or side. Setting each of these factors, or *mediating variables*, to different values may affect our inference about the glass breaking. For example, if we know that the glass was dropped from the 45th story of a building, we might strongly believe that the glass indeed broke, but if it was nudged from a coaster already placed on the ground, it may not have. Similarly, if the glass is a champagne flute, it may be thinner and less durable, making it more likely to break, but if it was a Red Solo Cup, it may not break at all.

Current reasoning datasets, including δ -NLI, contain many *independent* examples, often precluding us from understanding whether models have a *holistic* understanding of a situation by way teasing apart the precise factors that contribute to drawing a particular inference. While δ -NLI does start to address this by including many strengtheners and weakeners for a particular premise and hypothesis pair, these sentences were constructed by *showing* annotators the premise-hypothesis pair and asking them to generate two free-text sentences: one that supports the hypothesis (strengtheners) and one that weakens it. This label-based elicitation often skews the kinds of update sentences that annotators are likely to generate towards particular influence factors. Recent work argues that the causal direction of data collection can influence model behavior (Jin et al., 2021). Since the δ -NLI dataset was constructed by *first* presenting *P-H* pairs to annotators and *then* eliciting update sentences, the causal direction may shape which types of influence factors annotators implicitly rely on when writing update sentences, potentially biasing examples in the dataset towards certain causal mechanisms while underrepresenting others. In this proposed work, we aim to discover and analyze a more exhaustive set of factors that mediate a particular inference using a semi-automatic pipeline.

4.3.2 Research Objectives

1. How can we discover factors that mediate the strengthening and weakening inferences in δ -NLI (*influence factors*)? How should we represent these factors?
2. What kinds of influence factors do annotators overwhelmingly identify during dataset creation? Do they span a diverse set of factors, or are they clustered around a few?
3. Do language models “understand” some influence factors better? In other words, do they perform better with drawing inferences based on pieces of evidence that fall under certain influence factors as opposed to others?
4. What happens when we construct evidence by *composing* pieces of information across different factors? Are humans still able to make a decision, and what does the variability look like across judgments?
5. **Do the influence factors exhibit hierarchical dependencies, where answering one factor partially constrains or determines others?** Can we automatically induce this dependency structure from the data?
6. **How sensitive are model predictions to interventions on specific influence factors?** If we systematically vary one factor while holding others constant, which factors cause the largest shifts in model confidence?

4.3.3 Preliminary Experimentation

We select *questions* as the construct with which to describe an influence factor (“*From what height was the glass dropped?*”). The answers to these questions serve to offer more information or evidence about the situation described by the premise. Consider the influence graph visualized in Figure 4.9. Using the methodology from work presented earlier in this chapter (Srikanth and Rudinger, 2025), we first decompose hypotheses into atomic propositions. Then, we *cluster* existing updates with respect to the question that the update sentence answers. In this example, the hypothesis is decomposed into many atoms, two of which are relevant here: one about the activity (repairing) and one about the object of repair (dam). Update sentences are leaf nodes in this graph that answer influence factor questions (shown in deep purple circles). Updates written by annotators in the δ -NLI dataset are colored in dark green. We see that these annotator-written updates tend to cluster around certain influence factors, failing to cover others. We include manually written updates that span other influence factors in dark orange for contrast. This visualization helps us understand that the updates included in the δ -NLI dataset may be biased to particular types of factors, and may preclude us from measuring whether a model has truly understood a particular inference.

To automatically generate new updates for a situation, we generate question and answer *pairs*, where questions serve as potential influence factors, and answers to those questions serve as “update” sentences in the defeasible NLI framework. This allows us to generate

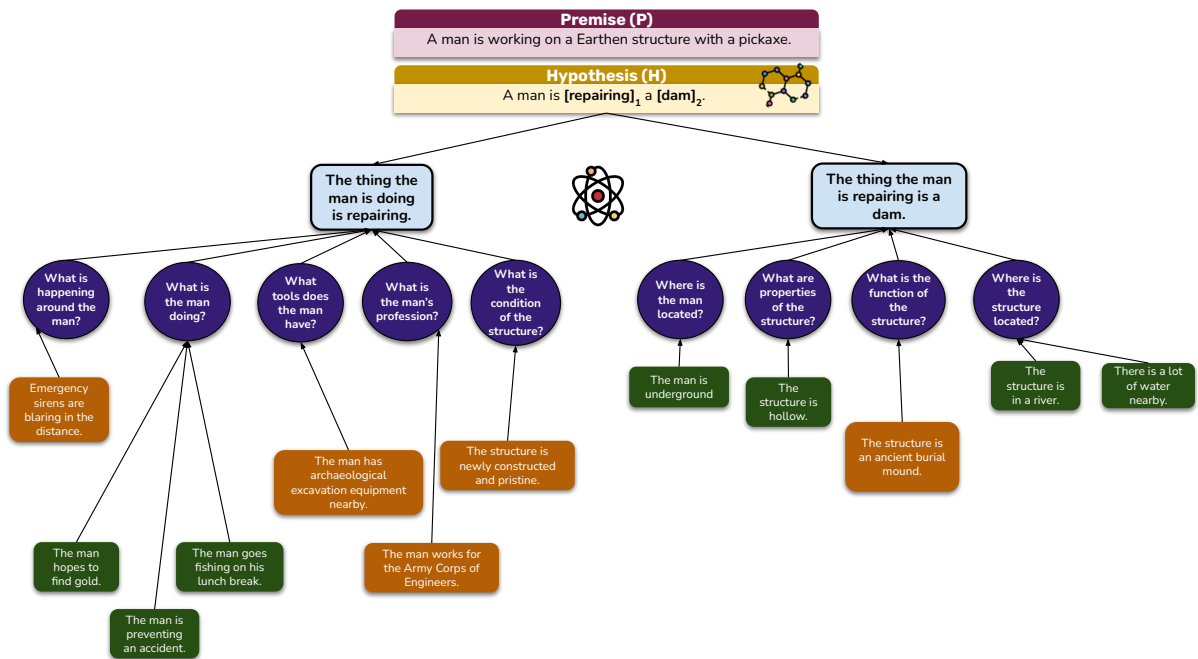


Figure 4.9: Visualizing the connections between different updates across influence factors.

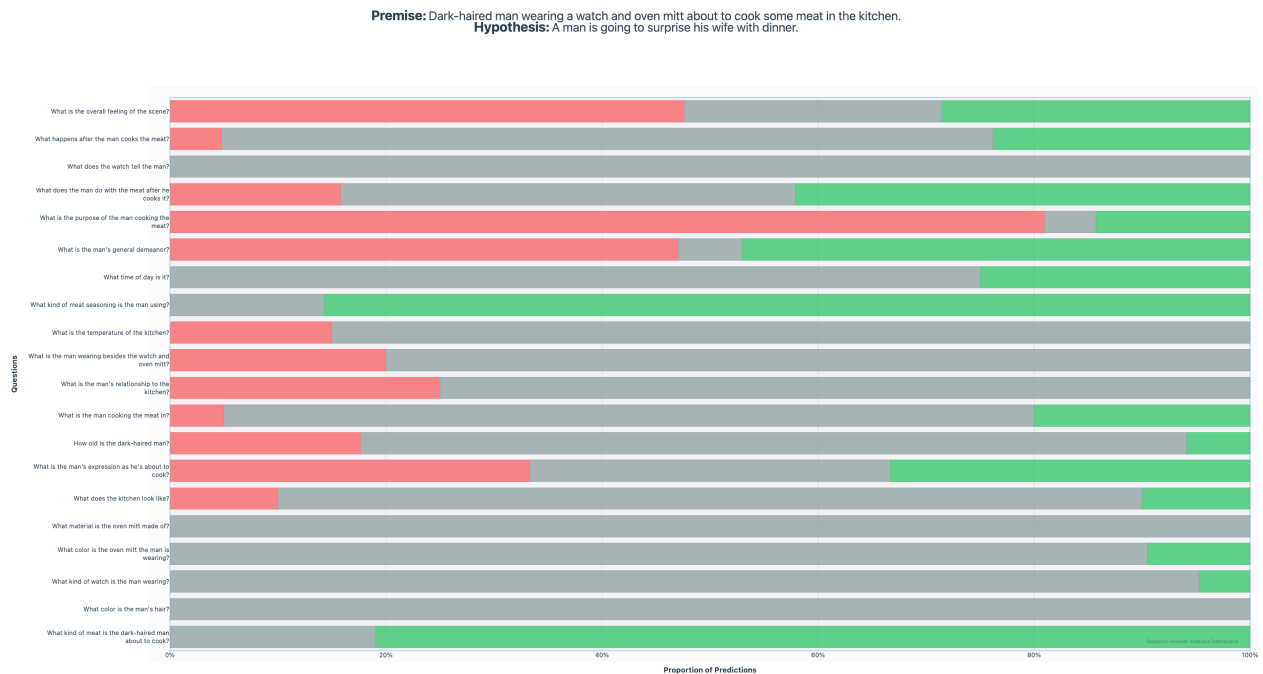


Figure 4.10: Visualizing generated questions for a particular premise and hypothesis pair.

update sentences that are *not* influenced by the hypothesis, eliminating the causal direction

of hypothesis to update. Generating these new examples helps us build out the set of possible influence factors in service of better modeling inference and situational dynamics.

We do this in a two stage pipeline. Examples in δ -NLI that are constructed on top of the SNLI dataset (Bowman et al., 2015). First, in a **question generation step** we use a vision-language model, **gemma-3** (Team et al., 2025) to generate situationally-grounded questions from the Flickr image corresponding to the premise, included in the metadata of SNLI examples. For each question, we prompt the model to generate an example answer as well. Then, we perform **answer set generation**, where the same LLM generates a set of alternate answers to the question. We filter out answers that directly contradict the premise, a fundamental constraint communicated to annotators during the construction of the δ -NLI dataset.

Figure 4.10 visualizes a particular set of generated questions and answers for a given premise and hypothesis pairs. We elicit judgments from **gpt-4o-mini** (OpenAI, 2024) on whether an answer strengthens, weakens, or has no effect on the hypothesis, and visualize the *distribution* of answers across the three labels. We see that certain influence factors (questions) are more likely to yield answers with some effect on the hypothesis.

Several questions *presuppose* information not present in the premise: future work must deal with the filtration of such questions.

4.3.4 Remaining Experimentation

1. **Better Question Generation:** We will refine the question generation process to produce higher quality questions that are (1) diverse (distinct from each other), (2) cover different *types* (events, cognitions, properties, etc.)
2. **Dependency Structure Generation:** We will use the questions and answers that were automatically generated, along with the existing updates in δ -NLI to generate structures similar to the one pictured in Figure 4.9. This will enable a number of downstream analyses listed above.
3. **Human Judgments:** After refining question and answer pairs, we will collect graded human judgments on particular question-answer pairs for different examples, allowing us to understand *human* reasoning for examples across different, sometimes conflicting, influence factors.
4. **Model Judgments:** Compute model judgments on different examples, and compute correlation with human judgments.

4.4 Proposed Work 2: δ -NLI++: A Dataset for Cascading Contextual Reasoning

Here, we discuss a proposal for the construction of a new contextual defeasible reasoning dataset, δ -NLI++. Again, this work targets all three contracts presented in this chapter: context sensitivity, logical consistency, and inferential consistency.

4.4.1 Motivation

Rudinger et al. (2020b) release δ -NLI, a dataset designed to evaluate the defeasible reasoning capabilities of large language models. The dataset is constructed on top of three primary sources: SNLI (Bowman et al., 2015), ATOMIC (Sap et al., 2019), and SOCIAL-CHEMISTRY-101 (Forbes et al., 2020b). Examples in this dataset typically require systems to reason about a singular piece of additional context. In contrast, reasoning in the real world often involves weighing *multiple* pieces of contextual information that may *interact* in complex ways. For example, an update may strengthen an inference, but subsequent information might weaken or even reverse that conclusion. Consider domains such as medical diagnostics, legal reasoning, or investigative journalism. In these settings, conclusions that we draw typically evolve as new evidence is presented, with each piece of information potentially reinforcing, contradicting, or adding nuance to previous updates. As it stands, δ -NLI is not equipped to evaluate the longer range defeasible inference capabilities of LLMs.

4.4.2 Proposed Experimentation

We propose the construction δ -NLI++, a dataset designed to capture the multi-step, interactive nature of defeasible reasoning by providing examples with multiple updates that must be reasoned about sequentially or jointly. We propose its construction on top of three data sources:

1. **NewsEdits** (Spangher et al., 2022): This dataset contains approximately 77K news article revisions from 22 English-language news outlets, collected over a 15-year period (2005-2019). It captures how journalists *update* news stories as events unfold and new information becomes available, making it well-suited for studying defeasible reasoning in real-world contexts. The authors of the dataset observe that “the addition and deletion of content in the articles are likely to contain fact-patterns associated with breaking news (quotes, events, or main ideas) than unchanged sentences.”
2. **Incremental Question Answering: Quizbowl** (Rodriguez et al., 2019): This dataset consists of questions from the academic quiz bowl competition, where questions are designed to be answered incrementally as more clues are revealed. These questions are structured as sequences of sentences that are meant to narrow down to a specific answer. The pyramid structure naturally supports defeasible reasoning, where players may form early hypotheses about properties or entities, and they must be abandoned or

strengthened as the players are presented with subsequent clues. For example, consider the inference “The entity is an invertebrate.” An early clue mentioning “eight legs” may strengthen this hypothesis, but a subsequent clue stating "this creature has an internal skeleton" would force a revision, as invertebrates lack internal skeletons.

3. **r/AmITheAsshole**: This dataset consists of posts from the Reddit community where users describe social situations or conflicts, and ask the community whether their actions were justified. The subreddit’s structure naturally elicits defeasible *moral* reasoning, as readers must evaluate whether the user is in the wrong based on the initial narrative, but may update their judgments additional emerging context such as follow-up comments, edits, or responses to clarifying questions. Situations often contain conflicting influence factors. For example, an initial post may suggest the poster acted reasonably, but they may subsequently reveal information about the past or hidden motivations that may reverse this initial assessment.

We will collect and clean raw data from these three dataset sources, identify particular *inferences* from which to construct defeasible NLI example *hypotheses*. Human annotators will see each piece of new information, and will be asked to provide a new rating for the hypothesis from -2 (strongly weakening) to 2 (strongly strengthening) based on the body of currently available evidence. This will allow us to *track* the progression of human judgments across evidence patterns. We will evaluate a series of state of the art language models in order to compare their judgment progression to human annotators.

4.5 Chapter Conclusion: Deep Context Understanding

This chapter introduced three new contracts: (1) **context sensitivity**, where models must modulate (increase or decrease) their likelihood of drawing an inference when the context in a reasoning problem changes, (2) **logical consistency**, where models’ individual predictions on sub-components of a reasoning problem must align with its overall prediction of the entire problem, and (3) **inferential consistency**, where the system must uphold its ability to reason about the same fact in different contexts. In service of our results that show that models still exhibit room for improvement with these contracts, we introduce two new proposed works designed to better evaluate and mitigate issues related to these contracts. First, we proposed a “*defeasible investigator*” to ask probing questions about a situation in order to understand the factors that may contribute to drawing an inference. Then, we proposed the construction of a more contextual defeasible reasoning dataset.

Now, we turn to a completely different NLP task, question answering, where we look at *naturalistic* user data. QA brings with it the need for different *kinds* of contracts focusing on better understand the user’s information need.

Chapter 5: Inferential Robustness with Presuppositionally Problematic Questions¹

Chapters 3 and 4 mainly examined the inferential robustness of natural language reasoning models in simplistic, toy-like setups. The data we use is largely constructed by researchers, and so does not naturally express the full range of linguistic inputs and complexity that NLP systems must handle. Moreover, natural language inference in its current formulation is a largely synthetic task, created by researchers (Bowman et al., 2015) to benchmark the general language understanding capabilities of models, but not necessarily intended to reflect the type of reasoning that arises naturally in human-AI interactions.

To assess the ability of language models to uphold inferential robustness-related contracts in naturalistic settings, we turn to the task of question answering, one of the foundational ways that users interact with LLMs such as ChatGPT (Hadi Mogavi et al., 2024; Zhao et al., 2024b). Large language models deployed as chatbots or QA systems *must* be able to handle all types of user input, including input that is flawed. The next two chapters (Chapters 5 and 6) examine model (and human) behavior in the face of such flawed input. We argue that models must emulate human ability to reason about such inputs *in context* (also known as *pragmatic* reasoning) to best support the user. This chapter focuses on one particular failure mode of input: false assumptions that are embedded in user input. In order for models to garner enough trustworthiness to be used in high-stakes domains such as health, users must be able to observe models upholding more complex or demanding contracts beyond simple language or context understanding. Namely, the system must critically reason about the user’s input *in the context of the real world* to draw inferences that help it satisfy the user’s information need, **and mitigate potential harm**. Here, we study model ability to uphold this contract in the domain of maternal and infant health.

Contract: **Making Critical Inferences.** When user input contains false assumptions, systems should detect and address these problematic beliefs in order to mitigate potential harm, rather than ignoring it or simply answering the surface-level question.

¹Neha Srikanth, Rupak Sarkar, Heran Mane, Elizabeth Aparicio, Quynh Nguyen, Rachel Rudinger, and Jordan Boyd-Graber. 2024b. [Pregnant questions: The importance of pragmatic awareness in maternal health question answering](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7253–7268, Mexico City, Mexico. Association for Computational Linguistics

5.1 Introduction

Humans have varying information needs when they ask questions (Taylor, 1962). Sometimes these needs are easily inferred from the surface form, such as in factoid questions (e.g. “Who is the 44th president of the United States”). However, in a question such as “Is there a good non-dairy baby milk I can supplement for my newborn?”, addressing the underlying false assumption “Newborns can safely drink non-dairy milk” **becomes part of satisfying the unexpressed information need**. Complete answers to these types of questions must not only address the surface question itself, but also “question the question”, critically examining its *pragmatic needs*.

These needs become magnified in sensitive domains, such as consumer health or the legal domain. In these settings, addressing pragmatic needs of questions involves proactively addressing false assumptions or implications in questions to ensure that the asker does not continue holding inaccurate beliefs that they may act on. For example, a complete answer to the question about non-dairy milk for newborns should address that while non-dairy milk is viable for older babies, newborns and infants need human breast milk or dairy-based formula because they offer complete nutrition.

Language models have been shown to exhibit sycophancy (Sharma et al., 2023), sometimes adjusting responses to align with a human user’s view. However, helpful QA systems should not only *challenge* false or subjective assumptions in questions (Kim et al., 2022) by verifying them against a vetted corpus, but also infer the asker’s intent to make sure that its answer satisfactorily addresses their deeper information needs (Taylor, 1962), just as humans do.²

We construct a dataset³ of 2,727 assumptions and implications in 500 questions (§5.2) collected from three diverse sources to study (1) how humans embed such assumptions and implications in questions, and (2) the extent to which they are *naturally* addressed in answers written by public health experts. We then ground assumptions and implications, two primary ways humans embed beliefs in questions, in existing linguistic theory of presuppositions and implicatures respectively (§5.3). We refer to presupposition and implicature collectively as *pragmatic inference*. While recent work has focused on the task of detecting and addressing false *presuppositions* in open-domain QA (Yu et al., 2022), we find that false beliefs of question askers are more likely to present as implicatures than presuppositions (§5.4). We experiment with *inducing* pragmatic behavior in existing QA pipelines with state-of-the-art retrieval and machine reading models (§5.5). On questions with at least one highly plausible false pragmatic inference, our expert annotators rated responses from our pragmatic QA system as more helpful and informative.

Thus, QA systems of the future must proactively address assumptions and implications in questions as they are increasingly deployed in sensitive domains.

²“Pregnant” in our title also refers to its secondary definition, “full of meaning” (as in “a pregnant pause”) alluding to the idea that questions are laden with implicit beliefs.

³<https://github.com/nehasrikn/pragmatic-inferences-qa>

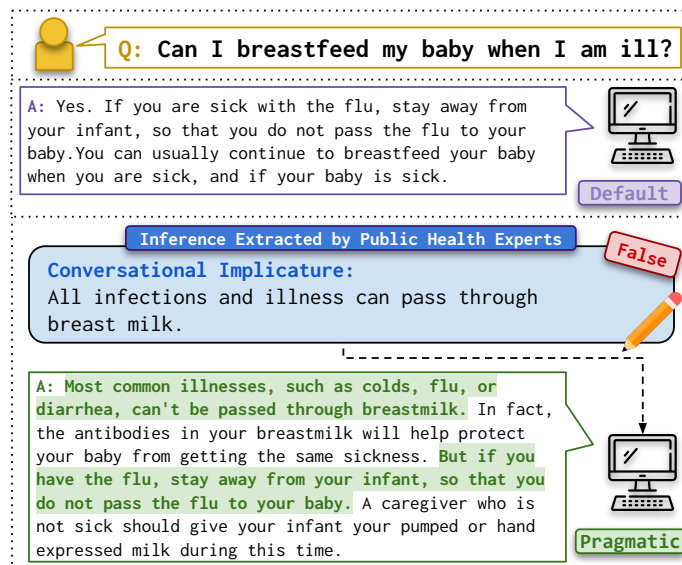


Figure 5.1: We ask public health experts to identify assumptions and implications from questions and find that incorporating them in a QA pipeline produces a more complete answer.

5.2 Collecting Assumptions and Implications in the Wild

In contrast with factoid QA, systems deployed in *sensitive* domains such as consumer health must proactively mitigate harm. In these settings, correcting false assumptions is *not optional*: systems must provide contextual answers that balance *information completeness* with brevity.

Access to high quality healthcare in the United States vastly differs across socioeconomic backgrounds (Becker and Newsom, 2003). Such users are often likely to turn to accessible internet resources and—as of late—general purpose chatbots (Palanica et al., 2019). This motivates us to focus on maternal and infant care, a challenging area of consumer health where patients are concerned with *both* their own physical health as well as the health of their child.

To effectively study and induce pragmatic behavior in QA systems, the evaluation questions we choose must reflect real-world experiences and situations for which there may not be a straightforward answer explicitly addressed in a single web document. For example, answers to Natural Questions (Kwiatkowski et al., 2019a, NQ)—a popular open-domain question-answering dataset—can be found directly in short extracted text snippets from Wikipedia (Table 5.1). In contrast, effectively answering the subjective questions sourced from Reddit requires commonsense reasoning and domain knowledge while identifying the asker’s intent.

We carefully construct a dataset of questions from three distributionally distinct sources: a domain-specific QA system we design and deploy to pregnant and postpartum participants

	Question	Expert-Annotated Pragmatic Inference & Veracity
ROSIE (Maternal Health QA System)	Is it okay for my to color my hair after giving birth?	Hair dye chemicals can pass through breast milk from mother to child. (False/Unsure)
	What is the advantage for not having an epidural during the labor?	Avoiding an epidural contributes to a more "natural" and unmedicated birthing experience. (False/Unsure)
	What cough medicine is appropriate for breastfeeding mothers?	Some cough medicines can be secreted in breast milk. (True)
Reddit	Is it safe to lay on my stomach at 28 weeks of pregnancy?	Sleeping on the stomach while pregnant may have potential risks (False/Unsure)
	Is it bad to use different bottles/nipples during feedings?	Using different bottles or nipples for feeding may compromise the baby's latch. (False/Unsure)
	How can I increase the time between feedings for my 3-month-old baby?	It may be possible to sleep through the night while still ensuring the baby is fed. (True)
Natural Questions	When does the fetus begin to develop memory?	Fetuses have the ability to form memories. (True)
	What causes a rupture in the amniotic sac?	There may be ways to prevent early amniotic sac rupture. (False/Unsure)
	When do the clinical manifestations of an ectopic pregnancy appear?	There may be clinical manifestations of an ectopic pregnancy that do not appear early on. (True)

Table 5.1: Health experts identify pragmatic inferences from questions from three sources: Reddit, Natural Questions (Kwiatkowski et al., 2019a), and questions asked to our domain-specific QA system, ROSIE (Mane et al., 2023). They also determine the veracity of each inference and provide supporting evidence from a trusted web document.

	Maternal Health QA	Reddit	Natural Questions
# questions	200	200	100
ans. length (# sent)	3.9	6.6	5.6
# inferences	1161	1114	452
% false/subjective	22.5	30.8	20.1
% true	77.5	69.2	79.9

Table 5.2: Dataset statistics stratified by question source.

we recruit (Mane et al., 2023), Reddit, and NQ. Then, we introduce an annotation scheme to elicit assumptions and implications from these questions, validate their plausibility, and finally collect supporting evidence to determine their veracity. Our final dataset contains 2,727 assumptions from 500 evaluation questions (Table A.6). We also include 150 development questions used to train annotators and develop our QA systems.

5.2.1 Gathering a Diverse Set of Maternal and Infant Health Questions

Maternal Health qa System. We source questions come from a maternal and infant health-specific question answering system that we build (Mane et al., 2023), henceforth referred to as ROSIE. Users ask questions pertaining to pregnancy or infant health and are instructed that the QA system does not have any personalized knowledge of their individual medical history or pregnancy.

This system operates over a corpus of web documents we construct⁴ from trusted sources including United States governmental and hospital organizations on maternal and infant health, and spans salient topics such as pregnancy and postpartum symptoms, developmental milestones, and infant safety. Our end-to-end QA system, ROSIE, uses a passage retriever and reranker to provide web passages as answers to study participants via a mobile application. We randomly sample 200 anonymized questions asked to ROSIE for our evaluation set and 50 questions for our development set.

Reddit. While the questions asked to ROSIE do reflect real-world experiences, they are asked *to an automatic system* and thus tend to include less situational detail or implicit content. We turn to Reddit⁵ to capture long-tail questions that are about the diverse set of unique situations a new or expecting parent goes through. Table 5.1 highlights some distributional differences between questions from Reddit and other data sources. Our questions come from four popular subreddits about maternal and infant health: r/BabyBumps, r/breastfeeding, r/NewParents, r/Mommit, and r/beyondthebump from the pushshift⁶ dump.

We develop a series of heuristics as a recall-oriented first step to identify questions with false or subjective assumptions. We begin by selecting questions where an upvoted comment shows assumption-correcting behavior or where a user invokes their medical expertise, identified by a select list of discourse markers (Appendix A.19). Of these, we only retain posts beginning with a “wh” word, filtering a few hundred thousand posts down to 2,858 questions.

As Reddit encourages community participation, many questions are “community seeking” as opposed to *information-seeking*. To identify information-seeking questions, we use GPT-3.5 (Ouyang et al., 2022) to filter medical questions from non-medical questions (Prompt A.19.1)

⁴Corpus available upon request. We use Barbaresi (2021) to scrape 408,000 web documents which we split into passages of 100 tokens following Karpukhin et al. (2020a).

⁵<https://www.reddit.com/>

⁶<https://github.com/Watchful1/PushshiftDumps>

then manually vet the final set of 297 questions. We randomly sample 200 questions for our evaluation dataset and 50 questions for our development set, discarding the rest.

Titles of Reddit posts are often a hook or a summary of the entire post. Using the 50 development questions, we use GPT-3.5 to minimally edit the titular question to include crucial details from the post description, providing a series of exemplars (Prompt A.19.2). These rewrites mainly include the age of a newborn or the stage of pregnancy from the description, but sometimes include small situational details that contextualize the question. Two authors validate all rewrites, keeping the original title wherever both authors agree that the rewrite changed the communicative goal of the asker.

Natural Questions. Lastly, we include maternal and infant health questions from NQ to study pragmatic aspects of factoid-style questions. We embed all questions in the train set of NQ using the `sentence-transformers` (Reimers and Gurevych, 2019) implementation of `all-mpnet-base-v2` (Song et al., 2020), including unanswerable questions (Asai and Choi, 2021). We identify 2500 answerable questions and 2500 unanswerable questions as maternal health-related by identifying the top 100 nearest neighbors of 50 randomly sampled questions from the development sets of Reddit and ROSIE.⁷ From this set, we randomly sample 100⁸ questions for our evaluation set and 50 for our development set. Though obtained with a nearest neighbors approach, these questions greatly differ from those obtained from our previous sources, as they reflect the factoid QA-oriented tasks and goals of the original dataset creators (Table A.6).

Collecting Human Answers from Health Experts. We recruit a team of twenty health experts using Upwork⁹ to annotate our data including obstetricians and gynecologists (OB/GYNs), nurses, lactation consultants, and public health experts, many of whom have experience with patients. In addition, many of these expert annotators are either currently pregnant or postpartum or have been in the past. We ask a subset of six experts to write helpful and informative long-form answers to all 500 questions in our dataset (Figure A.16, bottom panel). While annotators write answers from scratch, they must provide supporting web documents from the same list of verified sources we use to build the corpus for ROSIE.

5.2.2 Identifying Assumptions and Implications

Inferring *possible* assumptions, implications, and asker beliefs from patient questions in our domain are challenging. In the past, others have extracted assumptions using shallow signals from the surface form of a question (Kim et al., 2021; Parrish et al., 2021b). While

⁷We tried several different filtering heuristics, including keyword-based detection, but the nearest neighbors approach yielded the most topical questions.

⁸NQ questions make up a smaller proportion of our evaluation dataset as we avoid diverting large amounts of annotation resources to factoid-style questions.

⁹<https://www.upwork.com/>

some assumptions or implications in our dataset can be inferred directly from the question expression, others require deeper domain or experiential knowledge (Table 5.1).

Eliciting these assumptions and implications from non-linguists is challenging as existing linguistic frameworks (§5.3.1) are inaccessible or cumbersome for those unfamiliar with the theoretical concepts behind them. As such, we operationalize large-scale data collection by asking five annotators from a *different* subset of our expert annotator pool to first write a list of subquestions that an answer to the original question would address (Figure A.16, top panel). Doing so primes annotators to reason about the intent behind a question as well as the information needs of an asker. Then, we ask them to write a set of sentences reflecting possible beliefs or assumptions that the patient may hold (or, alternatively, beliefs that any complete answer to the question must address). We emphasize that the assumptions they write can be either medically or scientifically true or false.

Then, we *consolidate* the set of subquestions and human-written assumptions and beliefs into a single set of assumptions and implications using GPT-3.5 (Prompt A.20.1).

5.2.3 Annotating Inference Veracity

Lastly, we ask a new subset of eight expert annotators to annotate whether each assumption and implication in our dataset is medically or scientifically true, false, or subjective and provide a supporting web document from our list of verified sources along with a passage from the document (Figure A.16, middle panel).

Validation. To verify that the assumptions and implications we extract are plausibly inferable from the question, we recruit an additional pair of health experts, which we refer to as expert *validators*, to rate inferences.¹⁰ We sample 100 assumptions and implications judged as false or subjective, and 100 true inferences and ask our expert validators to rate the plausibility of an inference on a 1–5 Likert scale based on how likely the question asker is to believe the assumption or implication. Henceforth, we refer to this sample of 200 inferences coming from 152 unique questions as INFERENCE-SAMPLE. Both annotators judge the majority of our inferences as plausible, with 80% and 95% rated with a score of at least 3 (see Figure A.14 for the rating scale). Spearman’s correlation between the two annotators is 0.69. See Appendix A.21 for more detail.

5.3 Grounding Assumptions and Implications in Linguistic Theory

Assumptions and implications in our dataset map to two well-studied phenomena in linguistic pragmatics: presupposition and implicature (Grice, 1975; Stalnaker et al., 1977). We begin with a short primer of both types of *pragmatic inference* (§ 5.3.1) and then discuss the implications of both types in a QA setting.

¹⁰These expert validators were not a part of our dataset construction.

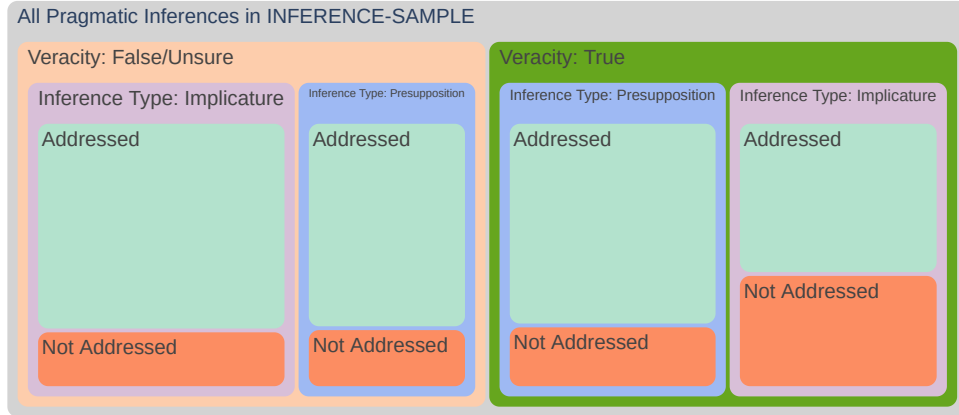


Figure 5.2: Tree-map (Shneiderman, 1992) visualizing the distribution of expert-annotated pragmatic inferences in INFERENC-SAMPLE with their veracity, inference type, and whether or not they were addressed or not in the expert-written answer to the question from which they came from. When users make false or subjective inferences, they are more likely to do so as an implicature. Moreover, when an inference is false, it is more likely to be naturally addressed in an answer by public health experts.

5.3.1 Two Types of Pragmatic Inference: Presupposition and Implicature

A sentence S is a *pragmatic inference* of a question Q if, depending on the *context* and *conversational goals* of discourse participants (Jeretic et al., 2020), a human would believe that the asker of Q believes or assumes S to be true. Henceforth, we refer to the assumptions and implications that we collect in our dataset as *pragmatic inferences*. See Chapter 2 for a discussion of these inference types.

5.4 How to ask and answer questions

Before we investigate the behavior of QA systems, we first study how humans embed pragmatic inferences in their questions (§5.4.1) as well as to what extent they are *naturally* addressed by human public health experts (§5.4.2).

5.4.1 Pragmatic Inference Type: Understanding Speaker Commitment

When users ask questions, how strongly are they committed to the inferences that experts identify in their questions? Presupposition is a phenomenon based on *mutual* acknowledgment of facts: when a human makes a presupposition, not only are they presuming the content of the inference, they are also signaling the belief that their *interlocutor* (here, a QA system) should believe it too.

On the other hand, implicatures are a softer way for humans to express uncertainty. For example, “Which immunity injections can I skip for my baby?” and “Is it sufficient if my

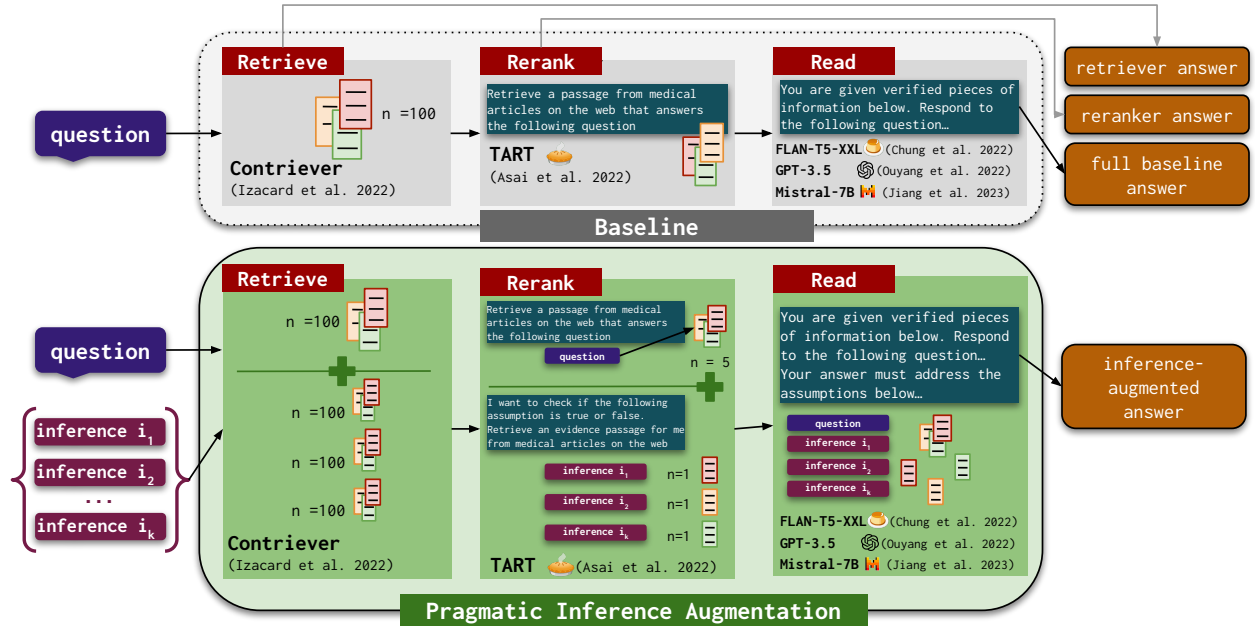


Figure 5.3: Our baseline and pragmatic inference-augmented QA pipelines. We experiment with retrieval, reranking, and reading stages and a variety of instruction-tuned and prompt-based models.

baby takes most immunity injections” have the same underlying inference (“*It is okay to pick and choose vaccines*”), but is taken for granted in the first (presupposition), whereas loosely suggested in the second (implicature). We want to distinguish inferences—separating implicatures from presuppositions—in our questions to better characterize *so that we can prioritize addressing stronger false inferences*.

Annotation Framework. A pair of authors independently annotate all inferences in INFERENCE-SAMPLE as a presupposition or implicature by first determining whether it is a proposition about the world that the asker believes to be true, without which the question would not be felicitous (presupposition) or whether it involves deriving asker belief through communicative principles (implicature). Between authors, Cohen’s kappa is $\kappa = 0.85$, indicating strong agreement. Author annotators adjudicated the final inference type (see Figure 5.2 for overall distribution), but individual annotator labels and adjudication rationales are preserved as a part of our dataset.

Findings. Presuppositions and implicatures are balanced in INFERENCE-SAMPLE (Figure 5.2), with a slight majority of inferences as implicatures, indicating that many inferences that health expert annotators identify are more subtle. When an inference is true, it is almost equally likely to be a presupposition or an implicature. However, when users make false or subjective (veracity marked “Unsure”) inferences, **they are more likely to do so via im-**

plicature (Figure 5.2). Past work has looked into generating and verifying *presuppositions* in open-domain QA, but identifying and addressing *implicatures* in an effort to make answers information-complete remains heavily underexplored. This finding highlights a key strength of our work: the ensuing context from our specific domain tests the usefulness of pragmatic inference in QA by allowing us to extract a greater range of inferences.

In settings that lack such context (e.g. single-turn open-domain QA), we are restricted to leveraging lexical or syntactic signals from the surface form of the question (Kim et al., 2021) since reasoning about asker belief is not possible without other contextual signals. For example, in the absence of additional context, the question “*Should I push grandparents for flu shot and tdap?*” may give rise to inferences involving the safety or effectiveness of these vaccines for the elderly. However, upon learning that this was asked in a web forum by a postpartum mother, we may reason that she believes her infant may be at risk for contracting the flu or other diseases if their grandparents handle them unvaccinated.

5.4.2 Addressing Inferences in Expert Answers

When health experts are tasked with *answering* questions, how likely are they to naturally address inferences that users implicitly make? Studying whether or not answers *naturally* address pragmatic inferences (§ 5.4.2) gives us better insight into the types of inference health experts, and in turn models, should prioritize when answering questions.

Annotation. We ask two annotators from our expert annotator pool to determine whether each inference in INFERENCE-SAMPLE is addressed, either implicitly or explicitly, by the human-written answer to its source question.

Findings. The majority of inferences in INFERENCE-SAMPLE are addressed by the human-written answer *naturally* (Figure 5.2). Importantly, **when an inference is false, it is more likely to be naturally addressed**. Moreover, a significant number of *true* inferences are also addressed by an answer, indicating that health experts not only aim to correct false or subject inferences but also prioritize completeness. This key finding supports one of the main arguments of this work: QA systems *must* address pragmatic inferences in their answers, just as humans do.

5.5 Inducing Pragmatic Behavior in QA

Inducing pragmatic behavior in QA systems is not straightforward. Existing systems are not trained to proactively reason about asker beliefs, since many popular QA datasets do not necessitate this type of behavior (e.g. factoid QA).

We experiment with eliciting model answers that address the pragmatic needs of questions, such as refuting false inferences, using the pragmatic inferences in our dataset. We inject inferences at each stage of the classic QA pipeline: passage retrieval, reranking, and

machine reading (§5.5.1) and evaluate outputs against expert-written answers with both automatic and human evaluation (§5.5.2).

5.5.1 Experimental Setup

Corpus. We use the corpus from Mane et al. (2023) of 408,000 documents from verified web sources on maternal health and infant care and augment the corpus with the sources that our expert annotators found while both writing answers and determining the veracity of inferences.

Baseline Models. As a baseline system, we use a retrieval, reranking, and reading-based QA pipeline. Contriever (Tizacard et al., 2022), an unsupervised dense passage retriever, identifies top relevant documents ($n = 100$) in our corpus given a question. Those documents are reranked using TART-full (Asai et al., 2022), a multi-task retrieval system with a cross-encoder architecture (Instruction A.23.1). TART is instruction-tuned, equipping it with the flexibility to redefine passage relevance for different tasks. We feed the top five reranked documents to three different reader models: FLAN-T5-XXL (Chung et al., 2022, 11 billion parameters), an instruction-tuned, *prompt*-based encoder-decoder model jointly trained on a multiple tasks with a standard answer extraction prompt from Mishra et al. (2022) (Instruction A.23.3), MISTRAL-7B (Jiang et al., 2023) (an open source large language model, Prompt A.23.4), and GPT-3.5 (Prompt A.23.4).

Augmenting Systems with Pragmatic Inferences. In addition to retrieving the top 100 passages using the question as input, we retrieve the top 100 passages for each pragmatic inference of the question ($i_1...i_k$) as well. Then, for each pragmatic inference i , we rerank the top 100 passages using a new inference-informed instruction (Instruction A.23.2) and select the top passage post-reranking. We augment the top five reranked passages from the question with these k top passages from each pragmatic inference to feed to each reader (Prompt A.23.5). During reading, we prompt MISTRAL-7B and GPT-3.5 to address all k assumptions when generating an answer.¹¹ To keep the same *number* of passages fed to readers in the baseline pipelines as in the inference-augmented pipeline, we add k extra passages to the top five existing ones. This ensures that while the volume of information presented to machine readers is the same in both pipelines, the nature of the content differs, allowing us to measure the utility of inference augmentation during retrieval and reranking. Figure 5.3 visualizes our baseline and inference-augmented QA pipelines.

5.5.2 Evaluation

We evaluate answers from seven pipeline variations against expert answers (Table 5.3): (1) The **No Reader** baselines consist of the top retrieved passage based on the input ques-

¹¹We do not use FLAN-T5-XXL here because it struggled with reading in the baseline setting.

	No Reader		Baseline Reader			Inference-Augmented	
	RETRIEVE	RERANK	FLAN-T5-XXL	MISTRAL-7B	GPT-3.5	MISTRAL-7B	GPT-3.5
ROUGE-L (F1)	15.6 _(10.3)	17.5 _(10.3)	15.89 _(13.7)	17.4 _(5.1)	18.7 _(6.0)	16.2 _(4.8)	18.6 _(5.9)
ROUGE-L (Recall)	23.0 _(15.7)	19.5 _(10.9)	36.9 _(21.9)	15.2 _(6.8)	17.4 _(8.1)	13.2 _(5.7)	16.6 _(7.6)
BLEURT	-0.72 _(0.34)	-0.61 _(0.29)	-0.79 _(0.45)	-0.38 _(0.22)	-0.37 _(0.22)	-0.36 _(0.21)	-0.38 _(0.22)
QaFactEval	0.69 _(1.05)	0.76 _(0.80)	1.16 _(1.5)	1.02 _(0.65)	1.15 _(0.73)	0.96 _(0.62)	1.17 _(0.75)
Human (/5)	-	-	-	-	4.43	-	4.45

Table 5.3: Mean and standard deviations of automatic (ROUGE, BLEURT, QAFactEval) and human evaluation metrics per question. We report results for the top retrieved passage and the top reranked passage, and two modes with and without access to human-written assumptions. Inference-augmented models perform competitively with baselines, indicating the promise of inducing pragmatic behavior in QA models to mitigate harm.

tion (RETRIEVE-ONLY) and the top reranked passage from the input question (RERANK-ONLY), (2) three **Baseline Readers** (BASELINE-FLAN-T5-XXL, BASELINE-MISTRAL-7B, and BASELINE-GPT-3.5), and (3) two **Inference-Augmented** pipelines INFERENCE-MISTRAL-7B, and INFERENCE-GPT-3.5).

Automatic Evaluation Metrics. Three automatic evaluation metrics measure the quality of generated answers: ROUGE (Lin, 2004) (both F1 and recall), BLEURT (Sellam et al., 2020), and QAFactEval (Fabbri et al., 2022), a more recent QA evaluation metric originally designed to measure the faithfulness of summaries. GPT-3.5 scores the strongest according to QAFactEval, our main evaluation metric because it—of the three metrics—most closely captures information content. However, automatic evaluation of generated answers does *not* capture several higher-level semantic and pragmatic aspects of the question. Thus, we still need experts to validate the answers.

Human Judgments. We ask our expert *validators* to score answers from the top-performing baseline and inference-augmented pipeline (BASELINE-GPT-3.5 and INFERENCE-GPT-3.5, on QAFactEval respectively). For each of the 152 questions in INFERENCE-SAMPLE, expert validators score both model outputs simultaneously from 1–5 based on completeness (instructions in Figure A.15). Answers typically received a score of 1 when they were off-topic and missing crucial information, a score of 2 when they were topical but still missing crucial information, 3 when containing all essential information to the question, 4 when most information was present for completeness, and a score of 5 when the answer was information complete. Judging the information completeness of an answer is a subjective task, as reflected by the Spearman rank correlation between their annotations ($\rho = 0.34$). While the mean score of inference-augmented examples is comparable to baseline answers (4.43 vs. 4.45), annotators rated the inference-augmented answer as equivalent or better than its baseline counterpart in 75% of questions in INFERENCE-SAMPLE (see Table A.10 for examples).

We further focus on annotator preferences on our original motivating population of

questions—those with highly plausible, false assumptions. Both annotators rate inference-augmented answers higher than the default answers in the subset of questions with at least one *highly plausible*, (plausibility=5) *false* pragmatic inference (Table A.9). We hypothesize that the similar ratings received by the two systems across *all* questions is due to shortcomings in the instruction-following capabilities of LLMs. Forcing the reader model to address pragmatic inferences distracts it from answering the question more completely, and does not always result in more helpful answers when the pragmatic inferences are true. This illustrates the promise of inducing pragmatic behavior in QA models and represents a lower bound: none of the models we experiment with were trained to optimize for addressing assumptions in questions.

5.6 Can inference extraction be automated?

While pragmatic inferences elicited from health experts are informed by their expertise, they are slow and costly to collect. Our QA experiments use *human-written inferences* to establish an upper bound of answer quality with existing models. However, a fully automatic pragmatic QA pipeline must first generate pragmatic inferences relevant to a question and *then* generate an answer that addresses the subset of false inferences. As such, we experiment with generating pragmatic inferences with GPT-3.5 (Ouyang et al., 2022) to understand to what extent automating the process is feasible with existing prompting and in-context learning.

Experimental Setup. We generate inferences with GPT-3.5 for all questions in INFERENCE-SAMPLE using seven in-context examples corresponding to 37 different pragmatic inferences, as more in-context examples yields diminishing returns (Liu et al., 2022b). We select pragmatic inferences written by multiple expert annotators from diverse user questions and randomly shuffle them to prevent unwanted effects emerging from example order (Si et al., 2022), including exemplars from all three sources (ROSIE, Reddit, and NQ) to capture distributional differences in their pragmatic inferences. As humans naturally did, we let GPT-3.5 generate *varied* numbers of inferences per question.

Evaluation: Can gpt-3.5 generate human-like pragmatic inferences? For each inference in INFERENCE-SAMPLE, a pair of authors annotate whether or not each human-written assumption is semantically equivalent to at least one inference generated by GPT-3.5 (Prompt A.24.1) with a Cohen’s kappa of $\kappa = 0.88$. Post-adjudication, 63% of inferences were not present among model generations. When stratifying by inference type, 53% of presuppositions and 71% of implicatures were not present. This illustrates that just as they are more difficult to detect, implicatures grounded in domain knowledge are more difficult for language models to generate.

5.7 Conclusion

This chapter introduced a new contract for QA evaluation, making “critical” inference, or those that express potentially problematic underlying beliefs of the user. It is especially important for systems to uphold this contract in sensitive domains such as the one we study here. We show that it is indeed possible to help systems uphold this contract by inducing pragmatic behavior in QA systems to correct latent false assumptions. Next-generation QA systems deployed in real-world settings *must* learn to address the pragmatics of user questions. Though we have shown the viability of *explicitly* inducing this pragmatic behavior in models in this work, directions for future work include training retrievers to inherently search for evidence to address pragmatic inferences and readers to reason on top of such evidence to tactfully and effectively challenge user beliefs.

Chapter 6: Inferential Robustness to Poorly-Posed Information-Seeking Questions¹

The previous chapter focused on inferential robustness in a domain-specific question answering setting, where users asked questions to a maternal health question answering system. In that setting, we defined one more contract towards inferential robustness to naturalistic user input: systems must examine user questions *in context* to produce answers that address false assumptions or potentially harmful beliefs embedded within questions. We showed that when systems increased their likelihood of computing pragmatic inferences about unstated assumptions, they could provide more complete responses that mitigated the propagation of harmful beliefs.

This chapter turns to an even more unconstrained setting: open-domain information-seeking question answering, where users may ask questions about any topic, without domain-specific constraints. Here, we examine a more general class of problematic user input that requires inferential robustness on the part of the system: questions that are poorly-posed. Unlike questions with problematic assumptions where the user’s information need *is* still often derivable from their question alone but must be augmented with content targeting their incorrect assumptions, poorly-posed questions present a slightly different challenge: they fail to clearly convey the user’s underlying information need due to ambiguity, vagueness, or other problematic properties that make direct answering difficult. For an NLP system to be robust in open-domain settings, it must be able to infer what users actually want to know, even when their questions are imperfectly formulated.

When users pose questions such as “*Can you have anxiety without feeling anxious?*”, or “*How does calling 911 work?*”, they are providing the system what we call *poorly-posed* input: questions that lack sufficient specificity, context, or clarity to enable a direct, helpful response. For a system to exhibit inferential robustness in such scenarios, it must go beyond attempting to answer the surface-level question and instead infer what the user *actually* wants to know. This requires the system to reason about the gap between what is explicitly asked and what is implicitly desired, a form of pragmatic reasoning that humans regularly engage with when faced with unclear questions in conversations. We study this capability using theories from question semantics to understand how both humans and models navigate

¹Neha Srikanth, Rachel Rudinger, and Jordan Lee Boyd-Graber. 2025. [No questions are stupid, but some are poorly posed: Understanding poorly-posed information-seeking questions](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3182–3199, Vienna, Austria. Association for Computational Linguistics

the interpretation process when questions exhibit various forms of poor posedness such as ambiguity, vagueness, or under-specification. Specifically, we explore what *kinds* of inferences humans seem to draw about user information need in order to assess whether models are similarly capable of such pragmatic reasoning. Our work is designed to lay the foundation for the creation of pragmatic question answering systems that *increase* the likelihood of reasoning deeply about information need.

Contract: Understanding Information Needs. When faced with poorly-posed questions that exhibit ambiguity, vagueness, or under-specification, systems should infer the user’s underlying information need rather than attempting to answer only the surface-level question.

6.1 Introduction

When an AI system is presented with a question by a user, it has an onus to answer. Sometimes, when information-seeking users know little about the topic they are asking about, the questions they present to the system are flawed. For example, the question “*How many universities are located in Cambridge?*” presents an ambiguity, as Cambridge, USA and Cambridge, England are both valid interpretations of the entity *Cambridge* in the question (Min et al., 2020; Stelmakh et al., 2022). Questions may also be flawed due to false presuppositions, as in the question “*Who is the king of France?*” (Kim et al., 2021; Yu et al., 2023). Ambiguity, false presupposition, and other such phenomena in questions are not new issues in NLP research (Section 6.7). However, rather than approaching each failure mode of questions independently, we provide a theoretical foundation to understand how imperfect questions are answered by humans and machines alike.

Consider the interaction in Figure 6.1, taken from the subreddit *r/NoStupidQuestions*. The asker (in orange) asks “*Are chiropractors considered doctors?*” in an attempt to articulate an information need. This need may have arisen from a variety of situations: perhaps the asker wants to know whether they should refer to their chiropractor as “Dr.” or maybe they are wondering if chiropractors undergo similar training as doctors. However, as written, it is difficult to identify the asker’s particular information need (Taylor, 1962), and in turn, determine the *resolution conditions* (Ciardelli et al., 2021) of their question. The space of interpretations of their question includes those about chiropractor schooling, title, or legal powers as compared to doctors, among others.

Humans often reason about the intent of conversational participants to smooth communication with others (Grice, 1975). When a hearer successfully reconstructs a speaker’s intent and responds appropriately, the communication can be considered as successful (Haugh, 2008). In question answering, this can include reasoning about *why* a speaker posed a question or reasoning about probable interpretations of a question. As pictured in Figure 6.1, the answerer tries to reason about the asker’s information need, ultimately selecting an interpretation and answering it, but as pictured, their chosen interpretation does not align with the asker’s original need.

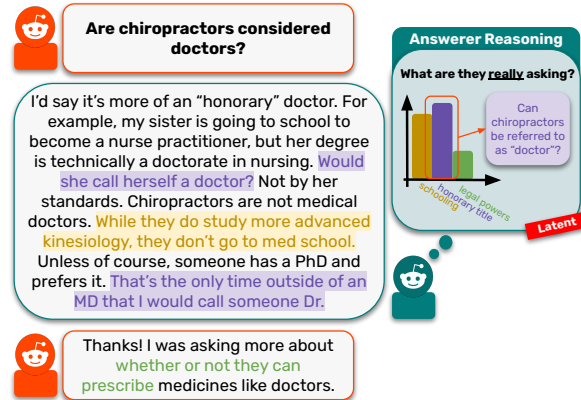


Figure 6.1: An *interaction* involving a poorly-posed question. After reasoning about the intent of the question, the answerer addresses two possible interpretations, but their answer did not satisfy the asker’s information need.

This interaction highlights the central concern of this work: what makes a particular expression of an information need *poorly posed*? How do humans fare when reconstructing information needs of askers from questions, and how does their behavior differ from model behavior on such questions?

We begin by providing theoretical structure to poorly-posed questions (§6.2) based on the linguistic theory of inquisitive semantics (Ciardelli et al., 2021), defining a poorly-posed question as one where—despite engaging in pragmatic reasoning—answerers cannot determine the asker’s information need, struggling to identify a probable interpretation. We collect questions² from the Reddit community *r/NoStupidQuestions*, focusing on QA *interactions* consisting of a question, answer, and a *reply* to the answer containing feedback on whether their information need was met (§6.3). Expert linguists annotate these interactions to identify interpretations chosen by answerers. We generate spaces of possible interpretations (§6.4) over which we induce a distribution representing the interpretations chosen by answerers (§6.5), and use this distribution to identify when a question is poorly posed. Lastly, we compare human and model behavior when answering poorly-posed questions (§6.6), studying cases when they converge or diverge on interpretations. We find that while humans adopt a precision-oriented strategy, engaging in *question negotiation* when answering poorly posed questions, models are recall-oriented.

6.2 What makes a question poorly posed?

Cooperative communication between humans hinges on our ability to draw inferences from the utterances of our conversational partners to identify their underlying intention or goal (Goodman and Frank, 2016). The space of possible interpretations or inferences one may draw from an utterance is not only large, but depends on conversational and contextual

²Code and data available at <https://github.com/nehasrikn/poorly-posed-questions>.

factors. Humans effortlessly engage in such reasoning processes daily. However, interactions may go awry when they cannot draw intended inferences or correctly identify a speaker’s intended interpretation.

When answerers’ attempts to answer a question cannot easily identify the intended interpretation, this may signal a poorly posed question. We describe our theoretical framework for analyzing questions, drawing on foundational work from information retrieval and theoretical linguistics to present our definition of a poorly posed information-seeking question.

The Process of Asking a Question. When someone wants to know something (an information need), they often ask a question. At times, this latent information need becomes distorted or muddled as the asker translates it into a question directed at another human or a QA system. Taylor (1962) formalizes this by describing four distinct stages of question formulation. Users start with an unexpressed need (the *visceral* need) for information, arising from some dissatisfaction. As they progress towards expressing this need, they may identify a description—possibly nebulous—of their dissatisfaction (the *conscious* need), but such a description may be rife with issues since they do not know much about what they are asking about. The user may continue to formalize their need as a question, ideally a more careful, clearer formulation with less ambiguity (the *formalized* need). Lastly, they may *reformulate* their question in the previous stage, taking into account whom it will be asked of (the *compromised* need): for example, if asking a search engine like Google, they may distill their question into keywords.

In practice, users may decide to ask their question to a QA system or a human **at any point throughout this process**. The burden then falls on agents answering the question to reconstruct the asker’s latent information need to provide a helpful answer. This is easy when the user has the appropriate knowledge to formulate their question *and* interpret a response (Miyake and Norman, 1979). In other cases, this reconstruction can prove difficult, and the answerer may fail to reconstruct the asker’s information need. **We consider a question poorly posed if, *even after an answerer’s additional reasoning*, they are unable to identify, and in turn address, the asker’s visceral or conscious information need.** This reasoning could involve computing a distribution over interpretations of the question (§6.5) or identifying and correcting false presuppositions (Yu et al., 2023).

Question Semantics. Another way to characterize a poorly posed question is using *resolution conditions* (Ciardelli et al., 2021). Classical accounts of semantics derive the meaning of statements by *truth conditions* (Heim and Kratzer, 1998). The semantic content of a sentence, or its *proposition*, represents the set of all worlds where the statement is true. However, it is difficult to analyze questions using these accounts, since questions are *inquiring* about the state of the world.

The framework of *inquisitive semantics* (Ciardelli, 2018) argues that question understanding is predicated on knowing what information is needed to resolve it (Ciardelli, 2017). An information state s , or a set of possible worlds that are compatible with the information s encodes, supports a question Q if the information in s resolves Q . Analogous to a proposition

expressed by a statement, the *issue* expressed by a question is the set of information states that support it. We draw on this framework to help define a poorly-posed question.

A question may be poorly posed if, as phrased, it is impossible to identify *what* information is needed to resolve it. Note that this does not commit to the *content* of the information.³ For example, consider the question “*Can literature not be about humans?*” There are many different interpretations of this question, each with different supporting information states, or sets of worlds where information holds true that would resolve the question, as in the two example interpretations below:

[Interpretation 1: “What is the criteria for a piece of text to be considered “literature” in the colloquial sense?”] Some example information states that may satisfy this interpretation include: (s_1) The set of worlds in which literature is defined by its artistic quality. A piece of text is considered “literature” if it demonstrates creativity, aesthetic, value, and artistic merit. (s_2) The set of worlds in which literature is defined by cultural or historical significance. (s_3) The set of worlds in which literature is defined by its ability to provoke thought or evoke emotion.

[Interpretation 2: “Can central characters in literature be non-human, such as animals?”] Some example information states that may satisfy this interpretation include: (s_1) The set of worlds in which all central characters in literature must be human. (s_2) The set of worlds in which non-human central characters are allowed, but only if they are anthropomorphized. (s_3) The set of worlds in which central characters in literature can be non-human, but only in specific genres.

An answer to the original question may include information encoded by any of the states described above, and *it is not necessary to know the answer to the question to identify which information states will resolve it*. However, we cannot clearly identify *which* information states resolve this original question, since there are many interpretations of the original question, a defining characteristic of poorly-posed questions.

Definition. Drawing on the groundwork above, we consider a question poorly posed if, even after an answerer engages in pragmatic reasoning (Grice, 1975) about an asker’s information need, they cannot identify a dominant interpretation of the question. This is not only a function of the asker’s utterance alone, but **a combination of their underlying information need and their chosen utterance**. It is not the presence of required reasoning by an answerer that makes a question poorly-posed, but rather that reasoning does not *bias* an answerer towards a particular set of resolution conditions.

6.3 Collecting Poorly-Posed Questions

Many QA datasets are collections of questions paired with gold answers (Rogers et al., 2023; Rodriguez and Boyd-Graber, 2021). From these pairs alone, it is not only difficult

³Poorly-posed questions are distinct from *open-ended questions*, those that are not answerable with a “yes” or “no.” See Appendix A.6 for more discussion.

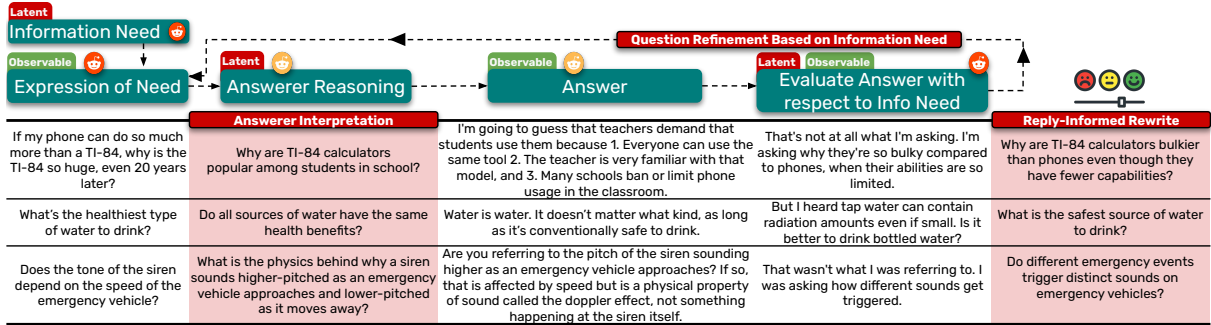


Figure 6.2: Example interactions through the question life cycle. Askers express latent information needs through questions to answerers who reason about those needs when answering. Askers then evaluate the answer with respect to their original information need, refine their question, and proceed through the cycle again. We intervene to collect answerer interpretations (Stage 1) and reply-informed rewrites (Stage 2) to study this cycle.

to understand the reasoning answerers engaged in when answering a question, but also *whether the answer satisfied the original asker’s information need*, a crucial part of improving QA: understanding what types of question interpretations, answerer reasoning, and dialog moves satisfy asker information needs can help inform QA systems faced with poorly-posed information-seeking questions.

We construct a dataset of information-seeking interactions between Reddit⁴ users, and introduce an annotation scheme designed to capture the latent reasoning answerers perform when answering a question. Our final dataset contains 500 questions annotated by expert linguists.

6.3.1 Question Selection

To maximize the ecological validity of our study, the questions we choose reflect organic instances of information-seeking interactions. We source our dataset from `r/NoStupidQuestions`, a popular Subreddit where users ask questions about any topic at any level, even if their questions are not fully formed (Appendix A.7). As such, these questions span all stages of the asking process (Figure 6.2). Unlike factoid QA datasets such as Natural Questions (Kwiatkowski et al., 2019b), questions in `r/NoStupidQuestions` are complex and open-ended, requiring long-form answers (Fan et al., 2019) with background information, examples, anecdotes, or analogies.

After filtering posts using a set of heuristics (Appendix A.8), we leverage the thread structure of Reddit by identifying **interactions**, or tuples of (question, answer, reply) where (1) a **question** (Q) is a post by a user to `r/NoStupidQuestions`; (2) an **answer** (A) is an individual top-level comment responding to the post which contains an answer to the question; and (3) a **reply** (R) is a comment from the original asker addressing answer.

⁴www.reddit.com

Collecting these interactions helps determine whether an answerer correctly interpreted the original question, as well as a more refined formulation of the asker’s original information need.

Question authors may confirm their need was **met** (*That’s exactly what I was looking for!*), **not met** (*“That was not what I meant”*), or provide **no useful feedback** (e.g reacting to a joke). To identify when askers do convey direct feedback, we elicit few-shot judgments from `llama-3-8b-instruct` (Dubey et al., 2024) on replies (R). For each reply R on posts that survived filtering, the model performs a three-way classification: (1) R indicates that an information need was met, (2) not met, or (3) not determinable.

We identify the subset of 142,878 comments from filtered posts containing R s (just 7% of all comments), and elicit judgments with Prompt A.8.1 on R s using eighteen manually selected in-context examples split evenly across the three labels. The majority of R s lack clear feedback: only 11.9% as indicate an information need was met and 9.4% indicate a need was not met. These judgments surface higher quality data that we then ask expert linguists to annotate.

6.3.2 Annotating QA Interactions

One of our primary goals is to analyze the relationship between an asker’s question Q and their underlying information need. Their reply R to an answer A gives us additional insight into their original need by correcting certain assumptions made by an answer, including feedback on the particular interpretation chosen by an answerer. We recruit seven expert linguists from Upwork⁵ to annotate 500 (Q, A, R) interactions, separating them into two groups for two stages of annotation.

Sampling Instances for Annotation. Using the judgments we elicited from `llama-3-8b-instruct` on asker replies (§6.3.1), we sample 400 (Q, A, R) interactions with an unmet information need and 100 interactions of a satisfied information need. Annotating both types of interactions allows us to study why questions lead to satisfactory or dissatisfactory answers.

Stage 1: Reconstructing Answerer Interpretations. Answerers make inferences about what askers are *intending* to ask or what askers are assuming. They then answer according to *their* interpretation of the asker’s question. Sometimes this interpretation is close to what the asker intended, and sometimes it is not. For example, the vague question asked in Row 1 of Figure 6.2 about why TI-84 calculators are so “huge” (Column 1) was interpreted by the answerer (Column 3) to mean “popular”. The first group (four annotators) reconstruct the interpretation answered in A (Column 2, Figure 6.2) from *only* Q and A (not R), with the option to indicate no change if there was strong alignment between the original question and the answerer’s interpretation of the question. They do not select from generated interpretations, but write from scratch to remain unbiased (full annotator instructions in Appendix A.9). Two annotators annotate each example.

⁵Annotators are paid \$15/hr on Upwork (www.upwork.com).

Stage 2: Reply-Informed Question Reformulation Next, the second group of three annotators use all three components of the interaction (Q, A, R) to rewrite Q to capture the intent or additional information expressed in R (Column 4, Figure 6.2) when R indicated an unmet information need (Column 5). Replies help us understand whether the interpretations adopted by answerers that annotators identify in Stage 1 matched the asker’s original information need. In the interaction in Row 1 of Figure 6.2, we learn from their reply (Column 4) that by “huge”, the asker wanted to understand why TI-84 calculators are so *bulky*, instead of “popular” as interpreted in A (Column 3). Annotators were instructed to eliminate any ambiguity, underspecificity, or vagueness in the original question. In some cases, annotator rewrites based on the asker reply were not answerable by the answer provided, since, in many cases, the answerer answered a different question than what the original poster intended. Again, two annotators annotate each example.

6.4 Question Interpretations

Poorly-posed questions often prevent answerers—despite reasoning about the asker’s intent—from reliably identifying a dominant interpretation (or set of interpretations) to address, making it difficult to satisfy the user’s needs. If many answerers attempt to determine the resolution conditions of the question, but arrive at different interpretations, this may signal that the asker’s information need was unclear. Conversely, answers *converging* on a particular interpretation *en masse* may indicate that a user’s information need was easily recoverable. Identifying likely interpretations of a question is the first step in building a distribution of which interpretations answerers choose.

We generate potential interpretations I of questions, or fine-grained natural language questions that capture a particular facet of the question. By definition, they must be free of ambiguity, vague, or underspecified language and their resolution conditions must be easily identifiable.⁶ These interpretations $I_1...I_n$ form a *space* (S_I) of valid interpretations of the question representing the range of plausible meanings that reasonable answerers might derive from the original question.

Interpretation Generation. We prompt gpt-4o and gemma-9b to generate interpretations for each original question.⁷ Models are only provided the question with no other context and are instructed to generate a variable number of interpretations in a zero-shot manner using Prompt A.9.1. Using models of different sizes and capacities allows for a wider range of interpretations that better cover the space.

Deduplicating Interpretations. Generated interpretations may be overlapping. For example, “*Why don’t cashiers sit in chairs when they perform their job?*” overlaps with the

⁶In practice, though some ambiguity may remain, most interpretations have clearer resolution conditions than original questions. Paraphrases or short clauses from original questions are not valid interpretations.

⁷Specifically, gpt-4o-2024-08-06 and gemma-2-9b. We experimented with many models, but found high overlap between interpretations generated by open-source models.

interpretation “*Why do cashiers perform their job standing?*”, since they share a subset of resolution conditions (§6.2). We deduplicate the set union of generated interpretations to form a set of *unique* interpretations, or the “space” of interpretations (S_I) we consider for each question (Table A.6) for use downstream. To deduplicate generated interpretations, we embed each I using NV-Embed-7B (Lee et al., 2024) and then compute pairwise cosine similarity between all interpretations (Instruction A.9.2). We then construct a graph G where nodes are generated interpretations $I_1 \dots I_n$ and edges $e(I_i, I_j)$ are drawn between interpretations with cosine similarity above $\theta = 0.85$. Maximal cliques of G (Tomita et al., 2006) then contain sets of highly similar or equivalent interpretations. We select clique representatives by choosing the interpretation least similar to other members, and together, these representatives form S_I . If an interpretation I is a member of multiple cliques, we assign it exclusively to its “strongest” clique by way of average similarity of clique nodes.

Validating Interpretations. To assess the *soundness* of interpretations, we randomly sample 100 (Q, I_n) pairs, and ask two external annotators to judge whether or not I_n reasonably follows from question Q .⁸ Annotators judge 76% of interpretations as sound, with a Cohen’s Kappa of $\kappa = 0.7$.

To assess *coverage* and alignment with human-generated interpretations, we ask a team of six undergraduate Quiz Bowl players⁹ to collaboratively enumerate interpretations on a random sample of 30 questions. Then, an author manually matched student-written interpretations with model-generated interpretations. On average, 78% of human-written interpretations were present in the generated interpretation space per question. Qualitatively, human interpretations tended to use less precise language than model-generated interpretations, sometimes propagating issues from original questions (Appendix A.10). Models include a higher number of interpretations per question, typically a superset of human interpretations. We elect to use model interpretations for replicability, scalability, and higher coverage.

6.5 Identifying Poorly-Posed Questions from Question-Answer Threads

Section 6.2 defines poorly-posed questions as those where answerers struggle to identify the asker’s information need despite attempting to reason about their intent. We operationalize this computationally by analyzing *many* answerers’ responses to each question in our dataset. We identify which interpretation(s) from S_I each top-level commenter adopts on the question’s Reddit post and examine the distribution of chosen interpretations. Convergence on one or few interpretations suggests a well-posed question, while high disagreement indicates poor posedness—answerers cannot reliably recover the asker’s intended information need.

⁸The annotators were provided no other context beyond the question and interpretation, nor were they told whether the interpretation was generated by a model or human.

⁹Quiz Bowl (Jennings, 2006) is a trivia competition designed to test knowledge on many topics including history, literature, and science.

6.5.1 Measuring Poorly-Posedness from Interpretation Distributions

Interpretation Matching. An answer A (here, a comment) *adopts* interpretation I if it contains the information necessary to answer I . Since it is impossible to obtain ground truth on an answerer’s chosen interpretation, this approximates their reasoning about what to address in their answer. This formulation also allows for an answer to match with multiple interpretations if it contains information addressing more than one.

We use an extractive QA setup to match answers with interpretations. For each comment A in the question’s corresponding Reddit thread, we consider each I in the interpretation space S_I , instructing a model to extract a span from the comment that supports I . If it cannot find a supporting span addressing a particular interpretation, the model is instructed to abstain from answering. We use three in-context examples along with Instruction A.9.3 to match interpretations to human comments using `flan-t5-xxl` (Chung et al., 2024). Using a relatively weaker QA model ensures fewer false positives when direct evidence is *not* present in an answer to support an interpretation.

We validate this matching by sampling 100 (I, A, span) judgments and judging whether the span identified by the model indeed supported I . 87% of the judgments from `flan-t5-xxl` were valid as annotated by an author, with a Cohen’s Kappa of $k = 0.82$ with an external annotator.

Computing a Distribution over Interpretations. Using this method, we construct an interpretation-comment matrix M for each question’s Reddit thread, where row i represents the i th interpretation in S_I , column j represents a top-level comment, and entry (i, j) is 1 if the model determines comment j supports interpretation i and 0 otherwise. To represent consensus, we multiply column j by the number of upvotes comment j receives, ensuring widely chosen interpretations are well-represented.¹⁰ We construct the chosen interpretation *distribution* by summing along rows in M to compute each interpretation’s net support across all comments and applying softmax, resulting in a distribution that approximates chosen answerer interpretations across S_I .

Measuring Poorly-Posedness. In line with our definition and intuition from §6.2, we want to compute a scalar score from these distributions in a way that assigns a low score when most of the mass is concentrated around a small number of interpretations (ideally one), indicating well-posedness, and a high score when that mass is more diffused (poorly-posedness). While there are many potential functions to achieve this, we choose to compute the *entropy* (Shannon, 1948) of the distribution, which **approximates how uncertain humans were in identifying the question asker’s information need**. *Low* entropy indicates a dominant interpretation that humans chose to address in an answer, while *high* entropy signals difficulty in deciding the original question’s resolution conditions, indicating a poorly-posed question.

¹⁰For example, if an answer receives 4 upvotes, the cells in M for that answer’s interpretation(s) change from 1 to 4.

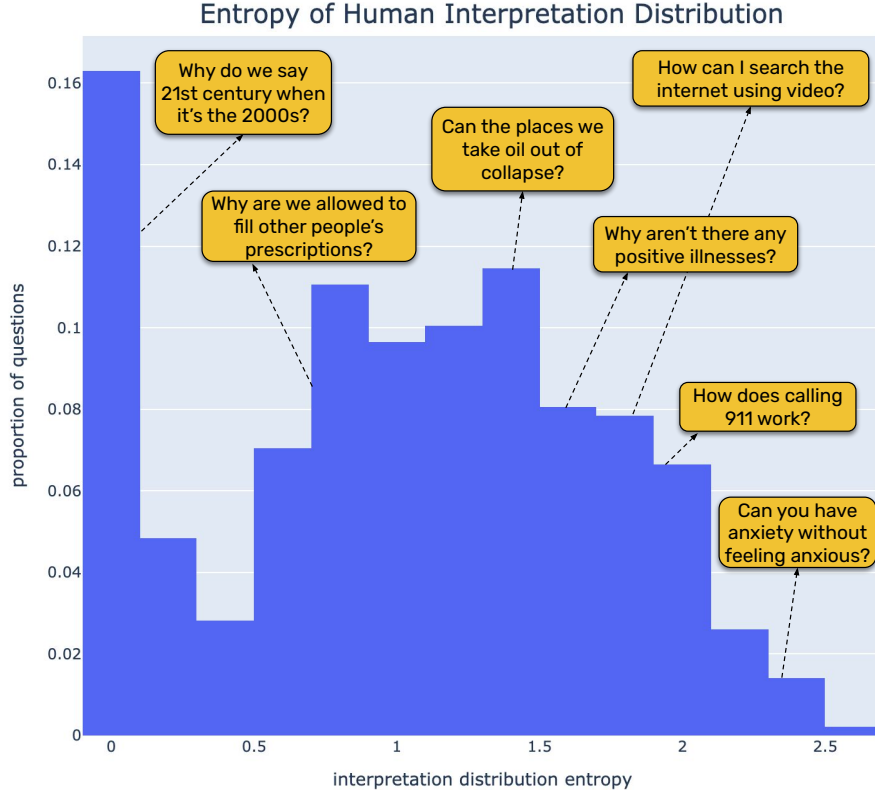


Figure 6.3: Entropy of human interpretation distributions. High entropy indicates a question is poorly posed.

To validate that this method meaningfully approximates poorly-posedness, we select the fifty questions with the lowest entropy interpretation distributions (LOW-ENTROPY) and the fifty questions with the highest entropy (HIGH-ENTROPY). Two native English speakers score each interpretation I in S_I on a 1–5 scale indicating the likelihood that I aligns with the question asker’s information need. This study controls for platform-specific factors that may influence inducing an interpretation distribution from comments in a thread.¹¹ Annotators received only the question and S_I , the interpretation space. Their judgments are positively correlated (Spearman’s $\rho = 0.57$).

For each question, we apply softmax to the distribution of annotator scores for all interpretations and compute its entropy (Appendix A.11). A two-sample t -test on the difference of mean entropy of *annotator distributions* between THREAD-LOW-ENTROPY and THREAD-HIGH-ENTROPY reveals that **thread-high-entropy has higher mean entropy than thread-low-entropy for both annotators ($p < 0.001$)**. These results validate our computational approach for estimating poorly-posedness using the *distribution* of chosen interpretations of human answers in the Reddit thread.

¹¹For example, the point in time that an answerer responds may affect the interpretation they choose since they may read comments posted earlier.

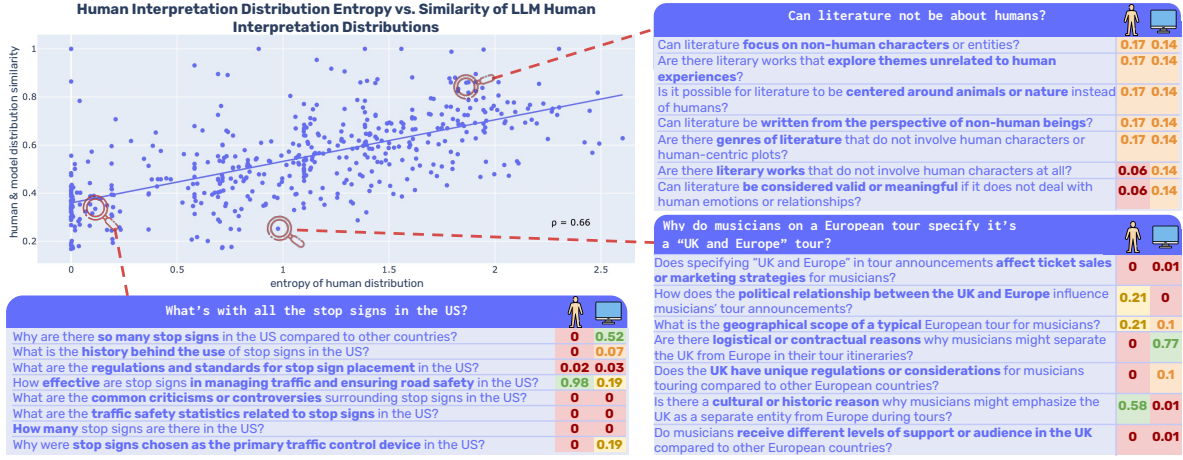


Figure 6.4: Entropy of human interpretation distributions plotted against the similarity of human and model interpretation distributions (1 - Jensen-Shannon distance). Humans and models struggle to converge on interpretations for poorly-posed questions with high entropy. Example callouts include both interpretation distributions.

6.5.2 Asker Replies and Poorly-Posed Questions

Aside from studying aggregate interpretation distributions from threads (constructed only from questions Q and sets of answers A), we also explore the relationship between asker replies R and questions identified as poorly posed by our method, specifically analyzing cases where askers indicate that their information need *was* met on high entropy (poorly-posed) questions.

Asker replies are more likely when questions are poorly posed. There is a statistically significant positive Pearson correlation ($\rho = 0.44$, $p < 0.0001$) between the entropy of the thread interpretation distribution and the proportion of comments that receive asker feedback. This indicates that question askers are likely early in the question asking process (Figure 6.2) and are attempting to define their conscious or formalized information need by engaging with answerers.

Asker replies are more likely to be positive when the question is poorly posed. Questions with entropy greater than one standard deviation above the mean entropy of all questions in our dataset have a *greater* likelihood of replying indicating that their information need was met than in questions with entropy less than one standard deviation *below* the mean. In these cases, answers are more packed with more information or reveal interesting information that the asker had not considered. For example, an answer to the question “*What is the resting position of the tongue?*” related it to the native language of a speaker, an interpretation that the asker had not considered nor was included in our generated interpretation space S_I .

Non-dominant interpretations are more likely to receive positive asker replies when the question is poorly-posed. Overall, when a comment A has a reply R , it is more likely to support a *non-dominant* interpretation (Figure A.5)—interpretations with lower likelihood in the softmax distribution. When entropy is high (greater than one standard deviation from the mean), asker replies to comments addressing non-dominant interpretations are 11% more likely to signal the satisfaction of an information need than in questions with low entropy (lower than one standard deviation from the mean), indicating that answerers strayed from popular interpretations and likely deeply reasoned to arrive at a lower-likelihood interpretation.

Reply-informed rewrites are semantically further away from dominant interpretations than original questions. Refined questions based on asker replies R are always semantically further away from dominant interpretations than original questions were, as measured by similarity of embeddings produced by NV-Embed-2 (Figure A.7). Interestingly, the reply-informed rewrite remains closer to the dominant interpretation in poorly posed questions than in other questions, underscoring the influence of answerers when users are still defining their exact information need.¹²

6.6 Contrasting Human and Machine Answers

Comparing human and model behavior in response to poorly-posed questions can help identify opportunities for improved alignment and inform the design of benchmarks that include poorly-posed questions. We experiment with computing distributions over interpretations chosen by a suite of LLMs on questions in our dataset to study divergences in crowdsourced human and model interpretations.

Experimental Setup. We include 12 instruction-tuned models of various sizes and configurations, including proprietary and open source models: 2B and 9B gemma-2-instruct (Team et al., 2024), 8B and 70B llama-3-instruct (Dubey et al., 2024), 2B and 8B granite-3.0-instruct (Granite Team, 2024), phi-3.5-mini-instruct and Phi-3-medium-128k-instruct (Abdin et al., 2024), ministral-8b-instruct-2410 (Mistral AI, 2024), athene-v2-chat (NexusFlow, 2024), gpt-4o-mini-2024-07-18, and gpt-4o-2024-08-06 (OpenAI, 2024). Models are prompted to respond in five sentences or less (Prompt A.9.4) in a zero-shot setup with greedy decoding, indirectly forcing them to adopt their likely interpretations instead of hedging or providing a deluge of information. We do not judge *accuracy* of answers since we decouple question *interpretation* from question *answering*.

We construct an interpretation-model answer matrix M by using the same interpretation matching method used for human comments (§6.5) and compute a *model* interpretation distribution by summing across rows in M and applying softmax.

¹²This is exemplified by the interaction: [Asker] Can literature be not about humans? [Answerer] Are you asking if human beings are capable of writing biology textbooks? [Asker Reply] I think I’m asking if biology textbooks can be literature, but I’m not sure myself...

	% of space addressed
granite-2b-instruct	75.24
granite-8b-instruct	75.4
Phi-3-medium-128k-instruct	75.42
gemma-2-2b-it	75.69
Meta-Llama-3-8B-Instruct	75.8
Ministral-8B-Instruct-2410	77.11
Phi-35-mini-instruct	77.43
Meta-Llama-3-70B-Instruct	77.96
gemma-2-9b-it	79.02
gpt-4o-2024-08-06	79.31
Athene-V2-Chat-70B	79.47
gpt-4o-mini-2024-07-18	79.64

Table 6.1: Models take a recall-oriented approach to questions, addressing significant portions of the interpretation space in a single answer.

Models, like humans, produce high entropy interpretation distributions on poorly-posed questions. We plot human interpretation distribution entropy (capturing question poorly-posedness) against human-model (h_i and m_i , respectively) distribution similarity (Figure 6.4) as measured by $1 - d(h_i, m_i)$ where d is the Jensen-Shannon distance between m_i and h_i . When questions were poorly posed (high human entropy), model answers were also distributed across the interpretation space S_I (e.g. *Can literature not be about humans?*). This high entropy pattern can flag questions in benchmarks where models or humans fail to converge on interpretations, indicating poor posedness.

When they do converge, models and humans converge on different interpretations. The entropy of human interpretation distributions is correlated with model interpretation distributions (Figure A.6). However, when humans converge, models tend to either converge on a *different* interpretation (Figure 6.4, lower left) or remain spread out over interpretations. When answering questions, humans reason about the intent, goal, or reason behind the question, ultimately affecting which interpretations they may choose to address. For example, the dominant selected interpretation of “*Why can’t I get drunk?*” by humans was “*Am I consuming alcohol in a way that does not lead to intoxication?*”, while models addressed interpretations related to the legal drinking age or medications that may interact with alcohol. It is unclear that models holistically consider the reason behind a question during QA to bias their response towards likely interpretations. Future multiturn QA datasets that include asker feedback on their information need may help produce well-aligned systems that engage in question negotiation to better pinpoint user need.

Models are recall-oriented answerers. In 84% of questions, the set of interpretations addressed by all models was a superset of those addressed by all humans. Investigating fur-

ther, we compute the average proportion of the interpretation space S_I addressed by each model’s responses across all questions in our dataset (Table 6.1): specifically, for each question, we divide the number of interpretations by each model adopted by $|S_I|$ to understand how comprehensive an average model answer is. `gpt-4o-mini` narrowly emerged as the model addressing the largest portion of S_I for questions (Table 6.1). We compute this metric for human answer threads as well to compare whether humans or models tend to address more interpretations per individual response. On average, individual human answerers addressed 44% of the interpretation space in an answer, while models addressed 77%. In doing so, models do tend to include at least *some* information targeting the dominant interpretation converged on by humans, but are “recall-oriented” with respect to the interpretation space, because they simply address a higher proportion of the space in an answer, in an attempt to satisfy the asker. This may be an artifact of human-AI alignment techniques that are based on human satisfaction with an answer.

6.7 Related Work

Many studies focus on a specific failure mode of questions, such as ambiguity (Min et al., 2020; Stelmakh et al., 2022; Cole et al., 2023; Piryani et al., 2024) or false presupposition (Yu et al., 2023; Kim et al., 2021, 2023; Hu et al., 2023; Kaur et al., 2024). Others study *unanswerable* questions with respect to a document (Zhao et al., 2024a; Rajpurkar et al., 2018). While our dataset contains questions presenting issues studied in these works, we present a *unified* definition of poorly-posed questions, abstracting away specific failure modes and focusing on *information need*. Pragmatic reasoning in QA is a relatively understudied area. Some study reasoning about speaker intent by constraining the domain (Wahlster et al., 1983) such that speakers share all referents (Fried et al., 2017). Qi et al. (2023) collect a dataset involving reasoning about speaker intent during QA, but their dataset *simulates* information-seeking interactions. In contrast, we study naturally-occurring interactions where users themselves are in the processing of defining questions for greater ecological validity (De Vries et al., 2020).

6.8 Conclusion

The contributions of this paper help detect problems that make questions less than perfect. Making the discourse moves that can lead to satisfying resolutions can make our interactions with AI more natural and successful while also improving our understanding of the subtle give and take two interlocutors engage in when answering an imperfect question.

6.9 Proposed Work 3: Answering Poorly-Posed Questions

Here, we describe a proposed work designed to deal with *answering* some of the poorly-posed questions we detected in the work above. We specifically target the same contract as above of *pragmatic interpretation*, or *information need-based inference*.

6.9.1 Motivation

We encounter questions all the time in our daily lives, and many of these questions are not perfect. So far, we have developed tools that have allowed us to begin to diagnose the issues present in questions, allowing us in turn to identify imperfect items in information-seeking benchmarks during dataset curation as well as *ex post facto*. The contributions of the work presented above help *detect* questions that may be less than perfect. However, when humans encounter these sorts of imperfect questions, how do they respond? While existing research has looked into the discourse format of long-form answers to *complex* questions (Xu et al., 2022), there remain open questions around the particular *strategies* that humans naturally take when dealing with information-seeking questions that may exhibit a variety of problems including ambiguity (Min et al., 2020), vagueness, and underspecificity (Ko et al., 2019).

Research has proposed quality long-form answers to ambiguous questions as those that contain all valid short answers (Stelmakh et al., 2022), but this is not always the case: in the work presented above, we observe cases where answers that adopt a single *likely* interpretation after rigorously reasoning about the asker’s information need also receive positive feedback from the original asker. Others have proposed a two step process to responding to ambiguous questions, where the language model first disambiguates the question before answering in order to clarify the interpretation (Cole et al., 2023). While most of this research has been done in the controlled setting of *factoid* question answering, disambiguating and answering is just one of *many* strategies that humans take to answer, and these strategies may vary depending on the context in which the question is asked. Consider the question and answer interaction below taken from the *r/asklinguistics* subreddit.

Question (reddituser123): Can language be described by math?

Answer 1: what exactly do you mean?

Answer 2: Yes, mathematical logic and discrete mathematics has often been used in describing and explaining language at various different levels. Two majors areas, syntax and semantics, have treated natural languages as being formal languages, in their structure and meaning respectively, which are *generated* or interpreted by mathematical functions, and operate according to certain rules (having a generating function is where the term generative grammar came from).

Answer 3: Is the question “can math usefully describe a range of linguistic phenomena” or is it “can math describe the totality of linguistic phenomena”? Does “describe” entail “accurately represents the causal sequence of production and comprehension processes that actually occur in humans” or does it just mean “produce similar output”? The answer could be “yes” for all of those, but it’s most transparently a yes for the combo of “a range” and “similar output”.

Inspecting the question further, we see that it is underspecified: the use of the word “can” could indicate that the asker is asking whether it is *theoretically* possible or whether it has

currently been done. There is also some ambiguity with the use of the word “described” as well as with the word “math”: does “described” mean formally modeled, accurately predicted, or causally explained? Does “math” refer to specific mathematical frameworks like logic and set theory, or to quantitative analysis more broadly? However, we do know that “language” most likely resolves to *natural language*, since this question was asked in a linguistics-related subreddit.

Interestingly, the interactions above present three fairly different strategies taken by answerers. The first answer stops at seeking clarification, requesting more information before attempting to provide *any* response. The second answerer chooses to assume a particular interpretation of the user’s question and then answer that interpretation; they assume the question is regarding formal descriptions of linguistic structure. The last answer presents a few possible interpretations, acknowledging that there are different ways to interpret the question, and notes that the answer critically depends on which interpretation the asker intended. These answering strategies are likely controlled by different answerer objectives, given the context of the question (asked in an expert QA subreddit). While existing research has studied answering strategies involving the asking of clarification questions (Rahmani et al., 2023; Kumar and Black, 2020), little work has enumerated different long-form answering strategies for *poorly-posed* questions exhibiting issues beyond ambiguity.

6.9.2 Research Objectives

In this work, we aim to satisfy a few research objectives:

1. Can we further define the parameters that contribute to the poorly-posedness of a question?
2. Assuming we know that a question is poorly posed, *what* are the observable strategies that humans employ, and to what degree are they successful?
3. When machines attempt to answer poorly-posed questions, what strategies do they employ, and are these strategies markedly different from humans?
4. How context-sensitive are answering strategies? When we vary different parameters related to context, do we observe a shift in the strategy distribution of humans?

6.9.3 Preliminary Development and Experimentation

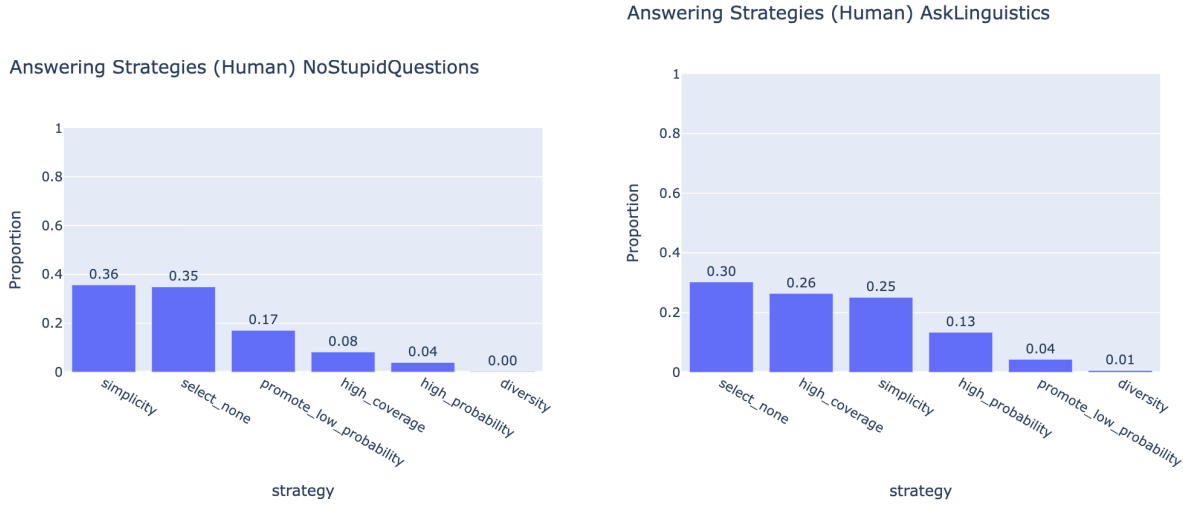
In Srikanth et al. (2025), we studied questions asked in the `r/NoStupidQuestions` subreddit, a forum designed for askers to ask information-seeking questions about any topic. Thus far, we have begun to build an ontology of the answering strategies we empirically observe in these communities. When choosing to respond to a question asked by a user, an answerer’s next discourse move lies on a spectrum: at one end, they may use strategies to **gather more information**, asking another question in return. On the other hand, the answerer may choose to respond directly with an answer or a statement.

We note that answerers often (1) select interpretation(s) to address, (2) perform some speech act (Searle, 1969) whose content possibly includes engaging with the interpretations.

A. Interpretation Selection		
A1	High Probability	Address most likely interpretation
A2	Simplest	Select interpretation with simplest answer
A3	High Coverage	Address as many possible interpretations
A4	Diversity	Address interpretations from distinct dimensions or groups
A5	Lower-Probability	Promote less obvious interpretation
A6	Practical	Select interpretation matching asker’s actual situation
A7	None	Avoid answering any interpretation
A8	Technicality/Pedagogical	Select interpretation that teaches relevant concepts or corrects technical misunderstandings
B. Discourse Moves		
<i>Seek Information</i>		
B1	Clarification	Ask follow-up to narrow interpretations
B2	Probe Context	Seek background/situational details
B3	Counter-Question	Challenge with a role-reversal
<i>Comment on Question</i>		
B4	Reject Presupposition	Point out logical flaw or premise of question
B5	Correct Fact or Terminology	Fix any non-presuppositional errors in question
B6	Comment on Question	Address question quality
B7	Redirect Formulation	Suggest better phrasing
<i>Answering</i>		
B8	Recontextualize	Place in different/broader framework
B9	Assert Answer	Commit to and answer particular interpretation(s)
<i>Redirect the Question</i>		
B10	Direct to Resource	Point to external answer source
B11	Recommend Expert	Suggest professional consultation

Table 6.2: Ontology of answering strategies.

Initial Detection of Strategies. We run the pipeline from Srikanth et al. (2025) on two Subreddit communities: r/NoStupidQuestions, and r/asklinguistics. Unlike communities surrounding a particular topic, we note that there is no assumed common ground in r/NoStupidQuestions, and if a question is vague or ambiguous, it is more likely it is because the question is posed poorly, and not because it is asked to a population that has a lot of common ground with the asker. Using the comment-interpretation matrix we built in Section 6.5, we develop automated strategies to detect a handful of the *interpretation* strategies from



(a) Distribution for r/NoStupidQuestions

(b) Distribution for r/AskLinguistics

Figure 6.5: Visualizing the strategy distribution of human answers across subreddits.

threads in both r/asklinguistics and r/NoStupidQuestions:

1. **Simplicity:** Comments adopt interpretations that lead to shorter, more concise answers. The strategy is detected when a comment's adopted interpretations have average response lengths significantly shorter than others, using a threshold of ϵ on normalized simplicity scores.
2. **High Probability:** Comments address only the most commonly adopted interpretations across the thread. Detection occurs when the average popularity of a comment's chosen interpretations exceeds $n\%$ adoption rate among all comments.
3. **High Coverage:** Comments attempt to address a large proportion of possible interpretations. A comment is flagged when it covers at least $n\%$ of all identified interpretations in the question's interpretation space.
4. **Diversity:** Comments address multiple interpretations that vary significantly in their popularity across the thread. Uses variance in interpretation popularity as a proxy, requiring a standard deviation of at least σ among adopted interpretations.
5. **Promote Low Probability:** Comments deliberately highlight less common or overlooked interpretations. Detected when a comment focuses on interpretations that fewer than $n\%$ of other comments address.
6. **Select None:** Comments that don't commit to answering any specific interpretation, instead offering meta-commentary, clarification requests, or other non-answering discourse moves.

Figure 6.5 shows the results of running these detection strategies over the comment-interpretation matrices on two different subreddits with different properties. For a community with very little shared common ground between asker and answerers, we see the dominant strategy in `r/NoStupidQuestions` as simplicity. On the other hand, `r/asklinguistics`, an expert-oriented community, opts for either selecting no interpretation (indicating the usage of non-answering discourse moves) or high coverage. This sort of analysis can reveal valuable insights about human behavior, as well as provide guidance for how we should guide models to behave.

6.9.4 Remaining Experimentation

Discourse Moves. We will build a discourse move classifier where we code answers according to Table 6.2 using an LLM and human verification. Then, we will use automatic metrics of interaction success (positive response from the original asker or high volumes of upvotes) to identify discourse moves and interpretation strategies that work best within a particular subcommunity, allowing us to make recommendations for how models should answer.

Model Behavior. We will run the same strategy analysis on answers from a suite of LLMs including open and closed source language models of different sizes. Doing so will allow us to compare and contrast the strategies that models currently take with those of humans.

Context-Sensitivity of Answering Strategies. We will extend this analysis to different contextual settings, or communities with different parameters and characteristics:

1. Question Type: Questions of different *types* may require different answering strategies. For example, opinion-seeking questions may differ from validation-seeking or information-seeking questions.
2. Domain-Specific vs. Domain General Forums: Question and answer dynamics may differ significantly when answerers and askers are part of a general open-domain community or a community centered around a particular topic or domain.
3. Expertise Asymmetry: In different communities, askers and answerers share different levels of similarity in expertise.

Chapter 7: Reading List

7.1 Question Answering

1. Question Negotiation and Information Seeking in Libraries (Taylor, 1968)
2. Progress in natural language understanding: an application to lunar geology (Woods, 1973)
3. Towards Identifying the Resolvability of Threads in MOOCs (Yang et al., 2014)
4. Reading Wikipedia to Answer Open-Domain Questions (Chen et al., 2017)
5. ELI5: Long Form Question Answering (Fan et al., 2019)
6. Dense Passage Retrieval for Open-Domain Question Answering (Karpukhin et al., 2020a)
7. AmbigQA: Answering Ambiguous Open-domain Questions (Min et al., 2020)
8. Challenges in Information-Seeking QA: Unanswerable Questions and Paragraph Retrieval (Asai and Choi, 2021)
9. Which Linguist Invented the Lightbulb? Presupposition Verification for Question-Answering (Kim et al., 2021)
10. ASQA: Factoid Questions Meet Long-Form Answers (Stelmakh et al., 2022)
11. QA Dataset Explosion: A Taxonomy of NLP Resources for Question Answering and Reading Comprehension (Rogers et al., 2023)

7.2 Natural Language Reasoning and Robustness

1. Interpretation as Abduction (Hobbs et al., 1993)
2. The pascal recognising textual entailment challenge (Dagan et al., 2005)
3. A large annotated corpus for learning natural language inference (Bowman et al., 2015)
4. Annotation Artifacts in Natural Language Inference Data (Gururangan et al., 2018)
5. Hypothesis Only Baselines in Natural Language Inference (Poliak et al., 2018)
6. Thinking Like a Skeptic: Defeasible Inference in Natural Language (Rudinger et al., 2020b)
7. Abductive Commonsense Reasoning (Bhagavatula et al., 2019)
8. Do Neural Language Models Overcome Reporting Bias? (Shwartz and Choi, 2020)
9. Measuring and Improving Consistency in Pretrained Language Models (Elazar et al., 2021b)
10. Natural Language Deduction with Incomplete Information (Sprague et al., 2022)

7.3 Discourse and Pragmatics

1. The process of asking questions ([Taylor, 1962](#))
2. What is a Speech Act? ([Searle, 1965](#))
3. Logic and Conversation ([Grice, 1975](#))
4. Common ground ([Stalnaker, 2002](#))
5. The CommitmentBank: Investigating projection in naturally occurring discourse ([De Marneffe et al., 2019](#))
6. Questions Under Discussion: From Sentence to Discourse ([Benz and Jasinskaja, 2017](#))
7. NOPE: A Corpus of Naturally-Occurring Presuppositions in English ([Parrish et al., 2021b](#))
8. How Do We Answer Complex Questions: Discourse Structure of Long-form Answers ([Xu et al., 2022](#))
9. PragmatiCQA: A Dataset for Pragmatic Question Answering in Conversations ([Qi et al., 2023](#))
10. Are Natural Language Inference Models IMPPRESSive? Learning IMPLicature and PRESupposition ([Jeretic et al., 2020](#))

Chapter 8: Timeline

1. October 2025: Proposal Defense
2. January 2026: Finish running experiments and analysis for Proposed Work 1 (Defeasible Investigator)
3. March-April 2026: Finish running experiments for Proposed Work 2 (Answering Strategies)
4. April 2026: Begin construction of new defeasible investigator dataset
5. August-September 2026: Starting running computational experiments using new dataset
6. October-November 2026: Finish computational experiments and submit paper
7. December 2026: Dissertation Defense

Appendix A: Appendix

Prompt A.1.1: Atom Generation

Prompt: You are an expert linguist. You are given a **sentence**. Generate a list of atomic facts that are strictly logically entailed from the given **sentence**. Keep each fact independent and self-contained. Each fact should make sense when read on its own. Only write facts that are directly described or supported by the **sentence**. End your response with [END].

SENTENCE: {**sentence**}

FACTS:

A.1.1 Coverage of Generated Atoms

Generated atoms must cover the information presented in the hypothesis. For example, for the hypothesis H in Figure 6.1, an atom generation model should produce atoms covering all three pieces of information: at least one mentioning *professional*, *actors*, and *summer*. Pruning should not reduce the coverage of the atom set. Here, we estimate the completeness, or coverage, of the generated atoms with respect to the hypothesis of NLI examples.

Completeness in the case of SNLI. We note that completeness of atoms only matters in particular cases. When the overall $P-H$ pair is predicted by the model to be entailed, the logical consistency check does not rely on completeness, as no single atom, if predicted by the model to be entailed, should be predicted as anything other than entailment. However, there are two possible scenarios of inconsistency when $P-H$ is predicted to be neutral: (a) All atoms were predicted to be entailed by P (this does necessitate ensuring completeness), or (b) one or more atoms is predicted to be a contradiction (this does not require ensuring completeness). Consistency when $P-H$ is predicted to be a contradiction does require checking that all atoms cover the hypothesis. Table A.2 shows the proportion of examples in δ -SNLI-TEST-SAMPLE that are dependent upon a completeness assumption as per the description for each label above. These help contextualize our results in Table 4.5 by estimating an upper bound of logical consistency.

To measure completeness in SNLI, we randomly sample 50 SNLI examples and annotate for completeness, and find that in 49/50 examples, all pieces of information from the original hypothesis project into at least one atom. The only example in our random sample that was missing an atom was for the hypothesis shown in Table A.4.

	δ -SNLI	δ - δ -SNLI
# test examples	1000	1837
label distr	36/32/32% (E/C/N)	50/50% (S/W)
# unique hypotheses	1000	203
# unique generated atoms	3263	475
# unique valid atoms	–	429
# unique atomic sub-problems	3263	4079
% strengtheners	–	46.0
% weakeners	–	31.4
% no effect	–	22.6
# unique critical atoms	–	349

Table A.1: Dataset statistics for both δ -SNLI and δ - δ -SNLI.

Model	Proportion
llama-3-8b-it	32.0
gpt-3.5-turbo	31.1
gemma-2-9b-it	25.5
gpt-4o-mini-2024-07-18	34.5
llama-3-70b-it	33.3
gpt-4o-08-06	27.6

Table A.2: Proportion of examples in δ -SNLI-TEST-SAMPLE (1000 examples) depending on a completeness assumption of atom generation.

We also sample another set of 50 random examples from δ -SNLI-TEST-SAMPLE where one of our models (**gpt-4o-2024-08-06**) predicted either contradiction or neutral *and* where consistency failed in order to understand of what percentage of errors are due to actual failures in consistency or are simply due to lack of completeness. Here, we study the atoms that the model deemed entailed from the hypothesis for each example, and annotate whether those cover all pieces of information in the hypothesis. In 6/50 examples, **gpt-4o-2024-08-06** incorrectly judged that at least one of the generated atoms was not entailed by the hypothesis, hence omitting it and causing a completeness issue for the set of atoms over which we measure its logical consistency. However, we find that of these six cases, only two examples were missing an atom in the set of generated atoms, again indicating that our *atom generation* process reliably projects all information from the hypothesis into the generated set of atoms.

Completeness in the case of δ -NLI. We manually validated all generated 4,079 atomic subproblems in the entire test set of δ - δ -SNLI-TEST. Lack of completeness is most likely

Example	Premise: A female within the foreground is heading towards a large white colored pillar that is apart of a large building with people are loitering or waiting on the steps of said building.
	Hypothesis The woman has an important meeting today in the building.
	Update The woman is wearing a janitor’s uniform (<i>weaker</i>)
Atoms	a_1 : The person has something (o, no effect).
	a_2 : The thing the person has is a meeting. (o, no effect)
	a_3 : The meeting is important (o, no effect)
	a_4 : The meeting is today (o, no effect)
	a_5 : The meeting is in a building (o, no effect)

Table A.3: An example of a lack of completeness in generated atoms.

when none of the atoms have a gold label in the direction of the gold label of the overall example, indicating that an atom may be missing. This happens in only 3% of examples (70/1761) in δ - δ -SNLI-TEST. We annotate these 70 examples to understand whether or not an atom was indeed missing after our automatic pruning step. We find that in only 28 of the 70 examples (representing only 1% of *all* examples), at least one atom was missing. Most of the cases where none of the atoms have a gold label in the direction of the gold label of the overall example are cases where the original example is flawed in some way (hinges on some stereotype, ambiguous language, faulty reasoning, or the original crowdworker who wrote the update did not understand the instructions) and our annotations do not propagate the flawed assumptions or reasoning from the original example. Table A.3 shows an example. The original gold label propagates a stereotype or attitude towards janitors (i.e it is less likely they have an important meeting if they are a janitor). We choose not to propagate such attitudes or stereotypes in our annotation, hence, all atoms have the “no effect” label.

Hypothesis	The man and woman are going to a movie in the city.
Atoms	a_1 : There are two people. a_2 : There are two people who are a man and a woman. a_3 : There are people going. a_4 : There are people going to a movie. a_5 : There is a destination for the people going. a_6 : The destination is a movie. a_7 : There is a city. a_8 : The people are going to the city.

Table A.4: An example of a lack of completeness in generated atoms for SNLI. Here the missing atom is “*The movie is in the city.*”

A.2 SNLI

Prompt A.2.1: Traditional NLI

Prompt: You will be given a **premise** and a **hypothesis** about that **premise**. You need to decide whether the **hypothesis** is entailed by the **premise** by choosing one of the following answers: 'e': The **hypothesis** follows logically from the information contained in the **premise**. 'c': The **hypothesis** is logically false from the information contained in the **premise**. 'n': It is not possible to determine whether the **hypothesis** is true or false without further information. Read the **premise** and **hypothesis** and select the correct answer from the three answer labels (e, n, c). Also provide a single line of explanation in a new line. End your response with [END] and output nothing after.

Premise: {premise}

Hypothesis:{hypothesis}

Is the hypothesis entailed by, contradicted by, or neutral with respect to the premise?

A.2.1 Inconsistency in SNLI

We analyze the set of randomly selected sample of 50 examples from Appendix A.1.1 where one of our models (gpt-4o-2024-08-06) was logically inconsistent. We find that inconsistencies arise for a number of reasons, some of which we discuss here.

Misjudgment of atom entailment from the hypothesis. Measuring logical consistency in SNLI examples happens in two phases (1) the model determines whether the atom entails from the hypothesis, and (2) the model predicts the relationship between the premise and an atom. Depending on its strength, we find that there are inconsistencies that arise from the model’s misjudgment of atom entailment. For example:

Premise: *A man on a bicycle, wearing cycle gear, riding at a fast past down paved trail surrounded by tree’s and grass.*

Hypothesis *The man is riding on the sidewalk.*

Generated Atoms (a₁) *There is a person.* (a₂) *There is a person who is a man.* (a₃) *There is a person riding.* (a₄) *There is a person riding on something.* (a₅) *The thing the person is riding on is a sidewalk.*

Based on the full premise and hypothesis, gpt-4o-2024-08-06 predicted that the hypothesis was *neutral* in relation to the premise. However, in its atomic predictions between the premise and each atom, the model incorrectly determined that the last atom was not entailed by the hypothesis, and hence it was not included in the set of atoms that were used to measure consistency. It judged all other atoms as entailed from the premise, hence leading to the model behaving inconsistently in this example.

Use of hypernyms in atoms. We make sure to include hypernyms of entities in the set of atomic facts. For example, “*the man dances*” is decomposed into “*there is a person*”, “*the person is a man*”, “*the person dances*”. In some cases, models have difficulty on premise-atom judgments where the premise uses the hyponym (“man”) and the hypothesis uses the hypernym (“person”).

Out of domain syntactic constructions. The syntactic structures of many of our atoms differ, in some cases significantly, from the constructions in the original dataset. For example “*the thing the person is eating is a sandwich*” is a pseudo-cleft construction that is very rare in the SNLI dataset. As such, some sentences are out of domain for encoder models that were trained on SNLI or prompt-based models that have inadvertently seen SNLI training data in pretraining corpora.

Weaker effects based on annotation elicitation. Both SNLI and defeasible NLI are datasets created by crowdworkers writing hypotheses and updates conditioned on a label. One of the consequences of this process is that hypotheses and updates tend to strongly express the desired label. In contrast, atoms tend to express labels in a softer way, and their

effects often compound when taken together. As such, these atoms may be out of distribution as compared to hypotheses that express the label with a higher magnitude. Since the effect of each atom is lighter than when they are taken together in a full sentence, atomic judgments are sometimes much more difficult than overall judgments.

Co-reference Effects. In some cases, atomic generations remove some of the implicit co-reference between the hypothesis and premise. We observe that changing the co-reference in atoms can result in inconsistencies between the overall example and each premise-atom judgment.

A.3 Defeasible NLI

Prompt A.3.1: Defeasible Inference

Prompt: You are a reasoning system. You are given a description of a **situation** and a **hypothesis** about that **situation** that may or may not be true. Given some more **evidence** about the **situation**, output 'more' if the **hypothesis** seems more likely to be true after learning the **evidence**, or output 'less' if the **hypothesis** seems less likely to be true after learning the **evidence**. Also provide a single line of explanation in a new line. End your response with [END] and output nothing after.

Situation: {context}

Hypothesis: {hypothesis}

Evidence: {evidence}

Does the evidence make the hypothesis about the situation more or less likely to be true?

Prompt A.3.2: Defeasible Inference Atoms

Prompt: You are a reasoning system. You are given a description of a **situation** and a **hypothesis** about that **situation** that may or may not be true. Given some more **evidence** about the **situation**, output 'more' if the **hypothesis** seems more likely to be true after learning the **evidence**, output 'less' if the **hypothesis** seems less likely to be true after learning the **evidence**, or output 'none' if the likelihood of the **hypothesis** remains unchanged after learning the **evidence**. Also provide a single line of explanation in a new line. End your response with [END] and output nothing after.

Situation: {context}

Hypothesis: {hypothesis}

Evidence: {evidence}

Does the evidence make the hypothesis about the situation more or less likely to be true?

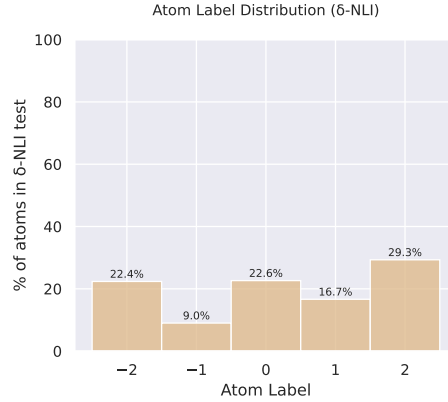


Figure A.1: Distribution of fine-grained labels across all atoms in δ - δ -SNLI-TEST.

Critical Atom	QUD
The kids are waiting.	What are the people doing?
The dog is brown.	What color is the dog?
The thing the grandpa is wearing is a shirt.	What is the person wearing?
The two women are sisters.	What is the relationship between the two people?
The bike is brand new.	What is the condition of the object?
The girls are posing.	What are the people doing?

Table A.5: QUDs generated from critical atoms in δ - δ -SNLI-TEST by gpt-4o-2024-08-06.

A.4 QUD Generation for Understanding Diversity

In order to generate QUDs from critical atom propositions, we use gpt-4o-2024-08-06 with a temperature of 0.01 and 15 exemplars of critical atom to QUD mappings along with Prompt A.4.1. For example, the critical atom *The dog is brown.* translates to the QUD *What color is the dog?* Exemplars include critical atoms that do not use generics (e.g. “girl”) mapped to QUDs that use generics (“person”). This is to make sure that similar inferences are correctly mapped to the same QUD. Table A.5 shows examples of QUDs generated by the model. We validate all generated QUDs by asking an external annotator if the QUD can be answered by the critical atom sentence, and find that in 92% of cases, the QUD generated by gpt4o correctly captures the critical atom.

Prompt A.4.1: QUD Generation

Prompt: You are an expert linguist. Given a short **sentence**, generate a **question** that is answered by the sentence. Read the whole sentence carefully before generating the question.
Sentence: critical atom
Question:

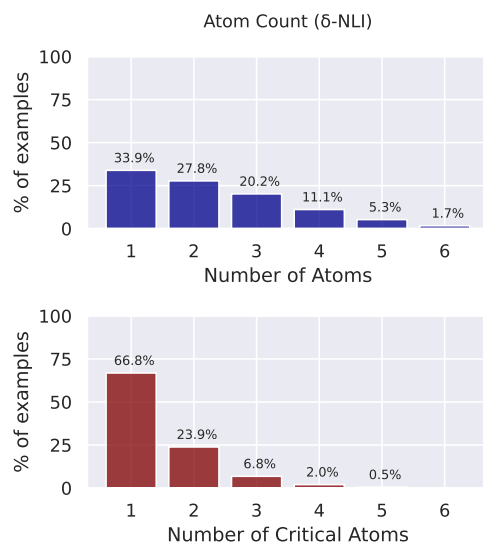


Figure A.2: Number of atoms (top) and number of critical atoms (bottom) per example in δ - δ -SNLI-TEST.

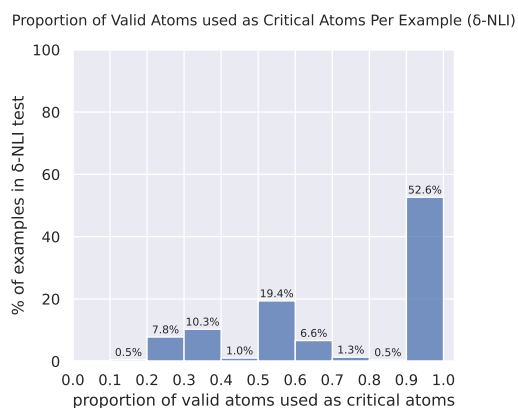


Figure A.3: Proportion of valid atoms used as critical atoms per update in δ - δ -SNLI-TEST. 52% of updates target all valid possible atoms at once in a single update sentence.

Comment Thread Length

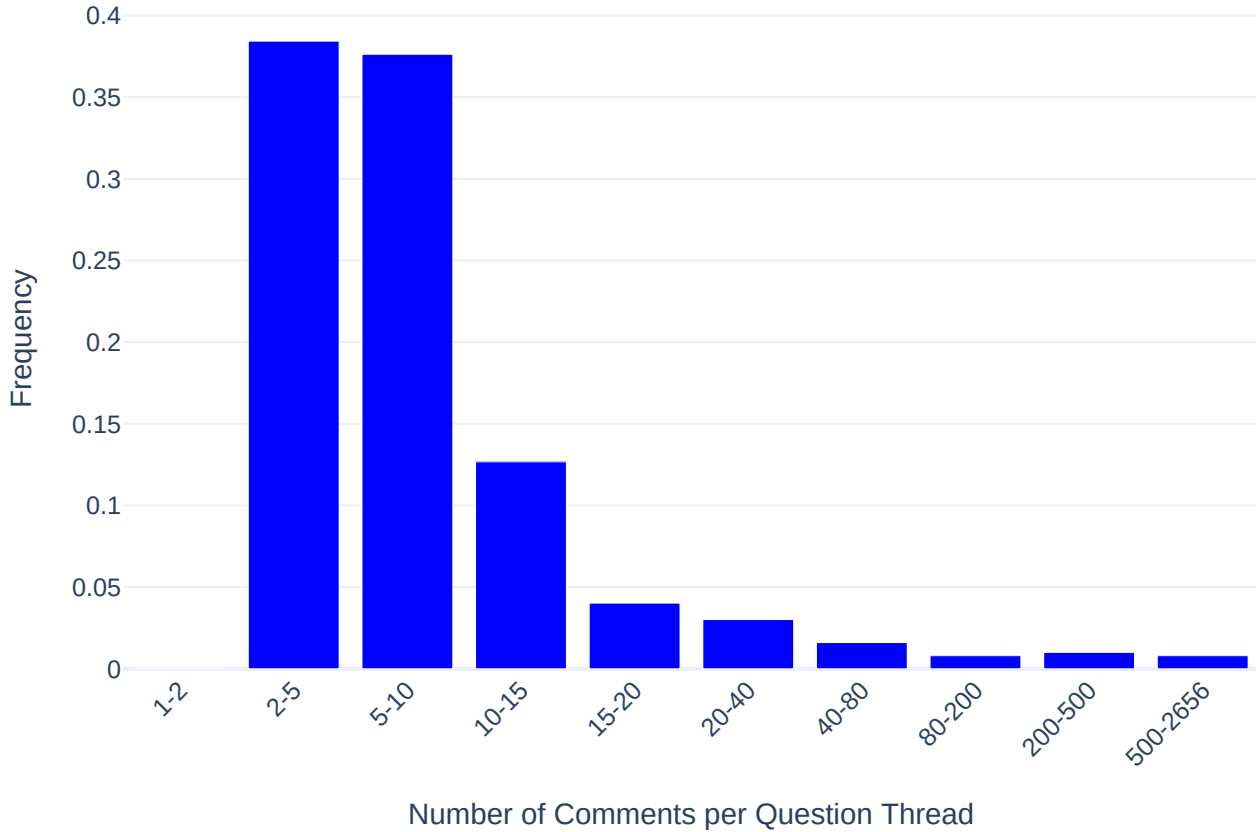


Figure A.4: Distribution of thread length of posts in our dataset. Most posts we study contain 2–10 comments.

A.5 Validation Instructions

Since atomic sub-problems mirror the original validation task for defeasible inference, we use the instructions provided to annotators from [Rudinger et al. \(2020b\)](#) to ensure alignment.

A.6 Open-Ended Questions

Open-ended questions are those that cannot be answered with binary “yes” or “no”, often requiring in-depth long form answers ([Chakrabarty et al., 2022](#)). This is distinct from poorly-posedness. We consider poorly-posed questions as those where answerers cannot identify a dominant interpretation of the question and in turn, what information or commentary is needed to satisfy the asker. See Table [A.7](#) for examples of open-ended questions that are poorly-posed and vice versa. Both questions in the right column of Table [A.7](#) can be

Interpretations from Comments with Asker Feedback

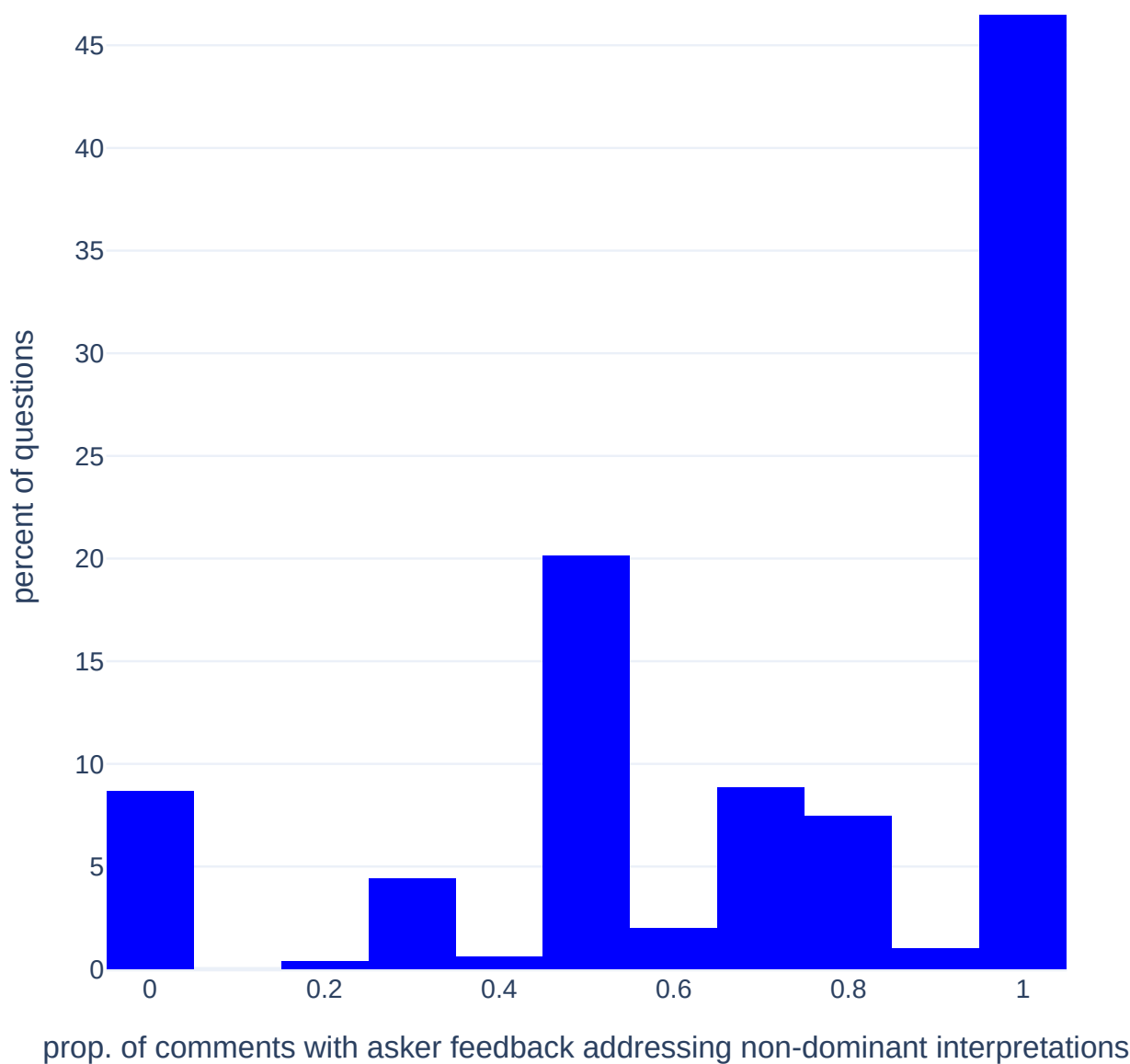


Figure A.5: Most comments with asker feedback address a non-dominant interpretation of the original question.

answered with a binary answer, and hence are not open-ended, but one is poorly-posed (e.g. it is difficult to understand what the user in the top right cell is asking) and the other is not. Subjective questions (a subcategory of open-ended questions) can also be well-posed. As long as an answerer can determine *what* information to include in an answer that would

Human Interpretation Distribution Entropy vs. Model Distribution Entropy

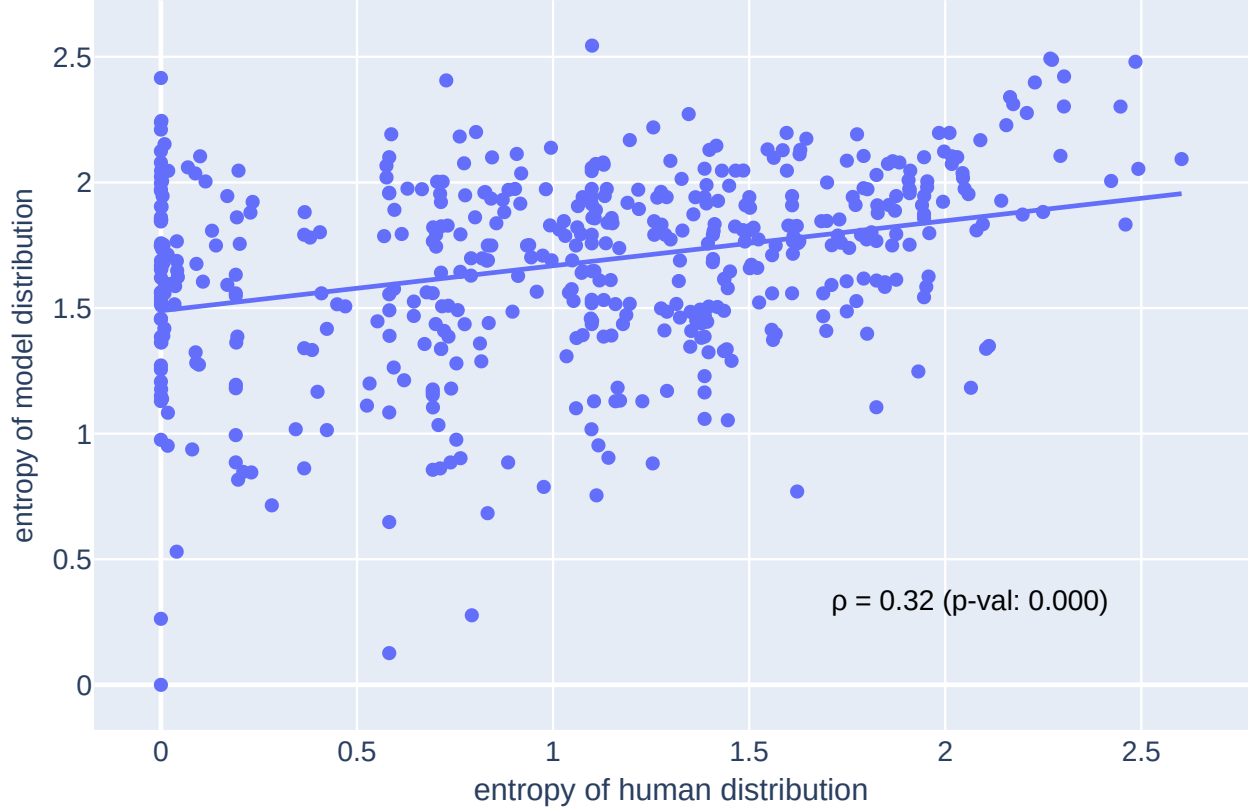


Figure A.6: Entropy of human interpretation distribution versus model interpretation distribution on questions in our dataset are positively correlated, indicating that they may be aligned in some cases when choosing interpretations.

satisfy the information need of the asker (regardless of whether the asker agrees or disagrees with the content), we consider the question well-posed.

A.7 r/NoStupidQuestions

r/NoStupidQuestions is a popular subreddit on the online platform Reddit that encourages users to post questions about any topic, with a few exceptions (e.g. loaded questions or medical questions) There are no particular expertise requirements for answerers.

We choose r/NoStupidQuestions as the backdrop for our study specifically because it is not specific to any topic, besides asking questions. Themed subreddits (i.e r/knitting or r/houseplants) induce richer contexts that can help disambiguate questions which would be harder for models to respond to without that additional contextual information. For example, consider the question “*How do you do happy hour?*” This question invites many potential

# questions	500
total generated interpretations	5715
total deduped interpretations	4680
avg. # generated interpretations	11.5
avg. # deduped interpretations	9.3
total deduplicated interpretations	4680
% gpt-4o	55.2
% gemma-2-9b	44.8

Table A.6: Dataset statistics for interpretations.

	Open-Ended	Not Open-Ended
Poorly-Posed	Why aren't there any positive illnesses?	Is it possible to track a child down from their address?
Not Poorly-Posed	How do people obtain the instrumental tracks of pop songs for karaoke even when those instrumental tracks are not publicly released?	Can a US President pardon themselves for a crime they have been convicted of?

Table A.7: Poorly-posed questions are distinct from open-ended questions, or those that cannot be answered with a binary response.

interpretations when asked in a (relatively) context-free setting like r/NoStupidQuestions. Potential interpretations include: (1) *What types of drinks or foods are typically included in a happy hour?* (2) *How do restaurants or bars structure happy hour deals or promotions?* (3) *What are some tips for hosting a happy hour at home?*, or (4) *How do different cultures or regions celebrate happy hour?* Consider two different scenarios: (1) this question was asked in r/Alcoholism, and (2) this question was asked in r/InternationalStudents. Learning that these questions were asked in topic-specific subreddits, allows us to suddenly identify very clear dominant interpretations. In the first scenario, a likely interpretation would be “*How can individuals in recovery navigate situations involving happy hour without compromising their sobriety?*” and in the second scenario, one dominant interpretation would be “*Are there cultural norms or etiquette that international students should be aware of when attending happy hour?*” r/NoStupidQuestions does not invite any particular bias towards or away from interpretations based on the common ground of subreddit members. r/NoStupidQuestions is similar to r/ELI5 (Fan et al., 2019), a subreddit and dataset that many studies on long-form QA use (e.g. Krishna et al. (2021) or Su et al. (2022)).

A.8 Data Filtering

Using the PushShift dump of `r/NoStupidQuestions` containing posts till 2022,¹ we filter out all posts with (1) deleted titles, (2) titles less than four characters and (3) less than five upvotes. We use surface-form features of post titles to identify questions that are likely to be information-seeking as opposed to community-seeking or other commentary, filtering out posts with titles that do not contain a question word, those that do not end with a question mark, and those with less than three tokens. These heuristics identify a subset of questions from the Subreddit that may be poorly posed for our study by reducing posts by nearly 92%, leaving 216,089 total posts that survived filtering. We do not include any PII when showing posts to annotators, nor do we plan to release post or user identifiers when releasing the data. All of the posts we consider for annotation are written in English.

Prompt A.8.1: Asker Reply Classification

Prompt: Community forum websites allows users to ask questions (askers), other users to answer them (answerers), and askers to reply to answerers. When askers reply to answerers, they may indicate whether their question was correctly interpreted, and then answered, by the answerer. Given an asker's **reply** to an answerer, determine if their question was correctly interpreted and answered (answered), not answered (unanswered), or if their **reply** is unrelated to whether or not the answerer correctly interpreted their question (unrelated). End your response with [END].

REPLY: {reply}

LABEL:

A.9 Annotator Instructions

A.9.1 Stage 1: Instructions

When humans formulate questions about topics or content they are unfamiliar with, they can end up posing questions that are unclear, vague, ambiguous, or underspecified. Consider the question: *Can I make my computer "Learn" new unicode symbols?*

In order to answer this question, we might make some assumptions, including about what exactly we believe the asker is really asking. For example, the word “learn” could mean render (e.g. can my computer render new unicode symbols), or it could mean generate (as in, can my computer generate new unicode symbols given a description). Sometimes these assumptions can lead to seamless question answering (a user may ask a poorly posed question, and an answerer makes the necessary inferences to answer well).

However, in some cases, askers may feel an answer to their question falls short if the answerer made incorrect or insufficient assumptions about the intent of the asker. Our goal is to study why questions posed by humans lead to satisfactory or dissatisfactory answers. To

¹<https://github.com/Watchfull/PushshiftDumps>

do this, we study interactions between users on Reddit, an online community forum-based platform. Specifically, we study interactions from `r/NoStupidQuestions`, a subreddit meant to openly ask questions about any topic.

Here, a user makes a post asking a question about a random topic. Sometimes, this question is clear and it is easy to infer the intent of the question, and in turn, answer it. Other times, the question is poorly formed, and it is difficult to understand what the user is asking. Then, a different user will make a comment on the post, trying to respond to the question. It is important to note that in order to answer the question, the answerer makes certain assumptions about the question and the asker, and picks out a likely interpretation of the question to answer. Sometimes this interpretation aligns with the question the user asked, and sometimes it is a departure from what the original asker intended. Your task is to simply identify the interpreted question that the answerer answered. We will show you a user question and a comment containing an answer to the question written by a different user. For example:

[**Original Post**] Can I make my computer "Learn" new unicode symbols?

[**Answer**] You can't really create your own unicode symbols, you can however submit an idea for future expansions at <http://unicode.org/pending/proposals.html>

Given a particular post and answer from another user, your task is to rewrite the question to capture what the answerer interpreted the question to be. It is okay to indicate "No Change" if there is strong alignment between the original question and the answerer's interpretation of the question. For example, in the example above, a rewrite may look like: *Can I create or propose new Unicode symbols myself?*

In the event that there is no change in the semantic meaning of the question, but the question itself uses colloquial/informal language or has a certain pragmatic interpretation, please rewrite to eliminate the informal language or make the pragmatic meaning explicit. For example:

[**Original Post**] Why is the enrollment period for insurance even a thing?

[**Answer**] The reason is so you don't go with cheap/no insurance while you are healthy and then sign up for expensive insurance as soon as you get sick.

[**Answerer's Interpretation**] Why are there designated enrollment periods for health insurance?

This rewrite removes the informal language "even a thing", as well as makes it clear that the answerer interpreted "insurance" to specifically mean "health insurance".

A.9.2 Stage 2: Instructions

We provide the same instructions as the first stage, but rephrase the statement of the task as follows: Given a particular post, answer, and a reply from the original poster, your task is to rewrite the question as clearly as possible to capture the intent or additional information expressed in the reply. Your rewrite should eliminate any ambiguity, underspecificity, or vagueness in the original question. It is okay for your rewrite not to be answerable by the

answer provided, since, in many cases, the answerer answered a different question than what the original poster intended. For example:

[**Original Post:**] Can I make my computer "Learn" new unicode symbols?

[**Answer:**] You can't really create your own unicode symbols, you can however submit an idea for future expansions at <http://unicode.org/pending/proposals.html>.

[**Original Poster's Reply:**] Create, no, Correctly render existing unicode symbols is what i mean

[**Rewrite:**] How do I install custom unicode symbols on my computer so that I can type custom glyphs and emojis and have my computer correctly render them?

Prompt A.9.1: Interpretation Generation

Prompt: Users typically try to express a need for information through a **question**. Sometimes, from the language of the **question** alone, it is not clear what their exact information need is. This leads to many distinct interpretations of their **question**, each representing different information needs. You will be given a **question** that may have many distinct interpretations. Output those interpretations as unambiguous questions in a numbered list. Make sure your interpretations are specific and as clear as possible, and are as diverse as possible. If the user's information need is already clear from their question, output 'NONE'. Otherwise, output the numbered list and nothing else.

QUESTION: {question}

Interpretations:

Instruction A.9.2: Duplicate Question Identification

Instruct: Given a question, identify other duplicate questions.

QUERY: {query}

Prompt A.9.3: Interpretation Support

Instruction: I will give you some context, and a question. Using the context, answer the question with a span from the context. If the question cannot be answered from the context, output **NONE**. Here are some examples:

Context: When you pull the string taut and talk into it your voice makes the string vibrate. The vibrations then travel through the string and into the air inside the receiving can, turning into audible sound waves.

Question: How does the tin can telephone work in terms of physics?

Answer Span: your voice makes the string vibrate. The vibrations then travel through the string and into the air

Context: You can't digest gum it's just the sugar that flavors it

Question: Are calories in gum counted even if you spit it out?

Answer Span: NONE

Context: I'd say it's more of an "honorary" doctor. My sister is in school to become a nurse practitioner, but her degree is technically a doctorate in nursing. She would not call herself a doctor by her standards. Chiropractors are not medical doctors. While they do study advanced kinesiology, they don't go to med school and thus should not be considered a doctor. Unless, someone has a PhD and prefers it. That's the only time outside of an MD that I would call someone Dr.

Question: Do patients use the title "doctor" when visiting professional chiropractors?

Answer Span: someone has a PhD and prefers it. That's the only time outside of an MD that I would call someone Dr.

Context: It's what the British call lines of people. If you're waiting in a line at a supermarket or something, that's a queue. From the dictionary: queue, British, a line or sequence of people or vehicles awaiting their turn to be attended to or to proceed.

Question: What is a queue in computer science?

Answer Span: NONE

Context: {context}

Question: {question}

Answer Span:

Prompt A.9.4: Question Answering

Prompt: Answer the following question in 5 sentences or less: {question}

Answer: {a}

A.10 Human-Written Interpretations versus Model-Generated Interpretations

Here, we provide examples of interpretations written by human trivia enthusiasts and model-generated interpretations. In some cases, humans performed deeper reasoning about

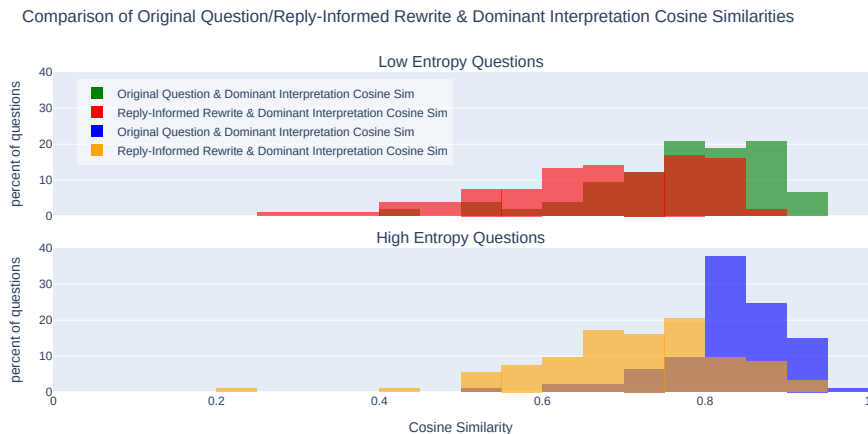


Figure A.7: Distance between original question and the dominant interpretation chosen by humans on the post thread and distance between reply-informed rewrite from Stage 2 of annotations and dominant interpretation for both low entropy (top) and high entropy questions (bottom).

why the asker may have posed their question in a particular manner, drawing on their own experiences or on commonsense reasoning. Figure A.8 shows three examples from the set of 30 that were annotated by humans.

A.11 Measuring and Validating Poorly-Posedness

Annotators from Section 6.5.1 are instructed to score each interpretation on a 1–5 scale based on likelihood of matching the asker’s information need. The scores are designed to be calibrated across the interpretations provided for the question: interpretations that the annotator felt better matched the asker’s information need received higher scores. If all interpretations seemed equally likely, scores reflected that uniformity. For example, if annotators felt *all* interpretations were equally unlikely to match the original question asker’s information need, they gave a score of 1 or 2 to all interpretations. For example, if all interpretations seemed unlikely, annotators gave scores of 1 or 2 to all; if most seemed to match, all received high scores.² Here, we explain the intuition behind our choice of scoring function to convert the rating vector of Likert scores for a question to a number that allows us to quantitatively compare the ambiguity (ill-posed nature) of questions.

Given a list of human ratings of interpretations, well-posed questions should have a small number of interpretations with high Likert scores, ideally just one. For instance, in the ideal case, the rating vector of $[1, 1, 1, 1, 5]$ indicates one well-matched interpretation of the question, with others poorly matched. Conversely, a rating vector of $[3, 3, 3, 3, 3]$ suggests all interpretations are *possible*, but no clear preference emerges. Similarly, a rating vector of $[5, 5, 5, 5, 1]$ indicates many well-matched interpretations, maintaining ambiguity about the

²We considered probability assignments or pairwise comparisons, but felt the 1-5 scale was most natural.

asker’s information need. These examples illustrate that any metric function that computes a score of ill-posed-ness should meet the following requirements: (1) accept a vector of Likert scores (or in the case of Reddit threads, counts representing the number of answers addressing the interpretation), (2) return a high value when there are no good matches (e.g., $[1, 2, 1, 2, 3]$), or many good matches (e.g., $[5, 5, 5, 1, 5]$), and (3) return a low value when there are a small number of very good matches (ideally one), e.g.,: $[1, 1, 1, 1, 5]$.

While there are a many ways to construct a metric function to achieve the three desiderata above, we choose one particular technique as a quantitative means to operationalize our definition of a poorly posed question. Our technique has two steps: 1) normalization, and 2) aggregation. We will now discuss the intuition behind the two steps of our technique to compute our metric, and our choice of mathematical operators: (1) normalization with softmax, (2) aggregation using entropy.

Normalization. First, we want to assign an amount of goodness (mass) to an interpretation based on how *unique* it is in its goodness. If we normalize every question to have a total mass of 1, then a question with only one good interpretation should have most of this mass concentrated on it. Alternatively, a question with many good interpretations should have the mass distributed uniformly across them. One could use many different normalization techniques: L_1 , L_n , or Softmax (with or without a temperature variable). L_n or Softmax normalization accentuates the difference between high scoring interpretations and low scoring ones, whereas L_1 or Softmax with a high temperature minimizes such differences. We find that the choice of normalization does not alter our findings, and so we pick softmax for illustrative purposes.

Aggregation. Here, we want to compute a single scalar score from this normalized vector to assign a low score when most of the mass was concentrated in a small number of interpretations (ideally one), and a high score when that mass was more diffused. Again, while there are a number of ways to construct such a function, we choose entropy since it is a well known way to determine if most of the mass is concentrated in a small number of options.

A.12 Model Infrastructure

All experiments were run on two NVIDIA RTX A6000s. Models greater than 11B parameters were run using 4 bit quantization.

A.13 SNLI and δ -NLI Dataset Sizes

To train and tune our neural and lexical models, we utilize the train and validation splits from SNLI and δ -NLI. We include train/validation/test split sizes in Table A.8. Our δ -NLI RoBERTa models were finetuned on examples from all 3 portions of the δ -NLI dataset (ATOMIC, SNLI, and SOCIAL-CHEM-101), however for our evaluation set and analysis, we

Question	Human-Written Interpretations	Model-Generated Interpretation Space
Do doctors (M.D., D.O., etc.) work outside their specialty under any circumstance?	(1) Do hospitals limit when and how doctors can work outside of their specialty? (2) Do insurance companies limit payment for specialties to doctors trained in that area? (3) Do doctors trained in one specialty have the skills to work outside of that specialty? (4) Why won't my (x specialty doctor) do (thing I want that y specialty doctors do)? (5) If I am studying (x branch of medicine), will I still be allowed to do (y thing I enjoy doing)? (6) Do doctors have knowledge outside of their specialties? (7) When can I trust my doctor?	(1) Are doctors (M.D., D.O., etc.) legally allowed to work outside their specialty in emergency situations? (2) Can doctors (M.D., D.O., etc.) choose to work outside their specialty if they have additional training or certification in another area? (3) Do doctors (M.D., D.O., etc.) ever work outside their specialty due to staffing shortages or in rural areas? (4) Are there any regulations or guidelines that permit doctors (M.D., D.O., etc.) to practice outside their specialty in certain healthcare settings? (5) How often do doctors (M.D., D.O., etc.) work outside their specialty in practice, and what are the common reasons for doing so? (6) Can a doctor who specializes in, for example, cardiology, ever perform surgery? (7) Do doctors ever collaborate with specialists outside their own area of expertise? (8) Are there instances where a doctor might be required to provide care outside their specialty due to an emergency situation? (9) Is it common for doctors to work in multidisciplinary teams that involve other specialties? (10) Can a doctor with a medical degree work in a general practice setting, even if they have specialized training?
Is it illegal to refuse to treat your sick child?	(1) Can parents be arrested if their kid is sick and they don't go to a hospital? (2) What religious exemptions exist for not treating a child? (3) Can I use medicinal practices from my culture/belief system rather than state-prescribed medicinal practices? (4) How much legal authority do parents have over their children's bodily autonomy? (5) Who gets to decide in the case of a medical dispute between a parent and a child?	(1) Is it illegal for parents to refuse medical treatment for their sick child in the state of California? (2) Is it illegal for parents to refuse medical treatment for their sick child in the United Kingdom? (3) Are there any laws that prohibit parents from refusing medical treatment for their sick children? (4) What are the legal consequences for a parent who refuses to seek medical treatment for their sick child? (5) In what situations is it legally permissible for a parent to refuse medical treatment for their sick child? (6) Are there any exceptions to the legal requirement for parents to provide medical treatment to their sick children? (7) What are the ethical considerations surrounding a parent's right to refuse medical treatment for their sick child?
Why don't cashiers just have chairs?	(1) Why don't cashiers sit in chairs when they perform their job? (2) Why do cashiers perform their job standing? (3) What cultural practices causes cashiers to stand? (4) Why does only (store x) allow their cashiers to sit?	(1) What are the ergonomic or health reasons for cashiers not having chairs? (2) Are there any legal or regulatory reasons that prevent cashiers from having chairs? (3) How does the absence of chairs for cashiers affect their job performance or customer service? (4) What are the historical reasons for cashiers not having chairs in stores? (5) Are there any stores or countries where cashiers are provided with chairs, and if so, why? (6) What are the financial implications for businesses if they were to provide chairs for cashiers? (7) How do cashiers feel about not having chairs, and what are their opinions on this matter? (8) What are the potential benefits and drawbacks of providing chairs for cashiers? (9) Do customers prefer interacting with cashiers who are standing or sitting? (10) Is there a cost-effectiveness analysis regarding the provision of chairs for cashiers? (11) What are the ergonomic considerations for cashier workstations, and how do they relate to the use of chairs?

Figure A.8: Human-written and model-generated interpretation spaces for question from r/NoStupidQuestions.

exclusively use the SNLI portion of the δ -NLI dataset, abbreviated as δ -SNLI. We include the split sizes for δ -SNLI in Table A.8 as well.

A.14 Model Training Setup

Neural Models. We use the Hugging Face library to train all of our RoBERTa models. We utilize `roberta-base`, which has 125M trainable parameters. All models were trained on a single NVIDIA 1080-TI GPU. After tuning on the validation set, all models were trained for two epochs with a learning rate of $2e-5$ and a batch size of 32. Tables 4.2 and 4.3 report best accuracy across five runs.

Lexical Models. We use the `fasttext` library to implement our bag-of-words models. `fasttext` is an off-the-shelf text classification library. All lexical models were trained for 5 epochs with 4-grams as the maximum length of word ngrams. We use the default learning rate of 0.1.

Dataset	Split	Size
SNLI	train	550,152
	valid	10,000
	test	10,000
δ -NLI	train	200,694
	valid	14,968
	test	15,414
δ -SNLI	train	88,676
	valid	1,785
	test	1,837

Table A.8: Split sizes across SNLI, δ -NLI, and δ -SNLI.

A.15 Visualizing Distribution Shifts in SNLI Edited Examples

We are able to visualize confidence distribution shifts for δ -NLI before and after editing using a 2D plane with ordered pairs, as in Figure 4.3, due to the label set containing only two update types—weakener and strengthener (so, a probability score > 0.5 results in the predicted label). Since the SNLI label set consists of *three* labels (*entailment*, *neutral*, *contradiction*), we choose to visualize shifts in the confidence distribution before and after editing contexts via ternary plots. For each directional label pair (*entailment* $\rightarrow$ $\{\textit{neutral}, \textit{contradiction}\}$, *neutral* $\rightarrow$ $\{\textit{entailment}, \textit{contradiction}\}$, *contradiction* $\rightarrow$ $\{\textit{entailment}, \textit{neutral}\}$), we plot a heatmap of probabilities, or confidences, in each of the three classes on the simplex with Gaussian smoothing using `python-ternary`, a ternary plotting package (Harper, 2015). Figures A.9, A.10, and A.11 show these visualizations. We include these plots mainly to help visualize information about the predicted labels of the incorrect examples (Table 4.2 only reports the accuracy on finer-grained buckets). We observe that for most classes of examples, the SNLI RoBERTa model utilizes the context, and correctly predicts the new induced gold label. However, the $\{\textit{neutral} \rightarrow \textit{entailment}\}$ class of examples in particular proved difficult for the model, as evidenced by a large chunk of a mass remaining in the neutral corner of the simplex.

A.16 Dataset Limitations

In this work, we choose to explore the role of context with respect to the SNLI and δ -NLI datasets. In particular, the proven presence of strong artifacts in SNLI made it an appealing dataset to explore a model’s behavior with respect to the utilization of context. We chose to include δ -NLI in our analysis, since ultimately, we’d like reasoning systems to operate in complex and *dynamic* contexts. The ability to be sensitive to shifting contexts and understand when default inferences should be overridden by additional context (i.e more nuanced inference) is both central to our exploration and central to the task of defeasible reasoning

itself. Our evaluation set does not include examples sourced from other NLI datasets such as MultiNLI (Williams et al., 2018b). It also does not contain datasets across domains, such as MedNLI (Romanov and Shivade, 2018). However, we note that while the datasets may be different, others have shown artifacts present in such datasets (i.e (Herlihy and Rudinger, 2021b; ?)). Our goal was to utilize datasets containing high amounts of artifacts.

Each story consists of a beginning sentence, an ending sentence, and a middle sentence that is either plausible or implausible. Plausible middle sentences make sense within the flow and context of the story and implausible ones do not necessarily fit within the flow of the story.

Provide three paraphrases for both middle sentences (one on each tab) such that the words or sentence structure substantially differ from the original sentence, but the meaning is retained.

If the sentence is labeled as "plausible", your paraphrase should still be plausible within the context of the story. If the sentence is labeled as "implausible", make sure that your paraphrase retains this property.

Make sure your paraphrases are as different as possible from the other paraphrases you enter and differ as much as possible from the original sentence. Your paraphrases should be **full, grammatical, coherent sentences**. They should **preserve the meaning of the sentence**. Make sure that sentence **meaning** does not differ from the original sentence and **refrain from introducing new information that violates the situation**. It is okay to introduce small bits of information as long as all the properties above are satisfied. For example, "The man obtained a discount" and "The man used a coupon" are permissible paraphrases if and only if they retain the plausibility of the sentence in the context of the story and they fit interchangeably with the flow.

Beginning

George decided to buy a TV.

Middle

Sentence 1 (Plausible Middle)

Sentence 2 (Implausible Middle)

Ending

Things had turned out just as he'd hoped.

Sentence 1 (Plausible Middle)

Complete paraphrases on both tabs!

He took it home, unpacked it, and placed it where he wanted it.

Paraphrase the sentence above in three different ways such that its meaning is retained, but the words themselves substantially differ from those in the original sentence.

Paraphrase #1
After bringing it home, he unpacked the TV and set it up where he intended.
Distance from Original Sentence: 63%

Paraphrase #2
Distance from Original Sentence:

Paraphrase #3
Distance from Original Sentence:

Distance from original sentence that is measured via word overlap. Try to **maximize** this value!
The color is relative to a target range word overlap.
→ Red (0-20%) - Too close to the original sentence
→ Yellow (20-40%) - Some overlap
→ Green (40%+) - Paraphrase is different enough!

Each scenario consists of context, an inference, and an evidence sentence that either *strengthens* or *weakens* the inference. The context and inference set the stage for the scenario. Strengthening evidence makes your belief in the inference stronger, and weakening evidence makes your belief in the inference weaker.

Provide three paraphrases for the evidence sentence such that the words or sentence structure substantially differ from the original sentence, but the meaning is retained.

If the evidence is labeled as "strengthening", your paraphrase should still strengthen your belief in the inference within the context of the scenario. If the sentence is labeled as "weakening" (i.e. it weakens your belief in the inference), make sure that your paraphrase retains this property.

Your paraphrases are as different as possible from the other paraphrases you enter and differ as much as possible from the original sentence.

Your paraphrases should be **full, grammatical, coherent sentences**. They should **preserve the meaning of the sentence**. Make sure that sentence **meaning** does not differ from the original sentence and **refrain from introducing new information that violates the situation**. It is okay to introduce small bits of information as long as all the properties above are satisfied. For example, "The man obtained a discount" and "The man used a coupon" are permissible paraphrases if and only if they retain the plausibility of the sentence in the context of the story and they fit interchangeably with the flow.

Scenario

Context
A group is walking between two giant rock formations.

Inference
A group is hiking.

Strengthening Evidence for Inference

They are wearing hiking gear.

Paraphrase the sentence above in three different ways such that its meaning is retained, but the words or sentence structure substantially differ from the original sentence. Make sure that your three paraphrases are **different from each other** as well.

Paraphrase #1
They are dressed in hiking attire.
Distance from Original Sentence: 63%

Paraphrase #2
Distance from Original Sentence:

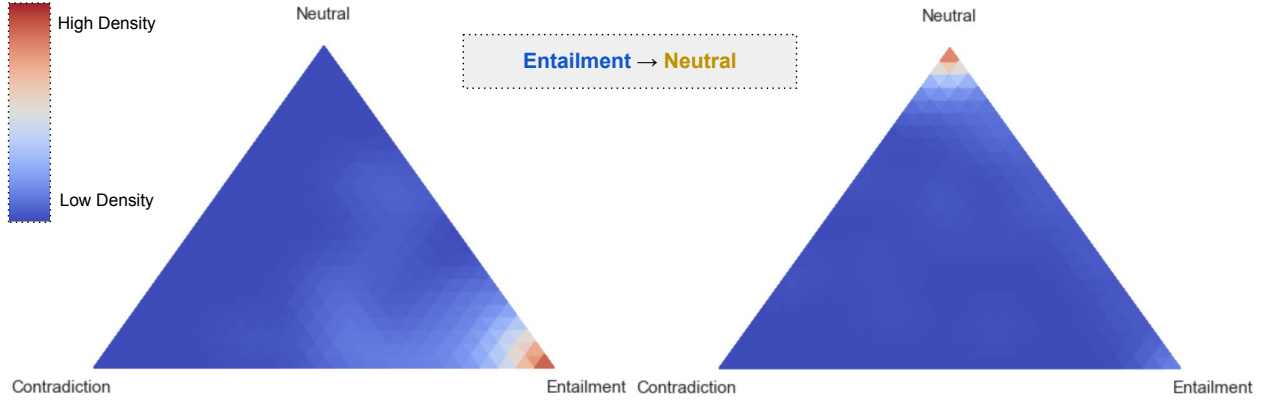
Paraphrase #3
Distance from Original Sentence:

Figure A.12: Annotation instructions and interface for collecting paraphrases of α -NLI (top) and δ -NLI (bottom) reasoning problems. Workers must write three label-preserving paraphrases.

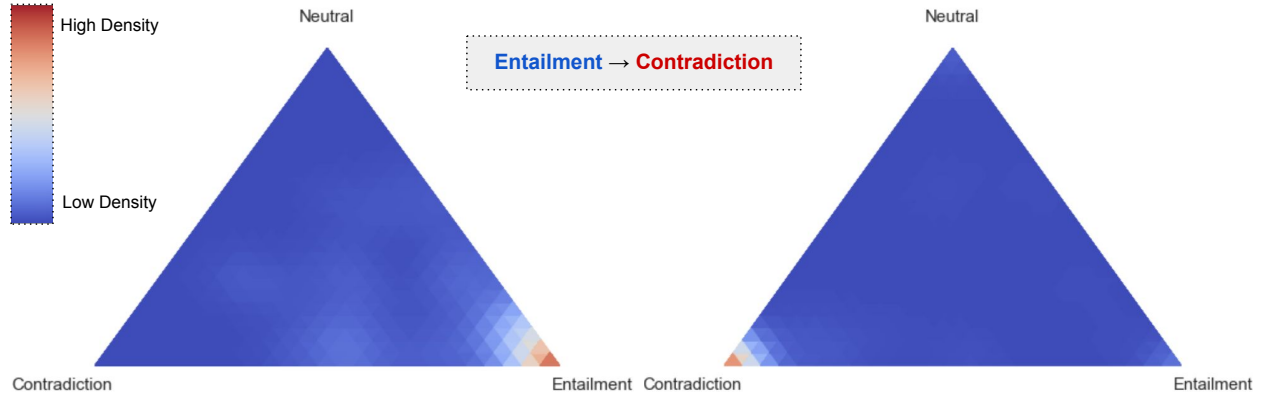
A.17 Crowdsourcing PARANLU

We collect paraphrases in PARANLU using Amazon Mechanical Turk. The instructions and annotation interface shown to crowdworkers is shown in Figure A.12.

Workers provide 3 paraphrases of both plausible and implausible hypotheses for α -NLI examples and 3 paraphrases of updates for δ -NLI examples. We include a distance widget in our interface that computes the Jaccard similarity between the entered paraphrase and the original text to encourage lexical diversity. Each example was annotated by 3 workers. Workers were paid US\$12/hour on average and were required to be native English speakers with a 95% or more HIT acceptance rate on at least 100 HITs.

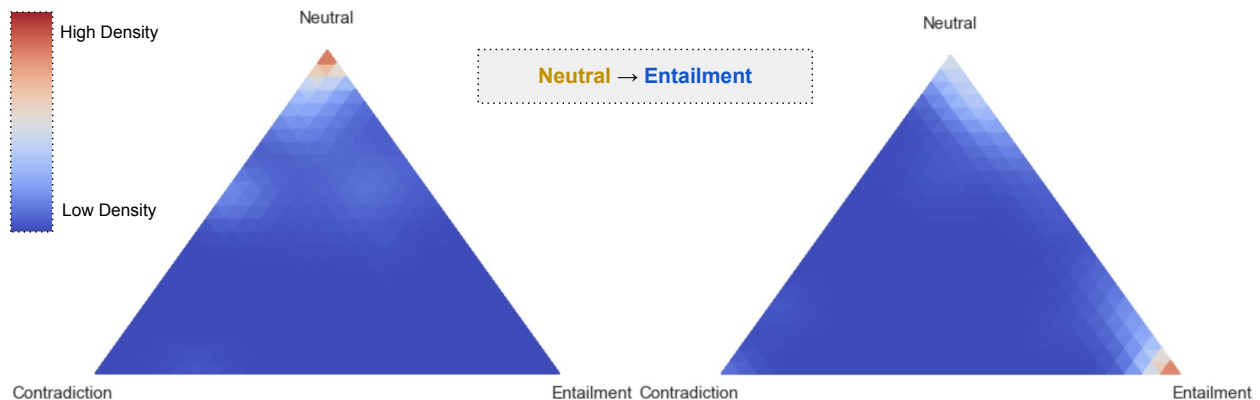


(a) Confidence distribution heatmap for examples with original relation as entailment, edited to induce a neutral relation.

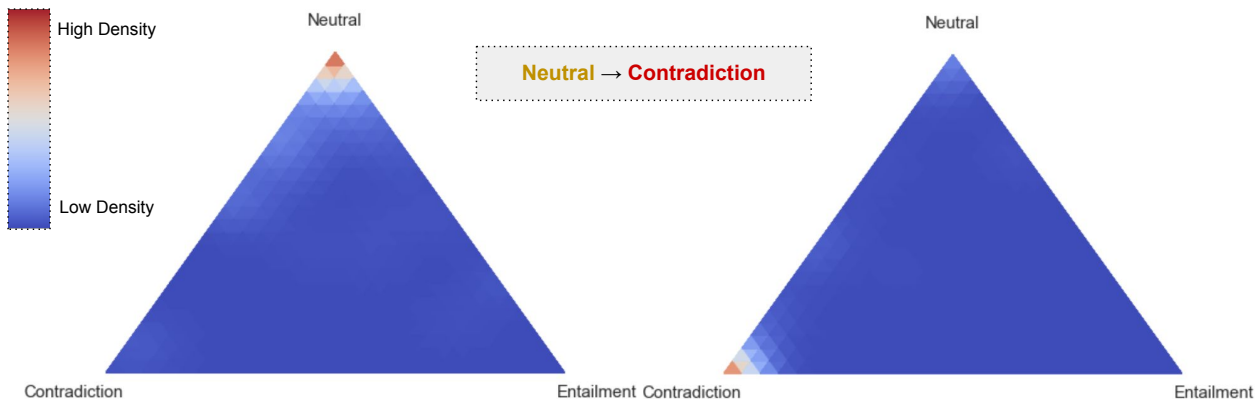


(b) Confidence distribution heatmap for examples with original relation as entailment, edited to induce a contradiction relation.

Figure A.9: Confidence distribution heatmaps for SNLI examples before and after editing examples with original relation as entailment in our evaluation set. A.9a shows entailment examples edited to induce a neutral relation, and A.9b shows examples edited to induce a contradiction relation. Each l and l' is shown in the center box. We note that these distributions help visualize the accuracies presented in Table 4.2.

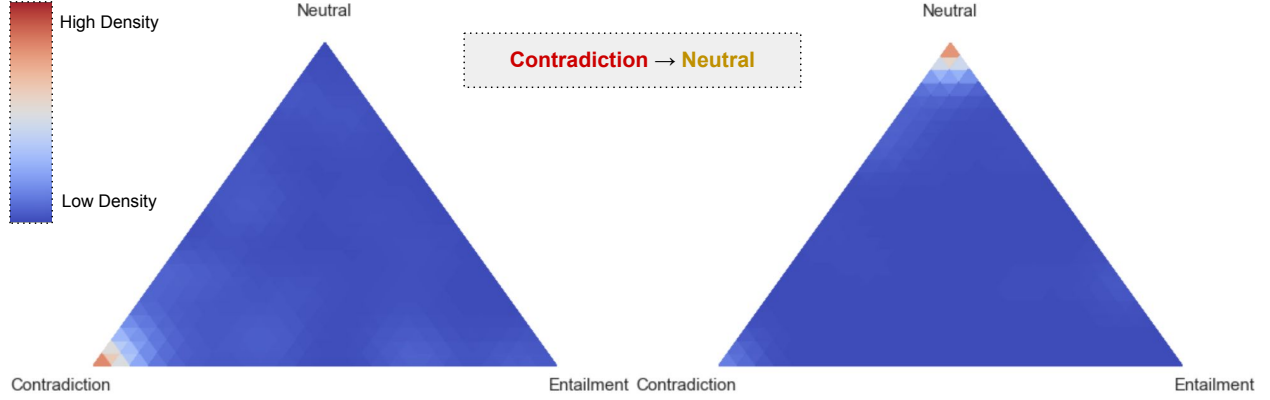


(a) Confidence distribution heatmap for examples with original relation as neutral, edited to induce an entailment relation. We note that this was the hardest edit category for RoBERTa models to flip, and draw attention to a substantial amount of mass still in the neutral corner of the simplex.

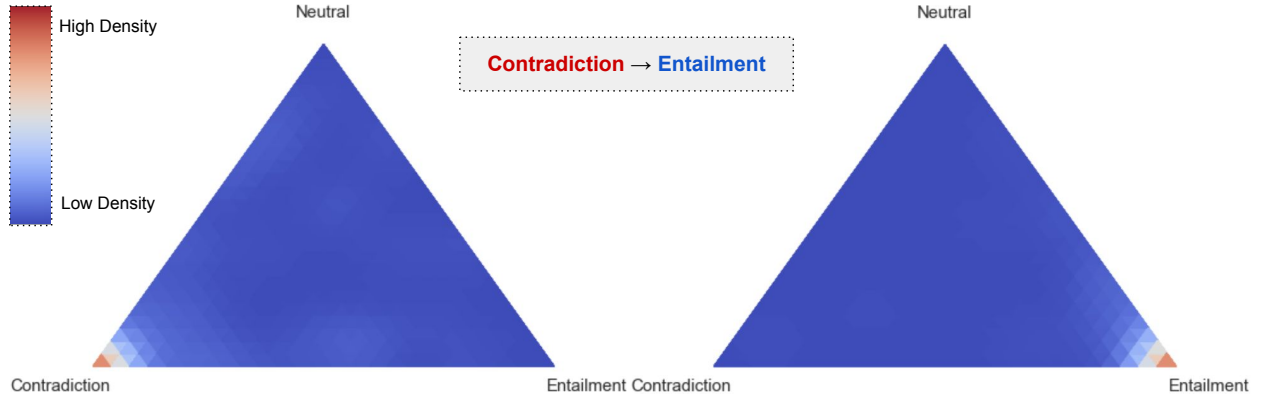


(b) Confidence distribution heatmap for examples with original relation as neutral, edited to induce a contradiction relation.

Figure A.10: Confidence distribution heatmaps for SNLI examples before and after editing examples with original relation as neutral in our evaluation set. [A.10a](#) shows examples edited to induce an entailment relation, and [A.10b](#) shows examples edited to induce a contradiction relation. Each l and l' is shown in the center box. These distributions aid in visualization of the performance metrics reported in Table [4.2](#).



(a) Confidence distribution heatmap for examples with original relation as contradiction, edited to induce a neutral relation.



(b) Confidence distribution heatmap for examples with original relation as contradiction, edited to induce an entailment relation. We note that this particular class of edited examples achieved the highest accuracy, reflected in the low density of examples away from the entailment corner of the simplex.

Figure A.11: Confidence distribution heatmaps for SNLI examples before and after editing examples with original relation as contradiction in our evaluation set. A.11a shows examples edited to induce a neutral relation, and A.11b shows examples edited to induce an entailment relation. Each l and l' is shown in the center box. These plots help visualize the performance metrics reported in Table 4.2.

A.18 Paraphrastic Consistency

Figure A.13 shows model accuracy plotted against corrected paraphrastic consistency of all models tested for the δ -ATOMIC and δ -SOCIAL splits of PARANLU.

A.19 Reddit Question Filtering

A.19.1 Discourse Markers

Assumption Correcting Markers: "however,", "actually,", "as a matter of fact", "in fact", "not true", "despite what you", "on the contrary", "common misconception", "not exactly", "just to clarify", "you're confusing", "correct me if i'm wrong", "correct me if i am wrong", "you're wrong", "we have to remember that", "while that's true", "could be dangerous", "might not be the best thing"

Expertise Invoking Markers: "as a doctor", "as a medical professional", "i'm a doctor", "being a doctor", "as a nurse", "i'm a nurse", "i'm a medical professional", "being a nurse"

Prompt A.19.1: Medical vs. Non-Medical Question Identification

Prompt: You are an expert in maternal and infant health who specializes in finding out whether a question posed by a new or expecting mother is seeking opinion/community participation, or whether it is a medical question. Given a question, you must answer whether it is question seeking medical advice or if it is seeking personal anecdotes and sharing experience. If it's seeking medical advice, answer with "medical". Otherwise, answer "non-medical". If a question is under-specified, answer with "non-medical".

Prompt A.19.2: Reddit Question Rewriting

Prompt: You will be shown questions about maternal and infant health asked by users. Each question contains a TITLE and DESCRIPTION that elaborates on it, containing details that are both relevant and irrelevant to answering the question. Given a question TITLE and a DESCRIPTION, your task is to incorporate only the relevant details from the DESCRIPTION and rewrite the TITLE into a REWRITE. If there are no relevant details, return the TITLE. As a general rule, keep the rewrite as similar to the original question as possible. The rewrite should be a question in a single sentence.

Title: How to Stop Co-Sleeping

Description: ...

Rewrite: How to wean my 11-month-old out of Co-Sleeping?

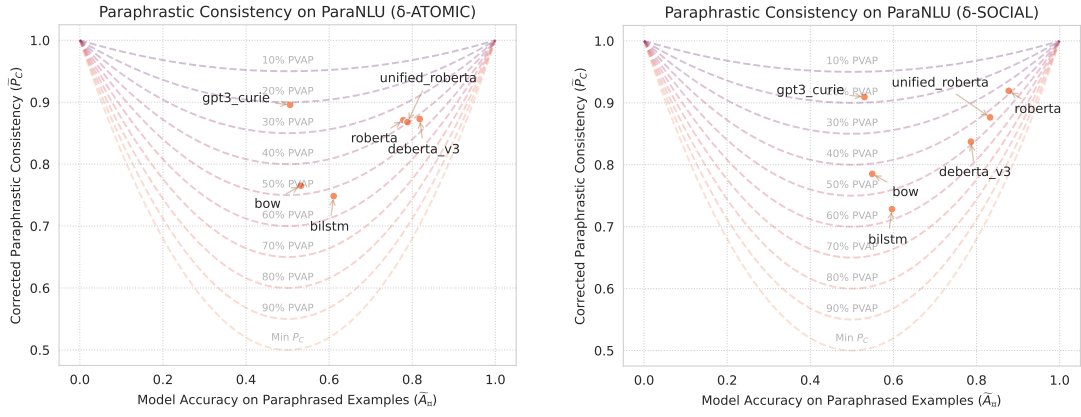


Figure A.13: Paraphrastic consistency ($\tilde{P}_C$) of different models on δ -ATOMIC and δ -SOCIAL paraphrased examples.

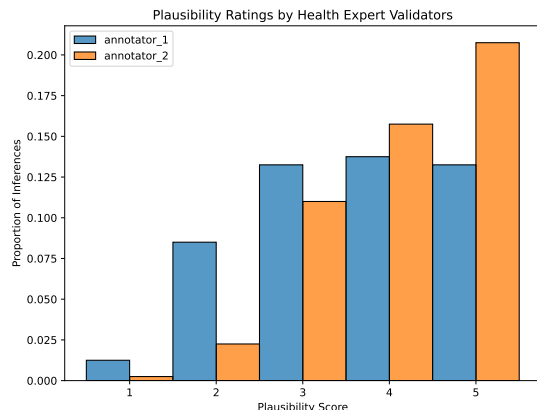


Figure A.14: Ratings of expert validators of the plausibility of inferences written by health experts in our dataset. The majority of inferences are plausible.

A.20 Consolidating Subquestions and Assumptions and Implications into Pragmatic Inferences

Prompt A.20.1: Question Consolidation

Prompt: Given questions asked by new or expecting mothers, your task is to identify the assumptions in them. For this task, you will be given a QUESTION asked by a new or expecting mother, some ASSUMPTIONS (as a list of beliefs or assumptions) in those questions identified by health experts, and some possible SUBQUESTIONS (as a list) that public health experts have identified to have the same information goals as the original question. Given all three of these, your task is to consolidate the SUBQUESTIONS and ASSUMPTIONS into a single, exhaustive list, called INFERENCES. Turning a SUBQUESTION into an inference may involve just turning it into a declarative sentence, or identifying the assumptions made in the SUBQUESTION. Finally, add the INFERENCES to the list of ASSUMPTIONS and remove any duplicates.

A.21 Further Details on Inference Validation

Plausibility scores are the outcome of a three-stage process: (1) a pregnant or new mother holding a belief that is latent while asking a question, (2) a maternal health expert reasoning about these latent beliefs of the mother from the question text, and finally (3) a different expert estimating the likelihood of the beliefs extracted in Step 2 of this process.

The plausibility distribution in Figure A.14 represents the results of Step 3. It is important to note that the humans involved in each step of the process are completely disjoint, and have little to no information about each other. In 90% of inferences, the validators agree that the extracted inferences in Step 2 are indeed plausible. The 10% of inferences that validators found less plausible is more a reflection of the subjectivity of pragmatic inference in QA in general (see lines 686-690) than our usage of GPT-3.5 as a textual transformation tool.

In fact, experts themselves can disagree about answers to questions or background infer-

Your Task

The question below is asked by a pregnant or postpartum mother on a topic related to pregnancy and postpartum symptoms, developmental milestones, infant care, etc. We will present you with two answers to this question. **Please read the two answers and provide a score for each on a discrete scale from 1 to 5 evaluating the quality of answer completeness.** If both answers are similarly poor, or similarly strong, you can give them similar scores. A high quality answer should identify the information needs of the question and make sure any assumptions are addressed without excluding any major content you feel is necessary in a response to this question. Higher quality answers may provide more comprehensive information or include cautionary caveats.

Figure A.15: Human evaluation instructions provided to two expert annotators.

	Mean Likert Score: Inference-	
	Default Answer	Augmented
Expert Annotator 1	4.45	4.6
Expert Annotator 2	4.37	4.45

Table A.9: Human preferences of answers on questions with at least one high-plausibility false assumption.

ences (a finding echoed in [Xu et al. \(2023\)](#)), so we include the plausibility study to reassure audiences that experts do find the inferences in our dataset plausible.

A.22 Human Evaluation Details

Human Evaluation Annotation Instructions. See Figure [A.15](#) for annotator instructions.

Annotator preferences on questions with a highly plausible, false assumption. Human annotators prefer inference-augmented answers for questions with a highly plausible, false assumption. See Table [A.9](#).

A.23 QA Model Prompts

A.23.1 Reranker Instructions

Prompt A.23.1: TART Baseline Instruction

Retrieve a passage from medical articles on the web that answers the following question.

Prompt A.23.2: TART Pragmatic Inference Instruction

I want to check if the following assumption is true or false. Retrieve an evidence passage for me from medical articles on the web.

A.23.2 Reader Prompts

Prompt A.23.3: Flan-T5 Reader Prompt

You are an expert in maternal and infant health. You are given a few passages below:

Context: {context}

Using information only from the passages above, respond to the following question with a helpful and complete answer. Use information from multiple passages if necessary to answer the question. Again, you are allowed to only use information from the passages above.

Question: {question}

Prompt A.23.4: Mistral/GPT Reader Prompt

You are an expert in maternal and infant health. You are given a few verified pieces of information below:

Context: {context}

Using evidence only from verified pieces of information above, respond to the following question with a helpful and complete answer. Use information from multiple sources above if necessary to answer the question. In your answer, do not reveal that you are fetching information from these pieces of evidence. Use information only from the verified sources above, and not from any other sources.

Question: {question}

Answer:

Prompt A.23.5: Inference-Augmented Reader Prompt

You are an expert in maternal and infant health. You are given a few verified pieces of information below:

Context: {context}

Using information only from verified pieces of information above, respond to the following question with a helpful and complete answer. As humans often make assumptions while asking questions, your answer must address assumptions made by the asker listed below.

Assumptions: {assumptions}

Use information from multiple sources above if necessary to answer the question below and address the ASSUMPTIONS. In your answer, do not reveal that you are fetching information from these pieces of evidence. Use information only from the verified sources above, and not from any other sources.

Question: {question}

Answer:

Question (Q)	Default Answer (A_D)	Pragmatic Inference-Augmented Answer (A_{IA})
When should I give my baby fever reducing meds after shots?	You should give your baby fever reducing medication after shots if they are experiencing discomfort or extreme discomfort due to the fever. However, it is important to note that in most cases, a child does not need to be medicated for a fever unless they are extremely uncomfortable. The most important thing to do when a child has a fever is to make sure they stay hydrated by drinking plenty of fluids. If you are unsure whether your child should receive medication to address the fever or other symptoms, it is recommended to consult with your child's healthcare provider.	It is not recommended to give your baby fever reducing medication before or immediately after receiving shots. Studies have shown that giving medication prior to a vaccine visit may decrease the child's immune response to the vaccine. Fevers after vaccination are actually a sign that the immune system is responding to the vaccine. If your baby develops a fever after receiving shots, it is generally not necessary to give them medication unless they are extremely uncomfortable. The most important thing to do when a child has a fever is to make sure they stay hydrated by drinking plenty of fluids. If you are unsure whether your baby should receive medication to address the fever or other symptoms, it is best to consult with your child's healthcare provider.
How warm can my bath be while staying safe for my 4-month-old baby?	To ensure the safety of your 4-month-old baby during bath time, it is recommended to fill the basin with water that feels warm, but not hot, to the inside of your wrist or elbow. The water temperature should be comfortable for your baby and should not pose a risk of burns or scalds. It is important to check the water temperature before placing your baby in the bath to prevent any potential harm.	The safe temperature for a baby may be different from that for an adult. It is important to ensure that the bathwater is warm, but not hot, for a 4-month-old baby. According to the provided information, the water should feel warm to the inside of your wrist or elbow. It is recommended to fill the basin with 2 inches of water at this warm temperature. This is to prevent the baby from getting burned or scalded by hot water. It is important to note that babies are not as adaptable as adults to temperature change, and they can lose heat rapidly. Therefore, it is crucial to maintain a safe and comfortable water temperature for the baby during bath time.
Can pregnancy tests be trusted?	According to pregnancy kit manufacturers, most at-home pregnancy tests are 98% to 99% accurate when used exactly as instructed. This means that positive results can generally be trusted. However, it is important to note that false negative results can occur if the test is taken too soon. Therefore, it is recommended to wait until a sufficient amount of time has passed since the possible conception before taking a pregnancy test to ensure more accurate results.	According to the information provided by pregnancy kit manufacturers, most at-home pregnancy tests are 98% to 99% accurate when used exactly as instructed. Positive results from these tests can be trusted. However, it is important to note that false negative results can occur if the test is taken too soon. Blood tests, on the other hand, can detect pregnancy sooner and are the only tests that can show specific amounts of HCG in the body...

Table A.10: Outputs generated by a standard QA system and a QA systems informed by pragmatic inferences. Answers from the pragmatic system better explicate their underlying inferences.

A.24 Inference Generation Prompt

Prompt A.24.1: Inference Generation Prompt

When humans ask questions, they often have certain assumptions or implications that are embedded in the questions. These assumptions and implications may be true or false, and they may or may not be present in the surface form of the question. Given a question asked by a new or expecting mother, your task is to identify all relevant assumptions and implications in these questions and write them in a list titled INFERENCES. Each inference under INFERENCES should be an independent and declarative assertion that represents an assumption or an implication that the speaker makes while asking the question.

Consider the question asked by an expectant mother to their doctor: **"What kind of music should I play my baby in the womb?"** This question implicitly makes some assumptions that may or may not be true:

1. Babies can hear sound in the womb.
2. Babies can differentiate music from other sounds.
3. Hearing music positively influences fetal development.
4. (Assuming #3) Certain genres of music are more beneficial than others.

5. There is a baby in my womb.
6. There are different kinds of music.
7. Music is something that can be played.

Unlike the assumptions in the blue box, the assumptions in the red box are linguistically *valid*, but are trivial. Identifying them doesn't help us write more helpful answers that address them. A complete answer to this question might address several sub-questions:

1. Does my baby hear sounds in the womb?
2. Can they differentiate music from other sounds?
3. Will listening to music somehow influence their fetal development?
4. Do particular genres of music influence fetal development more than others?

When patients ask questions to their doctors, they may implicitly make problematic, subjective, or false assumptions that are then explicitly corrected by their doctor. This process is often seamlessly carried out in human conversation:

Patient: "How many glasses of wine a day can I drink while pregnant?"
Doctor: "Actually, there's no safe amount of alcohol to drink during pregnancy! Growing babies are exposed to the same amount of alcohol as you are, and the alcohol will pass through your placenta to your baby."

Because this question starts with "how many", the doctor inferred that the patient believed that there was an alcohol amount that was safe to consume. Rather than answering "0 glasses are safe", the doctor corrected this assumption, and explained the consequences of drinking any alcohol while pregnant. On the other hand, chatbots may take questions at face value, failing to correct potentially harmful assumptions in questions. As a first step, we want to ensure that chatbots can identify assumptions or implications inherent in user questions.

Your Task

- 1 Write a list of sub questions that an answer to the original question would answer. Answers to these sub questions should reflect a complete and exhaustive answer to the original question.
- 2 Write a set of sentences reflecting possible beliefs or assumptions that the patient may hold (or, alternatively, beliefs that a complete answer to the question must address). The assumptions you write can be either medically or scientifically true or false. Please write all assumptions of both types you can identify. There is no set number of assumptions you can extract: many questions may contain more than one assumption.

Consider the question asked by an expectant mother to their doctor: **"What should I consume to increase breast milk?"** This question implicitly makes some assumptions that may or may not be true:

1. Breast milk production is affected by certain foods.
2. Increasing milk production is desirable.
3. Consumption of particular food or beverages is the optimal way to increase milk production.

When patients ask questions to their doctors, they may implicitly make these sorts of problematic, subjective, or false assumptions that are then explicitly corrected by their doctor. In this task you will be shown:

- **a patient question:** this question was asked by a pregnant or postpartum mother and is related to experiences around pregnancy, postpartum, or caring for an infant.
- **a set of possible assumptions:** these sentences reflect possible beliefs that the question asker may hold, or, alternatively, beliefs that any complete answer to the question must address.

Your Task

You will be given a question and a set of possible assumptions. You must first verify whether or not the assumption is medically or scientifically true (or sound). Some assumptions may be true, and others may be false. In order to mark an assumption as true or false, you must find accompanying evidence from a web article from a domain listed below.

For each assumption associated with a given patient question, first, label whether the assumption is medically or scientifically true by finding a web document. Then, find one or more pieces of text from that web document that explicitly provides supporting or refuting the assumption. This span of text can be a phrase, a sentence, or an entire paragraph.

Your evidence to support or refute a particular assumption must come from a trusted source document. Do not use sources like Wikipedia, personal blogs, or popular consumer health websites like WebMD or Healthline. Viable and reputable sources include US government websites (left) or US Hospitals or University Clinics (right):

Center for Disease Control (CDC): <https://www.cdc.gov/index.htm>
US Department of Health and Human Services: <https://health.gov/>
Office of Women's Health (from Health and Human Services): <https://www.womenshealth.gov/>
usa.gov
National Library of Medicine: <https://www.nlm.nih.gov/>
National Institute of Health: <https://www.nih.gov/>
National Institute of Child Health and Human Development: <https://www.nichd.nih.gov/>

Mayo Clinic: <https://www.mayoclinic.org/>
Seattle Children's: <https://www.seattlechildrens.org/>
Hopkins Children's: <https://www.hopkinsmedicine.org/johns-hopkins-childrens-center>
Yale Medicine: <https://www.yalemedicine.org>
Children's Hospital of Philadelphia: <https://www.chop.edu/>

A larger list of trusted knowledge sources websites can be found [here](https://www.nlm.nih.gov/healthcare/providers/clinicalandresearch/clinicaltrials/clinicaltrials.html). Note that you may use sources outside the above-mentioned sources as long as they are similarly reputable. **Please abstain from using blogs or for-profit websites such as WebMD, Healthline, or other consumer health websites.** Your answer may contain text verbatim from the sources you visit to write a complete answer to the question, but you can also paraphrase or summarize text across multiple sources.

Please do not use ChatGPT or other generative AI systems to write your answers, extract evidence, or summarize long documents.

Figure A.16: Instructions given to each annotator for each phase of annotation. First, we show questions to annotators and ask them to write sub questions and the assumptions present (top panel). Then, after passing these outputs to a prompt-based model to extract consolidated inferences, we ask a different set of annotators to *verify* the veracity of the inferences along with supporting evidence (middle panel). Simultaneously, we ask a third set of annotators to write answers to questions without any inference supervision (bottom panel).

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